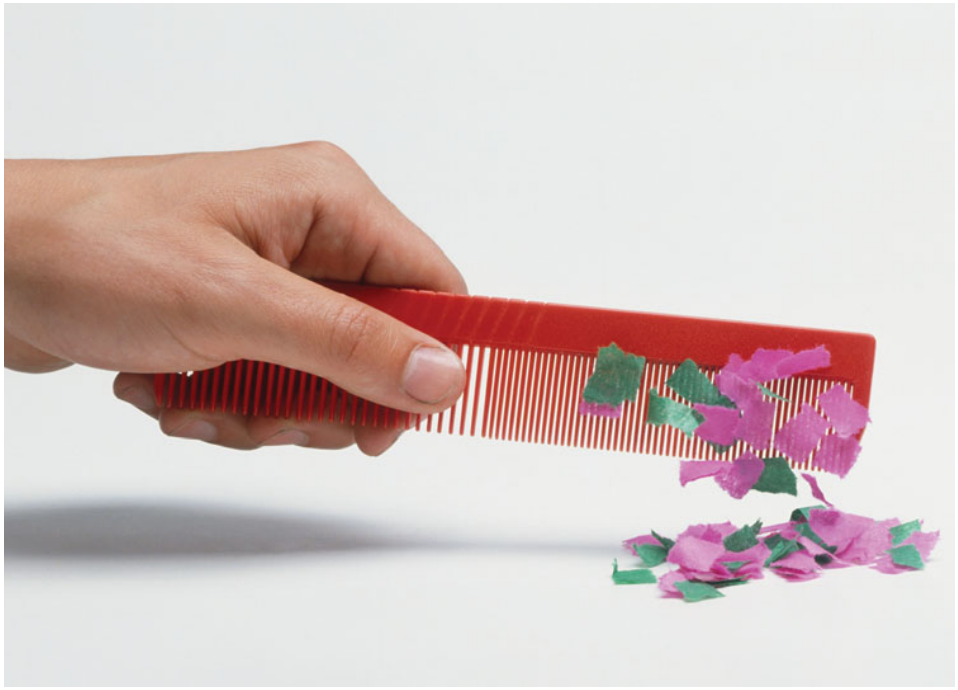


This comb has acquired a static electric charge, either from passing through hair, or being rubbed by a cloth or paper towel. The electrical charge on the comb induces a polarization (separation of charge) in scraps of paper, and thus attracts them.

Our introduction to electricity in this Chapter covers conductors and insulators, and Coulomb's law which relates the force between two point charges as a function of their distance apart. We also introduce the powerful concept of electric field.

CHAPTER 16

Electric Charge and Electric Field

CHAPTER-OPENING QUESTION—Guess now!

Two identical tiny spheres have the same electric charge. If their separation is doubled, the force each exerts on the other will be

(a) half.
(b) double.
(c) four times larger.
(d) one-quarter as large.
(e) unchanged.

The word "electricity" may evoke an image of complex modern technology: lights, motors, electronics, and computers. But the electric force plays an even deeper role in our lives. According to atomic theory, electric forces between atoms and molecules hold them together to form liquids and solids, and electric forces are also involved in the metabolic processes that occur within our bodies. Many of the forces we have dealt with so far, such as elastic forces, the normal force, and friction and other contact forces (pushes and pulls), are now considered to result from electric forces acting at the atomic level. Gravity, on the other hand, is a separate force.[†]

[†]As we discussed in Section 5–9, physicists in the twentieth century came to recognize four different fundamental forces in nature: (1) gravitational force, (2) electromagnetic force (we will see later that electric and magnetic forces are intimately related), (3) strong nuclear force, and (4) weak nuclear force. The last two forces operate at the level of the nucleus of an atom. Recent theory has combined the electromagnetic and weak nuclear forces so they are now considered to have a common origin known as the electroweak force. We discuss the other forces in later Chapters.

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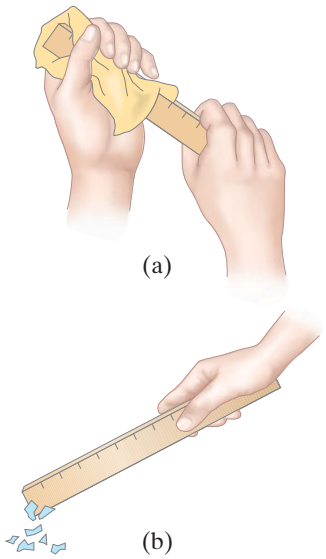
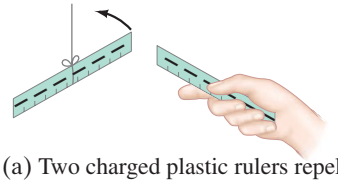
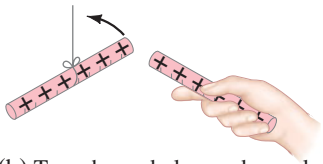


FIGURE 16–1 (a) Rub a plastic ruler with a cloth or paper towel, and (b) bring it close to some tiny pieces of paper.

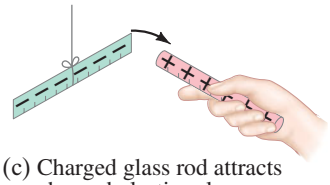
FIGURE 16–2 Like charges repel one another; unlike charges attract. (Note color coding: we color positive charged objects pink or red, and negative charges blue-green. We use these colors especially for point charges, but not always for real objects.)



(a) Two charged plastic rulers repel



(b) Two charged glass rods repel



(c) Charged glass rod attracts charged plastic ruler

LAW OF CONSERVATION OF ELECTRIC CHARGE

The earliest studies on electricity date back to the ancients, but only since the late 1700s has electricity been studied in detail. We will discuss the development of ideas about electricity, including practical devices, as well as its relation to magnetism, in the next seven Chapters.

16–1 Static Electricity; Electric Charge and Its Conservation

The word *electricity* comes from the Greek word *elektron*, which means "amber." Amber is petrified tree resin, and the ancients knew that if you rub a piece of amber with a cloth, the amber attracts small pieces of leaves or dust. A piece of hard rubber, a glass rod, or a plastic ruler rubbed with a cloth will also display this "amber effect," or **static electricity** as we call it today. You can readily pick up small pieces of paper with a plastic comb or ruler that you have just vigorously rubbed with even a paper towel. See the photo on the previous page and Fig. 16–1. You have probably experienced static electricity when combing your hair or when taking a synthetic blouse or shirt from a clothes dryer. And you may have felt a shock when you touched a metal doorknob after sliding across a car seat or walking across a synthetic carpet. In each case, an object becomes "charged" as a result of rubbing, and is said to possess a net **electric charge**.

Is all electric charge the same, or is there more than one type? In fact, there are *two* types of electric charge, as the following simple experiments show. A plastic ruler suspended by a thread is vigorously rubbed with a cloth to charge it. When a second plastic ruler, which has been charged in the same way, is brought close to the first, it is found that one ruler *repels* the other. This is shown in Fig. 16–2a. Similarly, if a rubbed glass rod is brought close to a second charged glass rod, again a repulsive force is seen to act, Fig. 16–2b. However, if the charged glass rod is brought close to the charged plastic ruler, it is found that they *attract* each other, Fig. 16–2c. The charge on the glass must therefore be different from that on the plastic. Indeed, it is found experimentally that all charged objects fall into one of two categories. Either they are attracted to the plastic and repelled by the glass; or they are repelled by the plastic and attracted to the glass. Thus there seem to be two, and only two, types of electric charge. Each type of charge repels the same type but attracts the opposite type. That is: **unlike charges attract; like charges repel**.

The two types of electric charge were referred to as **positive** and **negative** by the American statesman, philosopher, and scientist Benjamin Franklin (1706–1790). The choice of which name went with which type of charge was arbitrary. Franklin's choice set the charge on the rubbed glass rod to be positive charge, so the charge on a rubbed plastic ruler (or amber) is called negative charge. We still follow this convention today.

Franklin argued that whenever a certain amount of charge is produced on one object, an equal amount of the opposite type of charge is produced on another object. The positive and negative are to be treated *algebraically*, so during any process, the net change in the amount of charge produced is zero. For example, when a plastic ruler is rubbed with a paper towel, the plastic acquires a negative charge and the towel acquires an equal amount of positive charge. The charges are separated, but the sum of the two is zero.

This is an example of a law that is now well established: the **law of conservation of electric charge**, which states that

the net amount of electric charge produced in any process is zero;

or, said another way,

no net electric charge can be created or destroyed.

If one object (or a region of space) acquires a positive charge, then an equal amount of negative charge will be found in neighboring areas or objects. No violations have ever been found, and the law of conservation of electric charge is as firmly established as those for energy and momentum.

16–2 Electric Charge in the Atom

Only within the past century has it become clear that an understanding of electricity originates inside the atom itself. In later Chapters we will discuss atomic structure and the ideas that led to our present view of the atom in more detail. But it will help our understanding of electricity if we discuss it briefly now.

A simplified model of an atom shows it as having a tiny but massive, positively charged nucleus surrounded by one or more negatively charged electrons (Fig. 16–3). The nucleus contains protons, which are positively charged, and neutrons, which have no net electric charge. All protons and all electrons have exactly the same magnitude of electric charge; but their signs are opposite. Hence neutral atoms, having no net charge, contain equal numbers of protons and electrons. Sometimes an atom may lose one or more of its electrons, or may gain extra electrons, in which case it will have a net positive or negative charge and is called an **ion**.

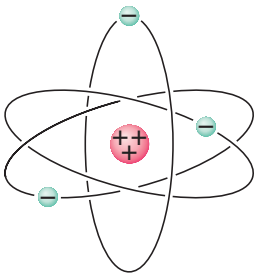
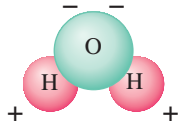


FIGURE 16–3 Simple model of the atom.

In solid materials the nuclei tend to remain close to fixed positions, whereas some of the electrons may move quite freely. When an object is *neutral*, it contains equal amounts of positive and negative charge. The charging of a solid object by rubbing can be explained by the transfer of electrons from one object to the other. When a plastic ruler becomes negatively charged by rubbing with a paper towel, electrons are transferred from the towel to the plastic, leaving the towel with a positive charge equal in magnitude to the negative charge acquired by the plastic. In liquids and gases, nuclei or ions can move as well as electrons.

Normally when objects are charged by rubbing, they hold their charge only for a limited time and eventually return to the neutral state. Where does the charge go? Usually the excess charge "leaks off" onto water molecules in the air. This is because water molecules are **polar**—that is, even though they are neutral, their charge is not distributed uniformly, Fig. 16–4. Thus the extra electrons on, say, a charged plastic ruler can "leak off" into the air because they are attracted to the positive end of water molecules. A positively charged object, on the other hand, can be neutralized by transfer of loosely held electrons from water molecules in the air. On dry days, static electricity is much more noticeable since the air contains fewer water molecules to allow leakage of charge. On humid or rainy days, it is difficult to make any object hold a net charge for long.

FIGURE 16–4 Diagram of a water molecule. Because it has opposite charges on different ends, it is called a "polar" molecule.



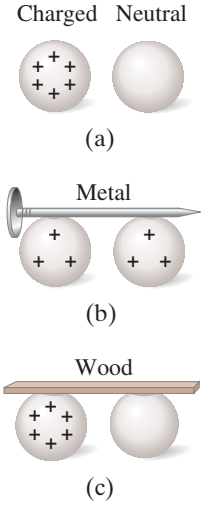
16–3 Insulators and Conductors

Suppose we have two metal spheres, one highly charged and the other electrically neutral (Fig. 16–5a). If we now place a metal object, such as a nail, so that it touches both spheres (Fig. 16–5b), the previously uncharged sphere quickly becomes charged. If, instead, we had connected the two spheres by a wooden rod or a piece of rubber (Fig. 16–5c), the uncharged ball would not become noticeably charged. Materials like the iron nail are said to be **conductors** of electricity, whereas wood and rubber are **nonconductors** or **insulators**.

Metals are generally good conductors, whereas most other materials are insulators (although even insulators conduct electricity very slightly). Nearly all natural materials fall into one or the other of these two distinct categories. However, a few materials (notably silicon and germanium) fall into an intermediate category known as **semiconductors**.

From the atomic point of view, the electrons in an insulating material are bound very tightly to the nuclei. In a good metal conductor, on the other hand, some of the electrons are bound very loosely and can move about freely within the metal (although they cannot *leave* the metal easily) and are often referred to as **free electrons** or **conduction electrons**. When a positively charged object is brought close to or touches a conductor, the free electrons in the conductor are attracted by this positively charged object and move quickly toward it. If a negatively charged object is brought close to the conductor, the free electrons in the conductor move swiftly away from it. In a semiconductor, there are many fewer free electrons, and in an insulator, almost none.

FIGURE 16–5 (a) A charged metal sphere and a neutral metal sphere. (b) The two spheres connected by a conductor (a metal nail), which conducts charge from one sphere to the other. (c) The original two spheres connected by an insulator (wood); almost no charge is conducted.



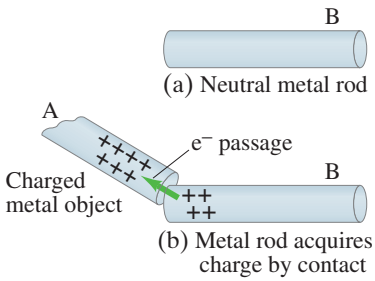


FIGURE 16–6 A neutral metal rod in (a) will acquire a positive charge if placed in contact (b) with a positively charged metal object. (Electrons move as shown by the green arrow.) This is called charging by conduction.

16–4 Induced Charge; the Electroscope

Suppose a positively charged metal object A is brought close to an uncharged metal object B. If the two touch, the free electrons in the neutral one are attracted to the positively charged object and some of those electrons will pass over to it, Fig. 16–6. Since object B, originally neutral, is now missing some of its negative electrons, it will have a net positive charge. This process is called **charging by conduction**, or "by contact," and the two objects end up with the same sign of charge.

Now suppose a positively charged object is brought close to a neutral metal rod, but does not touch it. Although the free electrons of the metal rod do not leave the rod, they still move within the metal toward the external positive charge, leaving a positive charge at the opposite end of the rod (Fig. 16–7b). A charge is said to have been *induced* at the two ends of the metal rod. No net charge has been created in the rod: charges have merely been *separated*. The net charge on the metal rod is still zero. However, if the metal is separated into two pieces, we would have two charged objects: one charged positively and one charged negatively. This is **charging by induction**.

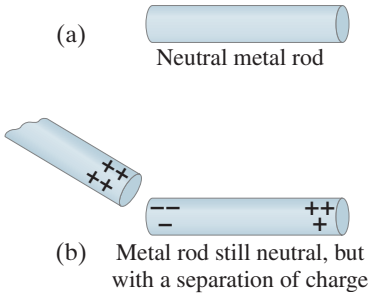


FIGURE 16–7 Charging by induction: if the rod in (b) is cut into two parts, each part will have a net charge.

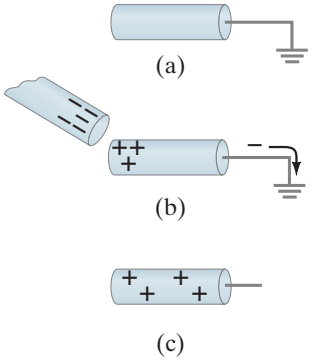


FIGURE 16–8 Inducing a charge on an object connected to ground.

FIGURE 16–9 A charged object brought near a nonconductor causes a charge separation within the nonconductor's molecules.

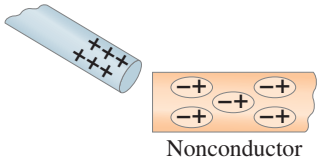
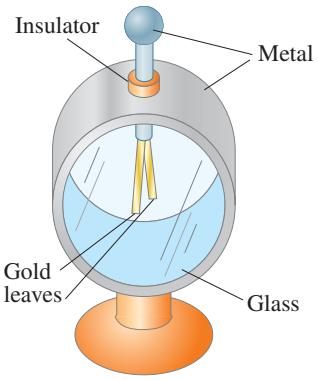


FIGURE 16–10 Electroscope.



Another way to induce a net charge on a metal object is to first connect it with a conducting wire to the ground (or a conducting pipe leading into the ground) as shown in Fig. 16–8a (the symbol ⏚ means connected to "ground"). The object is then said to be **grounded** or "earthed." The Earth, because it is so large and can conduct, easily accepts or gives up electrons; hence it acts like a reservoir for charge. If a charged object—say negative this time—is brought up close to the metal object, free electrons in the metal are repelled and many of them move down the wire into the Earth, Fig. 16–8b. This leaves the metal positively charged. If the wire is now cut, the metal object will have a positive induced charge on it (Fig. 16–8c). If the wire is cut *after* the negative object is moved away, the electrons would all have moved from the ground back into the metal object and it would be neutral again.

Charge separation can also be done in nonconductors. If you bring a positively charged object close to a neutral nonconductor as shown in Fig. 16–9, almost no electrons can move about freely within the nonconductor. But they can move slightly within their own atoms and molecules. Each oval in Fig. 16–9 represents a molecule (not to scale); the negatively charged electrons, attracted to the external positive charge, tend to move in its direction within their molecules. Because the negative charges in the nonconductor are nearer to the external positive charge, the nonconductor as a whole is attracted to the external positive charge (see the Chapter-Opening Photo, page 443).

An **electroscope** is a device that can be used for detecting charge. As shown in Fig. 16–10, inside a case are two movable metal leaves, often made of gold foil, connected to a metal knob on the outside. (Sometimes only one leaf is movable.)

If a positively charged object is brought close to the knob, a separation of charge is induced: electrons are attracted up into the knob, and the leaves become positively charged, Fig. 16–11a. The two leaves repel each other as shown, because they are both positively charged. If, instead, the knob is charged by conduction (touching), the whole apparatus acquires a net charge as shown in Fig. 16–11b. In either case, the greater the amount of charge, the greater the separation of the leaves.

Note that you cannot tell the sign of the charge in this way, since negative charge will cause the leaves to separate just as much as an equal amount of positive charge; in either case, the two leaves repel each other. An electroscope can, however, be used to determine the sign of the charge if it is first charged by conduction: say, negatively, as in Fig. 16–12a. Now if a negative object is brought close, as in Fig. 16–12b, more electrons are induced to move down into the leaves and they separate further. If a positive charge is brought close instead, the electrons are induced to flow upward, so the leaves are less negative and their separation is reduced, Fig. 16–12c.

The electroscope was used in the early studies of electricity. The same principle, aided by some electronics, is used in much more sensitive modern **electrometers**.

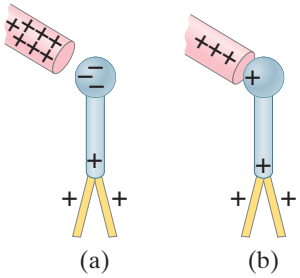
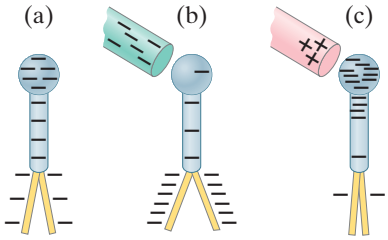


FIGURE 16–11 Electroscope charged (a) by induction, (b) by conduction.

FIGURE 16–12 A previously charged electroscope can be used to determine the sign of a charged object.



16–5 Coulomb's Law

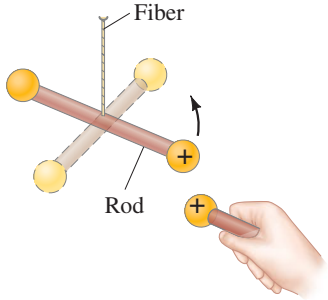
We have seen that an electric charge exerts a force of attraction or repulsion on other electric charges. What factors affect the magnitude of this force? To find an answer, the French physicist Charles Coulomb (1736–1806) investigated electric forces in the 1780s using a torsion balance (Fig. 16–13) much like that used by Cavendish for his studies of the gravitational force (Chapter 5).

Precise instruments for the measurement of electric charge were not available in Coulomb's time. Nonetheless, Coulomb was able to prepare small spheres with different magnitudes of charge in which the *ratio* of the charges was known.[†] Although he had some difficulty with induced charges, Coulomb was able to argue that the electric force one tiny charged object exerts on a second tiny charged object is directly proportional to the charge on each of them. That is, if the charge on either one of the objects is doubled, the force is doubled; and if the charge on both of the objects is doubled, the force increases to four times the original value. This was the case when the distance between the two charges remained the same. If the distance between them was allowed to increase, he found that the force decreased with the *square of the distance* between them. That is, if the distance was doubled, the force fell to one-fourth of its original value. Thus, Coulomb concluded, the magnitude of the force F that one small charged object exerts on a second one is proportional to the product of the magnitude of the charge on one, Q_1, times the magnitude of the charge on the other, Q_2, and inversely proportional to the square of the distance r between them (Fig. 16–14). As an equation, we can write **Coulomb's law** as

$$F = k\frac{Q_1 Q_2}{r^2}, \qquad \text{[magnitudes]} \quad \textbf{(16–1)}$$

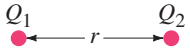
where k is a proportionality constant.[‡]

FIGURE 16–13 Coulomb's apparatus: when an external charged sphere is placed close to the charged one on the suspended bar, the bar rotates slightly. The suspending fiber resists the twisting motion, and the angle of twist is proportional to the force applied. With this apparatus, Coulomb investigated how the electric force varies as a function of the magnitude of the charges and of the distance between them.



COULOMB'S LAW

FIGURE 16–14 Coulomb's law, Eq. 16–1, gives the force between two point charges, Q_1 and Q_2, a distance r apart.



[†]Coulomb reasoned that if a charged conducting sphere is placed in contact with an identical uncharged sphere, the charge on the first would be shared equally by the two of them because of symmetry. He thus had a way to produce charges equal to $\frac{1}{2}$, $\frac{1}{4}$, and so on, of the original charge.

[‡]The validity of Coulomb's law today rests on precision measurements that are much more sophisticated than Coulomb's original experiment. The exponent 2 on r in Coulomb's law has been shown to be accurate to 1 part in 10^{16} [that is, $2 \pm (1 \times 10^{-16})$].

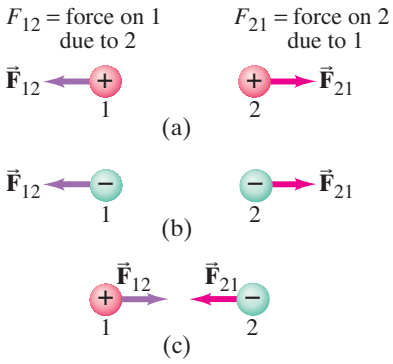


FIGURE 16–15 The direction of the static electric force one point charge exerts on another is always along the line joining the two charges, and depends on whether the charges have the same sign as in (a) and (b), or opposite signs (c).

As we just saw, Coulomb's law, Eq. 16–1,

$$F = k\frac{Q_1 Q_2}{r^2}, \qquad \text{[magnitudes]} \quad \textbf{(16–1)}$$

gives the *magnitude* of the electric force that either charge exerts on the other. The *direction* of the electric force *is always along the line joining the two charges*. If the two charges have the same sign, the force on either charge is directed away from the other (they repel each other). If the two charges have opposite signs, the force on one is directed toward the other (they attract). See Fig. 16–15. Notice that the force one charge exerts on the second is equal but opposite to that exerted by the second on the first, in accord with Newton's third law.

The SI unit of charge is the **coulomb** (C). The precise definition of the coulomb today is in terms of electric current and magnetic field, and will be discussed later (Section 20–6). In SI units, the constant k in Coulomb's law has the value

$$k = 8.988 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2$$

or, when we only need two significant figures,

$$k \approx 9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2.$$

Thus, 1 C is that amount of charge which, if placed on each of two point objects that are 1.0 m apart, will result in each object exerting a force of $(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(1.0\,\text{C})(1.0\,\text{C})/(1.0\,\text{m})^2 = 9.0 \times 10^9\,\text{N}$ on the other. This would be an enormous force, equal to the weight of almost a million tons. We rarely encounter charges as large as a coulomb.[†]

Charges produced by rubbing ordinary objects (such as a comb or plastic ruler) are typically around a microcoulomb $(1\,\mu\text{C} = 10^{-6}\,\text{C})$ or less. Objects that carry a positive charge have a deficit of electrons, whereas negatively charged objects have an excess of electrons. The charge on one electron has been determined to have a magnitude of about 1.6022×10^{-19} C, and is negative. This is the smallest charge observed in nature,[‡] and because it is fundamental, it is given the symbol e and is often referred to as the **elementary charge**:

$$e = 1.6022 \times 10^{-19}\,\text{C} \approx 1.6 \times 10^{-19}\,\text{C}.$$

Note that e is defined as a positive number, so the charge on the electron is $-e$. (The charge on a proton, on the other hand, is $+e$.) Since an object cannot gain or lose a fraction of an electron, the net charge on any object must be an integral multiple of this charge. Electric charge is thus said to be **quantized** (existing only in discrete amounts: $1e$, $2e$, $3e$, etc.). Because e is so small, however, we normally do not notice this discreteness in macroscopic charges ($1\,\mu$C requires about 10^{13} electrons), which thus seem continuous.

Coulomb's law looks a lot like the *law of universal gravitation*, $F = Gm_1 m_2/r^2$, which expresses the magnitude of the gravitational force a mass m_1 exerts on a mass m_2 (Eq. 5–4). Both are **inverse square laws** $(F \propto 1/r^2)$. Both also have a proportionality to a property of each object—mass for gravity, electric charge for electricity. And both act over a distance (that is, there is no need for contact). A major difference between the two laws is that gravity is always an attractive force, whereas the electric force can be either attractive or repulsive. Electric charge comes in two types, positive and negative; gravitational mass is only positive.

The constant k in Eq. 16–1 is often written in terms of another constant, ϵ_0, called the **permittivity of free space**. It is related to k by $k = 1/4\pi\epsilon_0$. Coulomb's law can be written

COULOMB'S LAW (in terms of ϵ_0)

$$F = \frac{1}{4\pi\epsilon_0}\frac{Q_1 Q_2}{r^2}, \qquad \textbf{(16–2)}$$

where

$$\epsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2.$$

[†]In the once common cgs system of units, k is set equal to 1, and the unit of electric charge is called the *electrostatic unit* (esu) or the statcoulomb. One esu is defined as that charge, on each of two point objects 1 cm apart, that gives rise to a force of 1 dyne.

[‡]According to the Standard Model of elementary particle physics, subnuclear particles called quarks have a smaller charge than the electron, equal to $\frac{1}{3}e$ or $\frac{2}{3}e$. Quarks have not been detected directly as isolated objects, and theory indicates that free quarks may not be detectable.

Equation 16–2 looks more complicated than Eq. 16–1, but other fundamental equations we haven't seen yet are simpler in terms of ϵ_0 rather than k. It doesn't matter which form we use since Eqs. 16–1 and 16–2 are equivalent. (The latest precise values of e and ϵ_0 are given inside the front cover.)[†]

Equations 16–1 and 16–2 apply to objects whose size is much smaller than the distance between them. Ideally, it is precise for **point charges** (spatial size negligible compared to other distances). For finite-sized objects, it is not always clear what value to use for r, particularly since the charge may not be distributed uniformly on the objects. If the two objects are spheres and the charge is known to be distributed uniformly on each, then r is the distance between their centers.

Coulomb's law describes the force between two charges when they are at rest. Additional forces come into play when charges are in motion, and will be discussed in later Chapters. In this Chapter we discuss only charges at rest, the study of which is called **electrostatics**, and Coulomb's law gives the **electrostatic force**.

When calculating with Coulomb's law, we usually use magnitudes, ignoring signs of the charges, and determine the direction of a force separately based on whether the force is attractive or repulsive.

PROBLEM SOLVING
Use magnitudes in Coulomb's law; find force direction from signs of charges

EXERCISE A Return to the Chapter-Opening Question, page 443, and answer it again now. Try to explain why you may have answered differently the first time.

EXAMPLE 16–1 **Electric force on electron by proton.** Determine the magnitude and direction of the electric force on the electron of a hydrogen atom exerted by the single proton $(Q_2 = +e)$ that is the atom's nucleus. Assume the average distance between the revolving electron and the proton is $r = 0.53 \times 10^{-10}\,\text{m}$, Fig. 16–16.

APPROACH To find the force magnitude we use Coulomb's law, $F = k\,Q_1 Q_2/r^2$ (Eq. 16–1), with $r = 0.53 \times 10^{-10}\,\text{m}$. The electron and proton have the same magnitude of charge, e, so $Q_1 = Q_2 = 1.6 \times 10^{-19}\,\text{C}$.

SOLUTION The magnitude of the force is

$$F = k\frac{Q_1 Q_2}{r^2} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(1.6 \times 10^{-19}\,\text{C})(1.6 \times 10^{-19}\,\text{C})}{(0.53 \times 10^{-10}\,\text{m})^2}$$
$$= 8.2 \times 10^{-8}\,\text{N}.$$

The direction of the force on the electron is toward the proton, because the charges have opposite signs so the force is attractive.

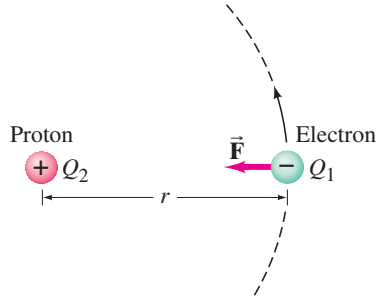


FIGURE 16–16 Example 16–1.

CONCEPTUAL EXAMPLE 16–2 **Which charge exerts the greater force?** Two positive point charges, $Q_1 = 50\,\mu\text{C}$ and $Q_2 = 1\,\mu\text{C}$, are separated by a distance ℓ, Fig. 16–17. Which is larger in magnitude, the force that Q_1 exerts on Q_2, or the force that Q_2 exerts on Q_1?

RESPONSE From Coulomb's law, the force on Q_1 exerted by Q_2 is

$$F_{12} = k\frac{Q_1 Q_2}{\ell^2}.$$

The force on Q_2 exerted by Q_1 is

$$F_{21} = k\frac{Q_2 Q_1}{\ell^2}$$

which is the same magnitude. The equation is symmetric with respect to the two charges, so $F_{21} = F_{12}$.

NOTE Newton's third law also tells us these two forces must have equal magnitude.

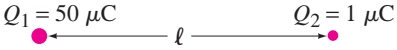
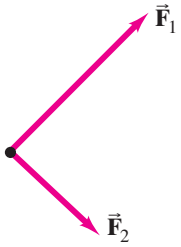


FIGURE 16–17 Example 16–2.

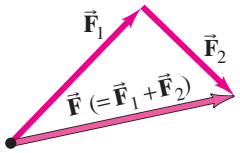
EXERCISE B In Example 16–2, how is the direction of F_{12} related to the direction of F_{21}?

[†]Our convention for units, such as $\text{C}^2/\text{N}\cdot\text{m}^2$ for ϵ_0, means m^2 is in the denominator. That is, $\text{C}^2/\text{N}\cdot\text{m}^2$ means $\text{C}^2/(\text{N}\cdot\text{m}^2)$ and does *not* mean $(\text{C}^2/\text{N})\cdot\text{m}^2 = \text{C}^2\cdot\text{m}^2/\text{N}$.

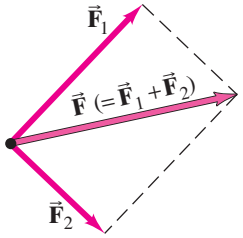
Keep in mind that Coulomb's law, Eq. 16–1 or 16–2, gives the force on a charge due to only *one* other charge. If several (or many) charges are present, the *net force on any one of them will be the vector sum of the forces due to each of the others*. This **principle of superposition** is based on experiment, and tells us that electric force vectors add like any other vector. For example, if you have a system of four charges, the net force on charge 1, say, is the sum of the forces exerted on charge 1 by charges 2, 3, and 4. The magnitudes of these three forces are determined from Coulomb's law, and then are added vectorially.



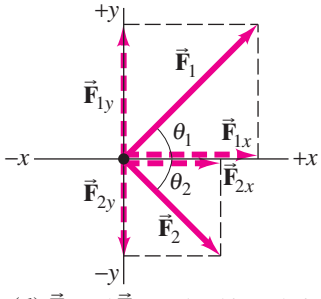
(a) Two forces acting on an object.



(b) The total, or net, force is $\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2$ by the tail-to-tip method of adding vectors.



(c) $\vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2$ by the parallelogram method.



(d) $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ resolved into their x and y components.

FIGURE 16–18 Review of vector addition.

16–6 Solving Problems Involving Coulomb's Law and Vectors

The electric force between charged particles at rest (sometimes referred to as the **electrostatic force** or as the **Coulomb force**) is, like all forces, a vector: it has both magnitude and direction. When several forces act on an object (call them $\vec{\mathbf{F}}_1$, $\vec{\mathbf{F}}_2$, etc.), the net force $\vec{\mathbf{F}}_{\text{net}}$ on the object is the vector sum of all the forces acting on it:

$$\vec{\mathbf{F}}_{\text{net}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 + \cdots.$$

This is the principle of superposition for forces. We studied how to add vectors in Chapter 3; then in Chapter 4 we used the rules for adding vectors to obtain the net force on an object by adding the different vector forces acting on it. It might be useful now to review Sections 3–2, 3–3, and 3–4. Here is a brief review of vectors.

Vector Addition Review

Suppose two vector forces, $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$, act on an object (Fig. 16–18a). They can be added using the tail-to-tip method (Fig. 16–18b) or by the parallelogram method (Fig. 16–18c), as discussed in Section 3–2. These two methods are useful for *understanding* a given problem (for getting a picture in your mind of what is going on). But for *calculating* the direction and magnitude of the resultant sum, it is more precise to use the method of adding components. Figure 16–18d shows the forces $\vec{\mathbf{F}}_1$ and $\vec{\mathbf{F}}_2$ resolved into components along chosen x and y axes (for more details, see Section 3–4). From the definitions of the trigonometric functions (Figs. 3–11 and 3–12), we have

$$\begin{aligned} F_{1x} &= F_1 \cos\theta_1 & F_{2x} &= F_2 \cos\theta_2 \\ F_{1y} &= F_1 \sin\theta_1 & F_{2y} &= -F_2 \sin\theta_2 . \end{aligned}$$

We add up the x and y components separately to obtain the components of the resultant force $\vec{\mathbf{F}}$, which are

$$\begin{aligned} F_x &= F_{1x} + F_{2x} = F_1 \cos\theta_1 + F_2 \cos\theta_2 , \\ F_y &= F_{1y} + F_{2y} = F_1 \sin\theta_1 - F_2 \sin\theta_2 . \end{aligned}$$

The magnitude of the resultant (or *net*) force $\vec{\mathbf{F}}$ is

$$F = \sqrt{F_x^2 + F_y^2}.$$

The direction of $\vec{\mathbf{F}}$ is specified by the angle θ that $\vec{\mathbf{F}}$ makes with the x axis, which is given by

$$\tan\theta = \frac{F_y}{F_x}.$$

Adding Electric Forces; Principle of Superposition

When dealing with several charges, it is helpful to use double subscripts on each of the forces involved. The first subscript refers to the particle *on* which the force acts; the second refers to the particle that exerts the force. For example, if we have three charges, $\vec{\mathbf{F}}_{31}$ means the force exerted *on* particle 3 *by* particle 1.

As in all problem solving, it is very important to draw a diagram, in particular a free-body diagram (Chapter 4) for each object, showing all the forces acting *on* that object. In applying Coulomb's law, we can deal with charge magnitudes only (leaving out minus signs) to get the magnitude of each force. Then determine separately the direction of the force physically (along the line joining the two particles: like charges repel, unlike charges attract), and show the force on the diagram. (You could determine direction first if you like.) Finally, add all the forces on one object together as vectors to obtain the net force on that object.

EXAMPLE 16–3 **Three charges in a line.** Three charged particles are arranged in a line, as shown in Fig. 16–19a. Calculate the net electrostatic force on particle 3 (the $-4.0\,\mu$C on the right) due to the other two charges.

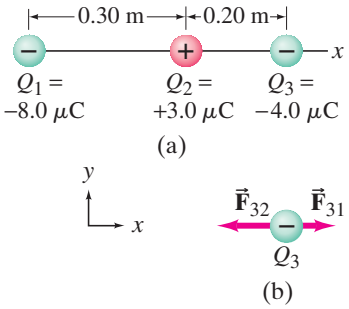


FIGURE 16–19 Example 16–3.

APPROACH The net force on particle 3 is the vector sum of the force $\vec{\mathbf{F}}_{31}$ exerted on particle 3 by particle 1 and the force $\vec{\mathbf{F}}_{32}$ exerted on 3 by particle 2:

$$\vec{\mathbf{F}} = \vec{\mathbf{F}}_{31} + \vec{\mathbf{F}}_{32}.$$

SOLUTION The magnitudes of these two forces are obtained using Coulomb's law, Eq. 16–1:

$$F_{31} = k\frac{Q_3 Q_1}{r_{31}^2} = \frac{(9.0 \times 10^9\,\mathrm{N\cdot m^2/C^2})(4.0 \times 10^{-6}\,\mathrm{C})(8.0 \times 10^{-6}\,\mathrm{C})}{(0.50\,\mathrm{m})^2} = 1.2\,\mathrm{N},$$

where $r_{31} = 0.50\,\mathrm{m}$ is the distance from Q_3 to Q_1. Similarly,

$$F_{32} = k\frac{Q_3 Q_2}{r_{32}^2} = \frac{(9.0 \times 10^9\,\mathrm{N\cdot m^2/C^2})(4.0 \times 10^{-6}\,\mathrm{C})(3.0 \times 10^{-6}\,\mathrm{C})}{(0.20\,\mathrm{m})^2} = 2.7\,\mathrm{N}.$$

Since we were calculating the magnitudes of the forces, we omitted the signs of the charges. But we must be aware of them to get the direction of each force. Let the line joining the particles be the x axis, and we take it positive to the right. Then, because $\vec{\mathbf{F}}_{31}$ is repulsive and $\vec{\mathbf{F}}_{32}$ is attractive, the directions of the forces are as shown in Fig. 16–19b: F_{31} points in the positive x direction (away from Q_1) and F_{32} points in the negative x direction (toward Q_2). The net force on particle 3 is then

$$F = -F_{32} + F_{31} = -2.7\,\mathrm{N} + 1.2\,\mathrm{N} = -1.5\,\mathrm{N}.$$

The magnitude of the net force is 1.5 N, and it points to the left.

NOTE Charge Q_1 acts on charge Q_3 just as if Q_2 were not there (this is the principle of superposition). That is, the charge in the middle, Q_2, in no way blocks the effect of charge Q_1 acting on Q_3. Naturally, Q_2 exerts its own force on Q_3.

CAUTION
Each charge exerts its own force. No charge blocks the effect of the others

EXERCISE C Determine the magnitude and direction of the net force on charge Q_2 in Fig. 16–19a.

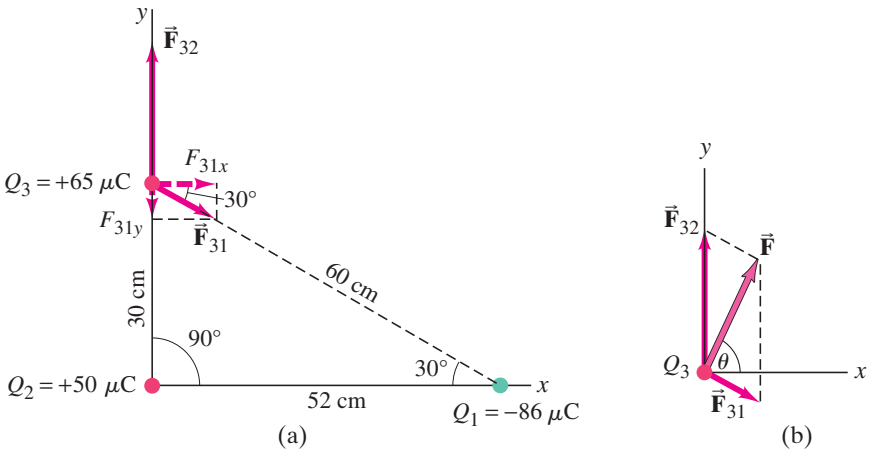


FIGURE 16–20 Determining the forces for Example 16–4. (a) The directions of the individual forces are as shown because $\vec{\mathbf{F}}_{32}$ is repulsive (the force on Q_3 is in the direction away from Q_2 because Q_3 and Q_2 are both positive) whereas $\vec{\mathbf{F}}_{31}$ is attractive (Q_3 and Q_1 have opposite signs), so $\vec{\mathbf{F}}_{31}$ points toward Q_1. (b) Adding $\vec{\mathbf{F}}_{32}$ to $\vec{\mathbf{F}}_{31}$ to obtain the net force $\vec{\mathbf{F}}$.

EXAMPLE 16–4 **Electric force using vector components.** Calculate the net electrostatic force on charge Q_3 shown in Fig. 16–20a due to the charges Q_1 and Q_2.

APPROACH We use Coulomb's law to find the magnitudes of the individual forces. The direction of each force will be along the line connecting Q_3 to Q_1 or Q_2. The forces $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ have the directions shown in Fig. 16–20a, since Q_1 exerts an attractive force on Q_3, and Q_2 exerts a repulsive force. The forces $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ are *not* along the same line, so to find the resultant force on Q_3 we resolve $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ into x and y components and perform the vector addition.

SOLUTION The magnitudes of $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ are (ignoring signs of the charges since we know the directions)

$$F_{31} = k\frac{Q_3 Q_1}{r_{31}^2} = \frac{(9.0 \times 10^9\ \mathrm{N \cdot m^2/C^2})(6.5 \times 10^{-5}\ \mathrm{C})(8.6 \times 10^{-5}\ \mathrm{C})}{(0.60\ \mathrm{m})^2} = 140\ \mathrm{N},$$

$$F_{32} = k\frac{Q_3 Q_2}{r_{32}^2} = \frac{(9.0 \times 10^9\ \mathrm{N \cdot m^2/C^2})(6.5 \times 10^{-5}\ \mathrm{C})(5.0 \times 10^{-5}\ \mathrm{C})}{(0.30\ \mathrm{m})^2} = 325\ \mathrm{N}.$$

(We keep 3 significant figures until the end, and then keep 2 because only 2 are given.) We resolve $\vec{\mathbf{F}}_{31}$ into its components along the x and y axes, as shown in Fig. 16–20a:

$$F_{31x} = F_{31}\cos 30^\circ = (140\ \mathrm{N})\cos 30^\circ = 120\ \mathrm{N},$$
$$F_{31y} = -F_{31}\sin 30^\circ = -(140\ \mathrm{N})\sin 30^\circ = -70\ \mathrm{N}.$$

The force $\vec{\mathbf{F}}_{32}$ has only a y component. So the net force $\vec{\mathbf{F}}$ on Q_3 has components

$$F_x = F_{31x} = 120\ \mathrm{N},$$
$$F_y = F_{32} + F_{31y} = 325\ \mathrm{N} - 70\ \mathrm{N} = 255\ \mathrm{N}.$$

The magnitude of the net force is

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(120\ \mathrm{N})^2 + (255\ \mathrm{N})^2} = 280\ \mathrm{N};$$

and it acts at an angle θ (see Fig. 16–20b) given by

$$\tan\theta = \frac{F_y}{F_x} = \frac{255\ \mathrm{N}}{120\ \mathrm{N}} = 2.13,$$

so $\theta = \tan^{-1}(2.13) = 65^\circ$.

NOTE Because $\vec{\mathbf{F}}_{31}$ and $\vec{\mathbf{F}}_{32}$ are not along the same line, the magnitude of $\vec{\mathbf{F}}_3$ is not equal to the sum (or difference as in Example 16–3) of the separate magnitudes. That is, F_3 is not equal to $F_{31} + F_{32}$; nor does it equal $F_{32} - F_{31}$. Instead we had to do vector addition.

CONCEPTUAL EXAMPLE 16–5 **Make the force on Q_3 zero.** In Fig. 16–20, where could you place a fourth charge, $Q_4 = -50\,\mu\text{C}$, so that the net force on Q_3 would be zero?

RESPONSE By the principle of superposition, we need a force in exactly the opposite direction to the resultant $\vec{\mathbf{F}}$ due to Q_2 and Q_1 that we calculated in Example 16–4, Fig. 16–20b. Our force must have magnitude 290 N, and must point down and to the left of Q_3 in Fig. 16–20b, in the direction opposite to $\vec{\mathbf{F}}$. So Q_4 must be along this line. See Fig. 16–21.

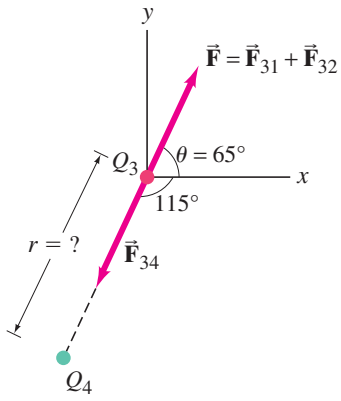


FIGURE 16–21 Example 16–5 and Exercise D: Q_4 exerts force $(\vec{\mathbf{F}}_{34})$ that makes the net force on Q_3 zero.

EXERCISE D In Example 16–5, what distance r must Q_4 be from Q_3?

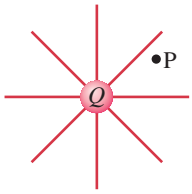
EXERCISE E (*a*) Consider two point charges, $+Q$ and $-Q$, which are fixed a distance d apart. Can you find a location where a third positive charge Q could be placed so that the net electric force on this third charge is zero? (*b*) What if the first two charges were both $+Q$?

16–7 The Electric Field

Many common forces might be referred to as "contact forces," such as your hands pushing or pulling a cart, or a tennis racket hitting a tennis ball.

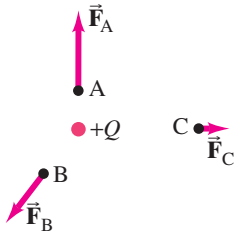
In contrast, both the gravitational force and the electrical force act over a distance: there is a force between two objects even when the objects are not touching. The idea of a force *acting at a distance* was a difficult one for early thinkers. Newton himself felt uneasy with this idea when he published his law of universal gravitation. A helpful way to look at the situation uses the idea of the **field**, developed by the British scientist Michael Faraday (1791–1867). In the electrical case, according to Faraday, an *electric field* extends outward from every charge and permeates all of space (Fig. 16–22). If a second charge (call it Q_2) is placed near the first charge, it feels a force exerted by the electric field that is there (say, at point P in Fig. 16–22). The electric field at point P is considered to interact directly with charge Q_2 to produce the force on Q_2.

FIGURE 16–22 An electric field surrounds every charge. The red lines indicate the electric field extending out from charge Q, and P is an arbitrary point.



We can in principle investigate the electric field surrounding a charge or group of charges by measuring the force on a small positive **test charge** which is at rest. By a test charge we mean a charge so small that the force it exerts does not significantly affect the charges that create the field. If a tiny positive test charge q is placed at various locations in the vicinity of a single positive charge Q as shown in Fig. 16–23 (points A, B, C), the force exerted on q is as shown. The force at B is less than at A because B's distance from Q is greater (Coulomb's law); and the force at C is smaller still. In each case, the force on q is directed radially away from Q. The electric field is defined in terms of the force on such a positive test charge. In particular, the **electric field**, $\vec{\mathbf{E}}$, at any point in space is defined as the force $\vec{\mathbf{F}}$ exerted on a tiny positive test charge placed at that point divided by the magnitude of the test charge q:

FIGURE 16–23 Force exerted by charge $+Q$ on a small test charge, q, placed at points A, B, and C.



$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q}. \qquad \textbf{(16–3)}$$

More precisely, $\vec{\mathbf{E}}$ is defined as the limit of $\vec{\mathbf{F}}/q$ as q is taken smaller and smaller, approaching zero. That is, q is so tiny that it exerts essentially no force on the other charges which created the field. From this definition (Eq. 16–3), we see that the electric field at any point in space is a vector whose direction is the direction of the force on a tiny positive test charge at that point, and whose magnitude is the *force per unit charge*. Thus $\vec{\mathbf{E}}$ has SI units of newtons per coulomb (N/C).

The reason for defining $\vec{\mathbf{E}}$ as $\vec{\mathbf{F}}/q$ (with $q \to 0$) is so that $\vec{\mathbf{E}}$ does not depend on the magnitude of the test charge q. This means that $\vec{\mathbf{E}}$ describes only the effect of the charges creating the electric field at that point.

The electric field at any point in space can be measured, based on the definition, Eq. 16–3. For simple situations with one or several point charges, we can calculate $\vec{\mathbf{E}}$. For example, the electric field at a distance r from a single point charge Q would have magnitude

$$E = \frac{F}{q} = \frac{kqQ/r^2}{q}$$

$$E = k\frac{Q}{r^2}; \qquad \text{[single point charge]} \quad \textbf{(16–4a)}$$

or, in terms of ϵ_0 as in Eq. 16–2 $\left(k = 1/4\pi\epsilon_0\right)$:

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}. \qquad \text{[single point charge]} \quad \textbf{(16–4b)}$$

Notice that E is independent of the test charge q—that is, E depends only on the charge Q which produces the field, and not on the value of the test charge q. Equations 16–4 are referred to as the electric field form of Coulomb's law.

If we are given the electric field $\vec{\mathbf{E}}$ at a given point in space, then we can calculate the force $\vec{\mathbf{F}}$ on any charge q placed at that point by writing (see Eq. 16–3):

$$\vec{\mathbf{F}} = q\vec{\mathbf{E}}. \qquad \textbf{(16–5)}$$

This is valid even if q is not small as long as q does not cause the charges creating $\vec{\mathbf{E}}$ to move. If q is positive, $\vec{\mathbf{F}}$ and $\vec{\mathbf{E}}$ point in the same direction. If q is negative, $\vec{\mathbf{F}}$ and $\vec{\mathbf{E}}$ point in opposite directions. See Fig. 16–24.

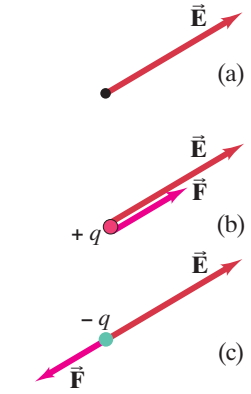


FIGURE 16–24 (a) Electric field at a given point in space. (b) Force on a positive charge at that point. (c) Force on a negative charge at that point.

PHYSICS APPLIED
Photocopier

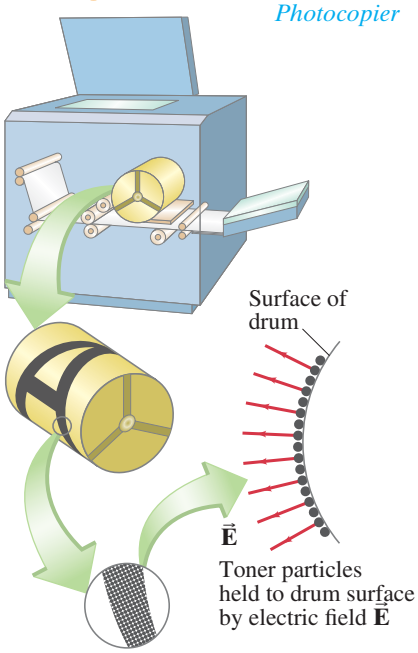


FIGURE 16–25 Example 16–6.

EXAMPLE 16–6 **Photocopy machine.** A photocopy machine works by arranging positive charges (in the pattern to be copied) on the surface of a drum, then gently sprinkling negatively charged dry toner (ink) particles onto the drum. The toner particles temporarily stick to the pattern on the drum (Fig. 16–25) and are later transferred to paper and "melted" to produce the copy. Suppose each toner particle has a mass of 9.0×10^{-16} kg and carries an average of 20 extra electrons to provide an electric charge. Assuming that the electric force on a toner particle must exceed twice its weight in order to ensure sufficient attraction, compute the required electric field strength near the surface of the drum.

APPROACH The electric force on a toner particle of charge $q = 20e$ is $F = qE$, where E is the needed electric field. This force needs to be at least as great as twice the weight (mg) of the particle.

SOLUTION The minimum value of electric field satisfies the relation

$$qE = 2mg$$

where $q = 20e$. Hence

$$E = \frac{2mg}{q} = \frac{2(9.0 \times 10^{-16}\,\text{kg})(9.8\,\text{m/s}^2)}{20(1.6 \times 10^{-19}\,\text{C})} = 5.5 \times 10^3\,\text{N/C}.$$

EXAMPLE 16–7 **Electric field of a single point charge.** Calculate the magnitude and direction of the electric field at a point P which is 30 cm to the right of a point charge $Q = -3.0 \times 10^{-6}$ C.

APPROACH The magnitude of the electric field due to a single point charge is given by Eq. 16–4. The direction is found using the sign of the charge Q.

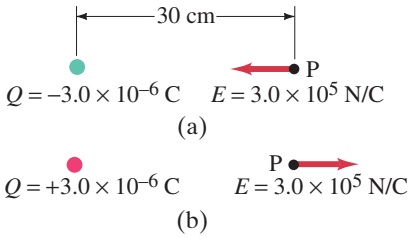
SOLUTION The magnitude of the electric field is:

$$E = k\frac{Q}{r^2} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(3.0 \times 10^{-6}\,\text{C})}{(0.30\,\text{m})^2} = 3.0 \times 10^5\,\text{N/C}.$$

The direction of the electric field is *toward* the charge Q, to the left as shown in Fig. 16–26a, since we defined the direction as that of the force on a positive test charge which here would be attractive. If Q had been positive, the electric field would have pointed away, as in Fig. 16–26b.

NOTE There is no electric charge at point P. But there is an electric field there. The only real charge is Q.

FIGURE 16–26 Example 16–7. Electric field at point P (a) due to a negative charge Q, and (b) due to a positive charge Q, each 30 cm from P.



This Example illustrates a general result: The electric field $\vec{\mathbf{E}}$ due to a positive charge points away from the charge, whereas $\vec{\mathbf{E}}$ due to a negative charge points toward that charge.

EXERCISE F Find the magnitude and direction of the electric field due to a $-2.5\,\mu$C charge 50 cm below it.

If the electric field at a given point in space is due to more than one charge, the individual fields (call them $\vec{\mathbf{E}}_1$, $\vec{\mathbf{E}}_2$, etc.) due to each charge are added vectorially to get the total field at that point:

$$\vec{\mathbf{E}} = \vec{\mathbf{E}}_1 + \vec{\mathbf{E}}_2 + \cdots.$$

The validity of this **superposition principle** for electric fields is fully confirmed by experiment.

EXAMPLE 16–8 **$\vec{\mathbf{E}}$ at a point between two charges.** Two point charges are separated by a distance of 10.0 cm. One has a charge of $-25\,\mu$C and the other $+50\,\mu$C. (*a*) Determine the direction and magnitude of the electric field at a point P between the two charges that is 2.0 cm from the negative charge (Fig. 16–27a). (*b*) If an electron (mass $= 9.11 \times 10^{-31}$ kg) is placed at rest at P and then released, what will be its initial acceleration (direction and magnitude)?

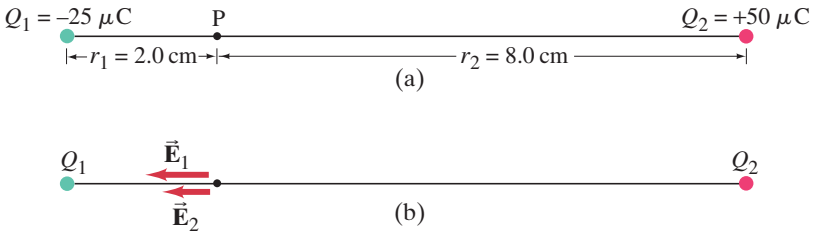


FIGURE 16–27 Example 16–8. In (b), we don't know the relative lengths of $\vec{\mathbf{E}}_1$ and $\vec{\mathbf{E}}_2$ until we do the calculation.

APPROACH The electric field at P will be the vector sum of the fields created separately by Q_1 and Q_2. The field due to the negative charge Q_1 points toward Q_1, and the field due to the positive charge Q_2 points away from Q_2. Thus both fields point to the left as shown in Fig. 16–27b, and we can add the magnitudes of the two fields together algebraically, ignoring the signs of the charges. In (*b*) we use Newton's second law ($\Sigma\vec{\mathbf{F}} = m\vec{\mathbf{a}}$) to find the acceleration, where $\Sigma\vec{\mathbf{F}} = q\Sigma\vec{\mathbf{E}}$.

SOLUTION (*a*) Each field is due to a point charge as given by Eq. 16–4, $E = kQ/r^2$. The total field points to the left and has magnitude

$$\begin{aligned} E &= k\frac{Q_1}{r_1^2} + k\frac{Q_2}{r_2^2} = k\left(\frac{Q_1}{r_1^2} + \frac{Q_2}{r_2^2}\right) \\ &= (9.0 \times 10^9\ \mathrm{N\cdot m^2/C^2})\left(\frac{25 \times 10^{-6}\ \mathrm{C}}{(2.0 \times 10^{-2}\ \mathrm{m})^2} + \frac{50 \times 10^{-6}\ \mathrm{C}}{(8.0 \times 10^{-2}\ \mathrm{m})^2}\right) \\ &= 6.3 \times 10^8\ \mathrm{N/C}. \end{aligned}$$

(*b*) The electric field points to the left, so the electron will feel a force to the *right* since it is negatively charged. Therefore the acceleration $a = F/m$ (Newton's second law) will be to the right. The force on a charge q in an electric field E is $F = qE$ (Eq. 16–5). Hence the magnitude of the electron's initial acceleration is

$$a = \frac{F}{m} = \frac{qE}{m} = \frac{(1.60 \times 10^{-19}\ \mathrm{C})(6.3 \times 10^8\ \mathrm{N/C})}{9.11 \times 10^{-31}\ \mathrm{kg}} = 1.1 \times 10^{20}\ \mathrm{m/s^2}.$$

NOTE By considering the directions of *each* field ($\vec{\mathbf{E}}_1$ and $\vec{\mathbf{E}}_2$) before doing any calculations, we made sure our calculation could be done simply and correctly.

EXERCISE G Four charges of equal magnitude, but possibly different sign, are placed on the corners of a square. What arrangement of charges will produce an electric field with the greatest magnitude at the center of the square? (*a*) All four positive charges; (*b*) all four negative charges; (*c*) three positive and one negative; (*d*) two positive and two negative; (*e*) three negative and one positive.

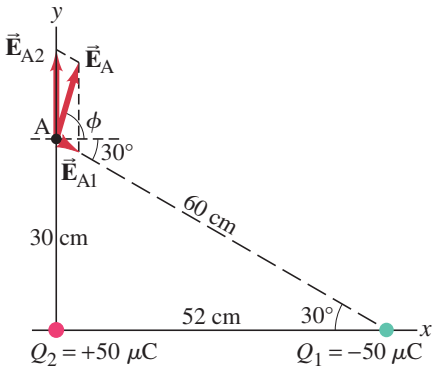


FIGURE 16–28 Calculation of the electric field at point A, Example 16–9.

PROBLEM SOLVING

Ignore signs of charges and determine direction physically, showing directions on diagram

EXAMPLE 16–9 **$\vec{E}$ above two point charges.** Calculate the total electric field at point A in Fig. 16–28 due to both charges, Q_1 and Q_2.

APPROACH The calculation is much like that of Example 16–4, except now we are dealing with electric fields instead of force. The electric field at point A is the vector sum of the fields $\vec{E}_{A1}$ due to Q_1, and $\vec{E}_{A2}$ due to Q_2. We find the magnitude of the field produced by each point charge, then we add their components to find the total field at point A.

SOLUTION The magnitude of the electric field produced at point A by each of the charges Q_1 and Q_2 is given by $E = kQ/r^2$, so

$$E_{A1} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\,\text{C})}{(0.60\,\text{m})^2} = 1.25 \times 10^6\,\text{N/C},$$

$$E_{A2} = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\,\text{C})}{(0.30\,\text{m})^2} = 5.0 \times 10^6\,\text{N/C}.$$

The direction of E_{A1} points from A toward Q_1 (negative charge), whereas E_{A2} points from A away from Q_2, as shown; so the total electric field at A, $\vec{E}_A$, has components

$$E_{Ax} = E_{A1}\cos 30° = 1.1 \times 10^6\,\text{N/C},$$
$$E_{Ay} = E_{A2} - E_{A1}\sin 30° = 4.4 \times 10^6\,\text{N/C}.$$

Thus the magnitude of $\vec{E}_A$ is

$$E_A = \sqrt{(1.1)^2 + (4.4)^2} \times 10^6\,\text{N/C} = 4.5 \times 10^6\,\text{N/C},$$

and its direction is ϕ (Fig. 16–28) given by $\tan\phi = E_{Ay}/E_{Ax} = 4.4/1.1 = 4.0$, so $\phi = 76°$.

It is worthwhile summarizing here what we have learned about solving electrostatics problems.

PROBLEM SOLVING

Electrostatics: Electric Forces and Electric Fields

Whether you use electric field or electrostatic forces, the procedure for solving electrostatics problems is similar:

1. **Draw** a careful **diagram**—namely, a free-body diagram for each object, showing all the forces acting on that object, or showing the electric field at a point due to all significant charges present. Determine the **direction** of each force or electric field physically: like charges repel each other, unlike charges attract; fields point away from a + charge, and toward a − charge. Show and label each vector force or field on your diagram.
2. **Apply Coulomb's law** to calculate the magnitude of the force that each contributing charge exerts on a charged object, or the magnitude of the electric field each charge produces at a given point. Deal only with magnitudes of charges (leaving out minus signs), and obtain the magnitude of each force or electric field.
3. **Add vectorially** all the forces on an object, or the contributing fields at a point, to get the resultant. Use **symmetry** (say, in the geometry) whenever possible.

EXAMPLE 16–10 **$\vec{E}$ equidistant above two point charges.** Figure 16–29 (top of next page) is the same as Fig. 16–28 but includes point B, which is equidistant (40 cm) from Q_1 and Q_2. Calculate the total electric field at point B in Fig. 16–29 due to both charges, Q_1 and Q_2.

APPROACH We explicitly follow the steps of the Problem Solving Strategy above.

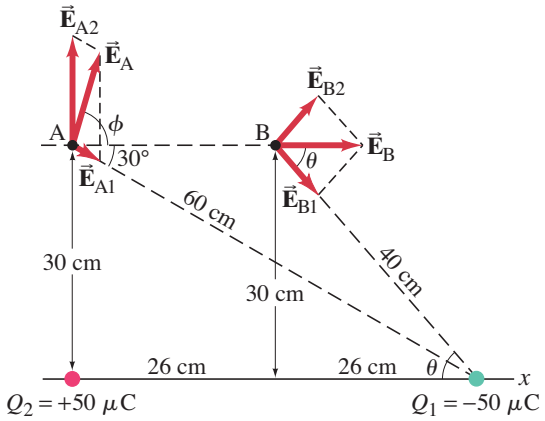


FIGURE 16–29 Same as Fig. 16–28 but with point B added. Calculation of the electric field at points A and B for Examples 16–9 and 16–10.

SOLUTION

1. **Draw** a careful **diagram**. The **directions** of the electric fields $\vec{E}_{B1}$ and $\vec{E}_{B2}$, as well as the net field $\vec{E}_B$, are shown in Fig. 16–29. $\vec{E}_{B2}$ points away from the positive charge Q_2; $\vec{E}_{B1}$ points toward the negative charge Q_1.
2. **Apply Coulomb's law** to find the magnitudes of the contributing electric fields. Because B is equidistant from the two equal charges (40 cm by the Pythagorean theorem), the magnitudes of E_{B1} and E_{B2} are the same; that is,

$$E_{B1} = E_{B2} = \frac{kQ}{r^2} = \frac{(9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\ \text{C})}{(0.40\ \text{m})^2} = 2.8 \times 10^6\ \text{N/C}.$$

3. **Add vectorially**, and use **symmetry** when possible. The y components of $\vec{E}_{B1}$ and $\vec{E}_{B2}$ are equal and opposite. Because of this symmetry, the total field E_B is horizontal and equals $E_{B1}\cos\theta + E_{B2}\cos\theta = 2E_{B1}\cos\theta$. From Fig. 16–29, $\cos\theta = 26\ \text{cm}/40\ \text{cm} = 0.65$. Then

$$E_B = 2E_{B1}\cos\theta = 2(2.8 \times 10^6\ \text{N/C})(0.65) = 3.6 \times 10^6\ \text{N/C},$$

and the direction of $\vec{E}_B$ is along the $+x$ direction.

PROBLEM SOLVING
Use symmetry to save work, when possible

16–8 Electric Field Lines

Since the electric field is a vector, it is sometimes referred to as a *vector field*. We could indicate the electric field with arrows at various points in a given situation, such as at A, B, and C in Fig. 16–30. The directions of $\vec{E}_A$, $\vec{E}_B$, and $\vec{E}_C$ are the same as for the forces shown earlier in Fig. 16–23, but the magnitudes (arrow lengths) are different since we divide $\vec{F}$ by q to get $\vec{E}$. However, the relative lengths of $\vec{E}_A$, $\vec{E}_B$, and $\vec{E}_C$ are the same as for the forces since we divide by the same q each time. To indicate the electric field in such a way at *many* points, however, would result in many arrows, which would quickly become cluttered and confusing. To avoid this, we use another technique, that of field lines.

To visualize the electric field, we draw a series of lines to indicate the direction of the electric field at various points in space. These **electric field lines** (or **lines of force**) are drawn to indicate the direction of the force due to the given field on a positive test charge. The lines of force due to a single isolated positive charge are shown in Fig. 16–31a, and for a single isolated negative charge in Fig. 16–31b. In part (a) the lines point radially outward from the charge, and in part (b) they point radially inward toward the charge because that is the direction the force would be on a positive test charge in each case (as in Fig. 16–26). Only a few representative lines are shown. We could draw lines in between those shown since the electric field exists there as well. We can draw the lines so that the *number of lines starting on a positive charge, or ending on a negative charge, is proportional to the magnitude of the charge*. Notice that nearer the charge, where the electric field is greater $(F \propto 1/r^2)$, the lines are closer together. This is a general property of electric field lines: *the closer together the lines are, the stronger the electric field in that region*. In fact, field lines can be drawn so that the number of lines crossing unit area perpendicular to $\vec{E}$ is proportional to the magnitude of the electric field.

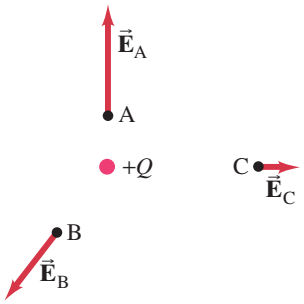
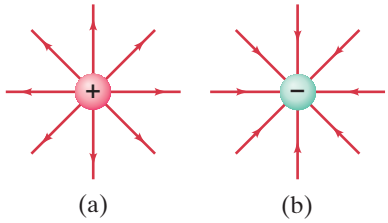


FIGURE 16–30 Electric field vector, shown at three points, due to a single point charge Q. (Compare to Fig. 16–23.)

FIGURE 16–31 Electric field lines (a) near a single positive point charge, (b) near a single negative point charge.



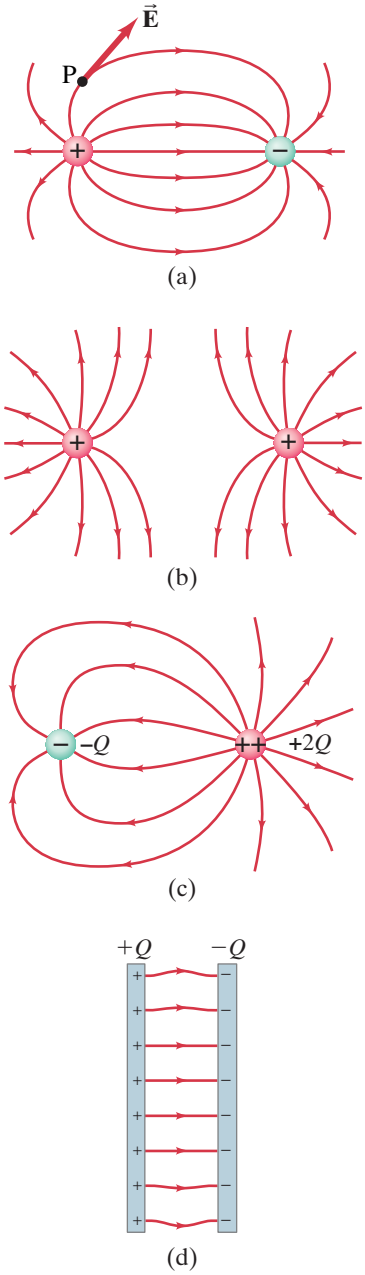


FIGURE 16–32 Electric field lines for four arrangements of charges.

FIGURE 16–33 The Earth's gravitational field, which at any point is directed toward the Earth's center (the force on any mass points toward the Earth's center).

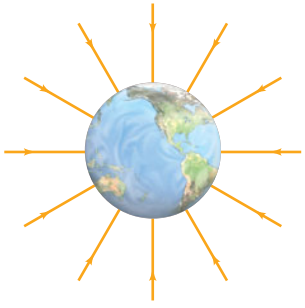


Figure 16–32a shows the electric field lines due to two equal charges of opposite sign, a combination known as an **electric dipole**. The electric field lines are curved in this case and are directed from the positive charge to the negative charge. The direction of the electric field at any point is tangent to the field line at that point as shown by the vector arrow $\vec{\mathbf{E}}$ at point P. To satisfy yourself that this is the correct pattern for the electric field lines, you can make a few calculations such as those done in Examples 16–9 and 16–10 for just this case (see Fig. 16–29). Figure 16–32b shows the electric field lines for two equal positive charges, and Fig. 16–32c for unequal charges, $-Q$ and $+2Q$. Note that twice as many lines leave $+2Q$ as enter $-Q$ (number of lines is proportional to magnitude of Q). Finally, in Fig. 16–32d, we see in cross section the field lines between two flat parallel plates carrying equal but opposite charges. Notice that the electric field lines between the two plates start out perpendicular to the surface of the metal plates (we will see why this is true in the next Section) and go directly from one plate to the other, as we expect because a positive test charge placed between the plates would feel a strong repulsion from the positive plate and a strong attraction to the negative plate. The field lines between two close plates are parallel and equally spaced in the central region, but fringe outward near the edges. Thus, in the central region, the electric field has the same magnitude at all points, and we can write

$$E = \text{constant.} \left[\begin{array}{l}\text{between two closely spaced, oppositely}\\ \text{charged, flat parallel plates}\end{array}\right] \quad \textbf{(16–6)}$$

The fringing of the field near the edges can often be ignored, particularly if the separation of the plates is small compared to their height and width.[†] We summarize the properties of field lines as follows:

1. Electric field lines indicate the direction of the electric field; the field points in the direction tangent to the field line at any point.
2. The lines are drawn so that the magnitude of the electric field, E, is proportional to the number of lines crossing unit area perpendicular to the lines. The closer together the lines, the stronger the field.
3. Electric field lines start on positive charges and end on negative charges; and the number starting or ending is proportional to the magnitude of the charge.

Also note that field lines never cross. Why not? Because it would not make sense for the electric field to have two directions at the same point.

Gravitational Field

The field concept can also be applied to the gravitational force (Chapter 5). Thus we can say that a **gravitational field** exists for every object that has mass. One object attracts another by means of the gravitational field. The Earth, for example, can be said to possess a gravitational field (Fig. 16–33) which is responsible for the gravitational force on objects. The *gravitational field* is defined as the *force per unit mass*. The magnitude of the Earth's gravitational field at any point above the Earth's surface is thus GM_E/r^2, where M_E is the mass of the Earth, r is the distance of the point from the Earth's center, and G is the gravitational constant (Chapter 5). At the Earth's surface, r is the radius of the Earth and the gravitational field is equal to g, the acceleration due to gravity. Beyond the Earth, the gravitational field can be calculated at any point as a sum of terms due to Earth, Sun, Moon, and other bodies that contribute significantly.

[†]The magnitude of the constant electric field between two parallel plates is given by $E = Q/\epsilon_0 A$, where Q is the magnitude of the charge on each plate and A is the area of one plate. We show this in the optional Section 16–12 on Gauss's law.

16–9 Electric Fields and Conductors

We now discuss some properties of conductors. First, *the electric field inside a conductor is zero in the static situation*—that is, when the charges are at rest. If there were an electric field within a conductor, there would be a force on the free electrons. The electrons would move until they reached positions where the electric field, and therefore the electric force on them, did become zero.

This reasoning has some interesting consequences. For one, *any net charge on a conductor distributes itself on the surface*. For a negatively charged conductor, you can imagine that the negative charges repel one another and race to the surface to get as far from one another as possible. Another consequence is the following. Suppose that a positive charge Q is surrounded by an isolated uncharged metal conductor whose shape is a spherical shell, Fig. 16–34. Because there can be no field within the metal, the lines leaving the central positive charge must end on negative charges on the inner surface of the metal. That is, the encircled charge $+Q$ induces an equal amount of negative charge, $-Q$, on the inner surface of the spherical shell. Since the shell is neutral, a positive charge of the same magnitude, $+Q$, must exist on the outer surface of the shell. Thus, although no field exists in the metal itself, an electric field exists outside of it, as shown in Fig. 16–34, as if the metal were not even there.

A related property of static electric fields and conductors is that *the electric field is always perpendicular to the surface outside of a conductor*. If there were a component of $\vec{\mathbf{E}}$ parallel to the surface (Fig. 16–35), it would exert a force on free electrons at the surface, causing the electrons to move along the surface until they reached positions where no net force was exerted on them parallel to the surface—that is, until the electric field was perpendicular to the surface.

These properties apply only to conductors. Inside a nonconductor, which does not have free electrons, a static electric field can exist as we will see in the next Chapter. Also, the electric field outside a nonconductor does not necessarily make an angle of 90° to the surface.

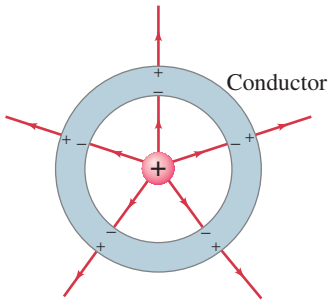
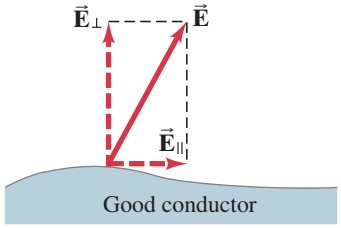


FIGURE 16–34 A charge inside a neutral spherical metal shell induces charge on its surfaces. The electric field exists even beyond the shell, but not within the conductor itself.

FIGURE 16–35 If the electric field $\vec{\mathbf{E}}$ at the surface of a conductor had a component parallel to the surface, $\vec{\mathbf{E}}_{\parallel}$, the latter would accelerate electrons into motion. In the static case, $\vec{\mathbf{E}}_{\parallel}$ must be zero, and the electric field must be perpendicular to the conductor's surface: $\vec{\mathbf{E}} = \vec{\mathbf{E}}_{\perp}$.



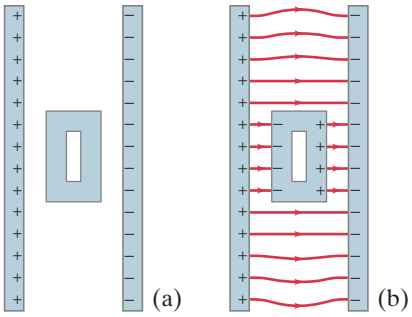


FIGURE 16–36 Example 16–11.

CONCEPTUAL EXAMPLE 16–11 **Shielding, and safety in a storm.** A neutral hollow metal box is placed between two parallel charged plates as shown in Fig. 16–36a. What is the field like inside the box?

RESPONSE If our metal box had been solid, and not hollow, free electrons in the box would have redistributed themselves along the surface until all their individual fields would have canceled each other inside the box. The net field inside the box would have been zero. For a hollow box, the external field is not changed since the electrons in the metal can move just as freely as before to the surface. Hence the field inside the hollow metal box is also zero, and the field lines are shown in Fig. 16–36b. A conducting box is an effective device for shielding delicate instruments and electronic circuits from unwanted external electric fields. We also can see that a relatively safe place to be during a lightning storm is inside a parked car, surrounded by metal. See also Fig. 16–37, where a person inside a porous "cage" is protected from a strong electric discharge. (It is not safe in a lightning storm to be near a tree which can conduct, or out in the open where you are taller than the surroundings.)

FIGURE 16–37 High-voltage "Van de Graaff" generators create strong electric fields in the vicinity of the "Faraday cage" below. The strong field accelerates stray electrons in the atmosphere to the KE needed to knock electrons out of air atoms, causing an avalanche of charge which flows to (or from) the metal cage. The metal cage protects the person inside it.



PHYSICS APPLIED
Electrical shielding

*16–10 Electric Forces in Molecular Biology: DNA Structure and Replication

PHYSICS APPLIED
Inside a cell: kinetic theory plus electrostatic force

The study of the structure and functioning of a living cell at the molecular level is known as molecular biology. It is an important area for application of physics. The interior of every biological cell is mainly water. We can imagine a cell as a thick soup of molecules continually in motion (kinetic theory, Chapter 13), colliding with one another with various amounts of kinetic energy. These molecules interact with one another because of the *electrostatic force* between molecules.

Indeed, cellular processes are now considered to be the result of *random ("thermal") molecular motion plus the ordering effect of the electrostatic force*. As an example, we look at DNA structure and replication. The picture we present is a model of what happens based on physical theories and experiment.

The genetic information that is passed on from generation to generation in all living cells is contained in the chromosomes, which are made up of genes. Each gene contains the information needed to produce a particular type of protein molecule, and that information is built into the principal molecule of a chromosome, DNA (deoxyribonucleic acid), Fig. 16–38. DNA molecules are made up of many small molecules known as nucleotide bases which are each *polar* (Section 16–2) due to unequal sharing of electrons. There are four types of nucleotide bases in DNA: adenine (A), cytosine (C), guanine (G), and thymine (T).

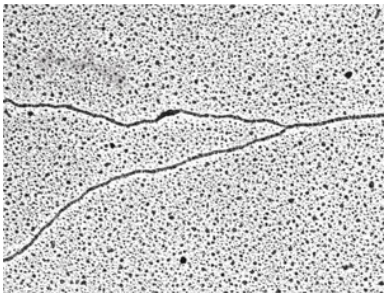
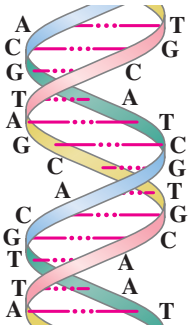


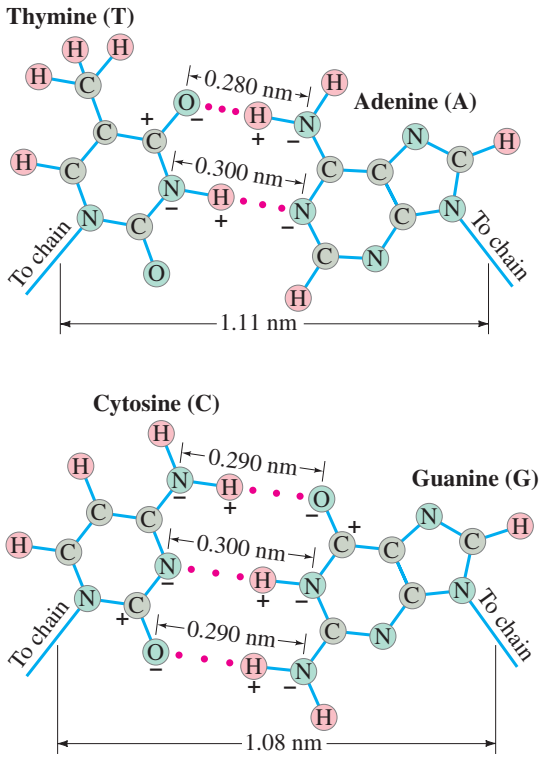
FIGURE 16–38 Image of DNA replicating, made by a transmission electron microscope.

PHYSICS APPLIED
DNA structure

The DNA of a chromosome generally consists of two long DNA strands wrapped about one another in the shape of a "double helix." The genetic information is contained in the specific order of the four bases (A, C, G, T) along each strand. As shown in Fig. 16–39, the two strands are attracted by electrostatic forces—that is, by the attraction of positive charges to negative charges that exist on parts of the molecules. We see in Fig. 16–39a that an A (adenine) on one strand is always opposite a T on the other strand; similarly, a G is always opposite a C. This important ordering effect occurs because the shapes of A, T, C, and G are such that a T fits closely only into an A, and a G into a C. Only in the case of this close proximity of the charged portions is the electrostatic force great enough to hold them together even for a short time (Fig. 16–39b), forming what are referred to as "weak bonds."



(a)



(b)

FIGURE 16–39 (a) Schematic diagram of a section of DNA double helix. (b) "Close-up" view of the helix, showing how A and T attract each other and how G and C attract each other through electrostatic forces. The + and − signs indicated on certain atoms represent net charges, usually a fraction of e, due to uneven sharing of electrons. The red dots indicate the electrostatic attraction (often called a "weak bond" or "hydrogen bond"—Section 29–3). Note that there are two weak bonds between A and T, and three between C and G.

The electrostatic force between A and T, and between C and G, exists because these molecules have charged parts. These charges are due to some electrons in each of these molecules spending more time orbiting one atom than another. For example, the electron normally on the H atom of adenine (upper part of Fig. 16–39b) spends some of its time orbiting the adjacent N atom (more on this in Chapter 29), so the N has a net negative charge and the H is left with a net positive charge. This H^+ atom of adenine is then attracted to the O^- atom of thymine. These net + and − charges usually have magnitudes of a fraction of e (charge on the electron) such as $0.2e$ or $0.4e$. (This is what we mean by "polar" molecules.)

[When H^+ is involved, the weak bond it can make with a nearby negative charge, such as O^-, is relatively strong (partly because H^+ is so small) and is referred to as a **hydrogen bond** (Section 29–3).]

When the DNA replicates (duplicates) itself just before cell division, the arrangement of A opposite T and G opposite C is crucial for ensuring that the genetic information is passed on accurately to the next generation, Fig. 16–40. The two strands of DNA separate (with the help of enzymes, which also operate via the electrostatic force), leaving the charged parts of the bases exposed.

PHYSICS APPLIED
DNA replication

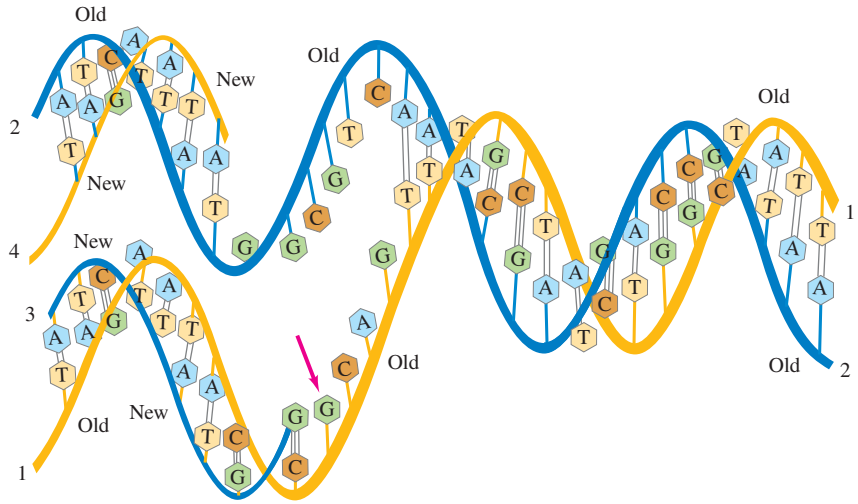


FIGURE 16–40 Replication of DNA.

Once replication starts, let us see how the correct order of bases occurs by looking at the G molecule indicated by the red arrow in Fig. 16–40. Many unattached nucleotide bases of all four kinds are bouncing around in the cellular fluid, and the only type that will experience attraction to our G, if it comes close to it, will be a C. The charges on the other three bases can not get close enough to those on the G to provide a significant attractive force—remember that the electrostatic (Coulomb) force decreases rapidly with distance $(\propto 1/r^2)$. Because the G does not attract an A, T, or G appreciably, an A, T, or G will be knocked away by collisions with other molecules before enzymes can attach it to the growing chain (number 3 in Fig. 16–40). But the electrostatic force will often hold a C opposite our G long enough so that an enzyme can attach the C to the growing end of the new chain. Thus electrostatic forces are responsible for selecting the bases in the proper order during replication. Note in Fig. 16–40 that the new number 4 strand has the same order of bases as the old number 1 strand; and the new number 3 strand is the same as the old number 2. So the two new double helixes, 1–3 and 2–4, are identical to the original 1–2 helix. Hence the genetic information is passed on accurately to the next generation.

This process of DNA replication is often presented as if it occurred in clockwork fashion—as if each molecule knew its role and went to its assigned place. But this is not the case. The forces of attraction are rather weak and become significant only when charged parts of the two molecules have "complementary shapes," meaning they can get close enough so that the electrostatic force $(\propto 1/r^2)$ is strong enough to form weak bonds. If the molecular shapes are not just right, there is almost no electrostatic attraction, which is why there are so few mistakes. Thus, out of the random motion of the molecules, the electrostatic force acts to bring order out of chaos.

The random (thermal) velocities of molecules in a cell affect *cloning*. When a bacterial cell divides, the two new bacteria have nearly identical DNA. Even if the DNA were perfectly identical, the two new bacteria would not end up behaving in exactly the same way. Long protein, DNA, and RNA molecules get bumped into different shapes, and even the expression[†] of genes can thus be different. Loosely held parts of large molecules such as a methyl group (CH_3) can also be knocked off by a strong collision with another molecule. Hence, cloned organisms are not identical, even if their DNA were identical. Indeed, there can not really be genetic determinism.

*16–11 Photocopy Machines and Computer Printers Use Electrostatics

Photocopy machines and laser printers use electrostatic attraction to print an image. They each use a different technique to project an image onto a special cylindrical drum (or rotating conveyor belt). The drum is typically made of aluminum, a good conductor; its surface is coated with a thin layer of selenium, which has the interesting property (called "photoconductivity") of being an electrical nonconductor in the dark, but a conductor when exposed to light.

PHYSICS APPLIED
Photocopy machines

In a **photocopier**, lenses and mirrors focus an image of the original sheet of paper onto the drum, much like a camera lens focuses an image on an electronic detector or film. Step 1, done in the dark, is the placing of a uniform positive charge on the drum's selenium layer by a charged roller or rod: see Fig. 16–41.

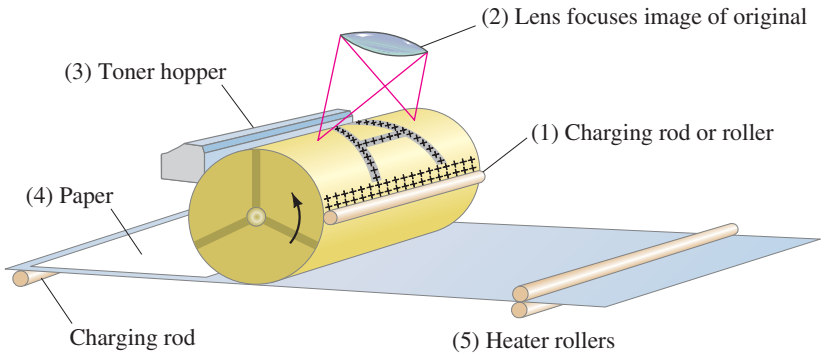


FIGURE 16–41 Inside a photocopy machine: (1) the selenium drum is given a + charge; (2) the lens focuses image on drum—only dark spots stay charged; (3) toner particles (negatively charged) are attracted to positive areas on drum; (4) the image is transferred to paper; (5) heat binds the image to the paper.

In step 2, the image to be copied is projected onto the drum. For simplicity, let us assume the image is a dark letter A on a white background (as on the page of a book) as shown in Fig. 16–41. The letter A on the drum is dark, but all around it is light. At all these light places, the selenium becomes conducting and electrons flow in from the aluminum beneath, neutralizing those positive areas. In the dark areas of the letter A, the selenium is nonconducting and so retains the positive charge already put on it, Fig. 16–41. In step 3, a fine dark powder known as *toner* is given a negative charge, and is brushed on the drum as it rotates. The negatively charged toner particles are attracted to the positive areas on the drum (the A in our case) and stick only there. In step 4, the rotating drum presses against a piece of paper which has been positively charged more strongly than the selenium, so the toner particles are transferred to the paper, forming the final image. Finally, step 5, the paper is heated to fix the toner particles firmly on the paper.

In a color copier (or printer), this process is repeated for each color—black, cyan (blue), magenta (red), and yellow. Combining these four colors in different proportions produces any desired color.

[†]The separate genes of a DNA double helix can be covered by protein molecules, keeping those genes from being "expressed"—that is, translated into the proteins they code for (see Section 29–3).

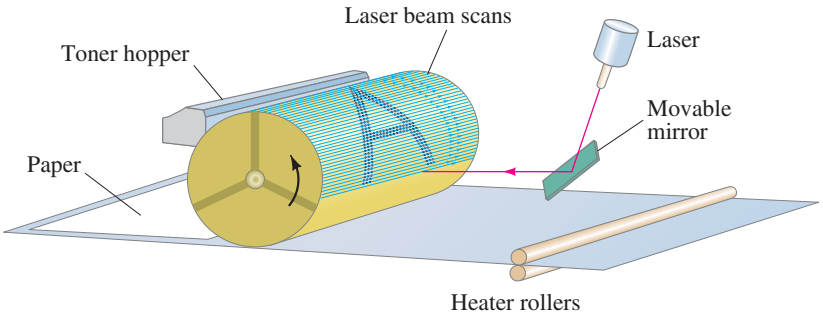


FIGURE 16–42 Inside a laser printer: a movable mirror sweeps the laser beam in horizontal lines across the drum.

A **laser printer** uses a computer output to program the intensity of a laser beam onto the selenium-coated drum of Fig. 16–42. The thin beam of light from the laser is scanned (by a movable mirror) from side to side across the drum in a series of horizontal lines, each line just below the previous line. As the beam sweeps across the drum, the intensity of the beam is varied by the computer output, being strong for a point that is meant to be white or bright, and weak or zero for points that are meant to come out dark. After each sweep, the drum rotates very slightly for additional sweeps, Fig. 16–42, until a complete image is formed on it. The light parts of the selenium become conducting and lose their (previously given) positive electric charge, and the toner sticks only to the dark, electrically charged areas. The drum then transfers the image to paper, as in a photocopier.

PHYSICS APPLIED
Laser printer

An **inkjet printer** does not use a drum. Instead nozzles spray tiny droplets of ink directly at the paper. The nozzles are swept across the paper, each sweep just above the previous one as the paper moves down. On each sweep, the ink makes dots on the paper, except for those points where no ink is desired, as directed by the computer. The image consists of a huge number of very tiny dots. The quality or resolution of a printer is usually specified in dots per inch (dpi) in each (linear) direction.

PHYSICS APPLIED
Inkjet printer

*16–12 Gauss's Law

An important relation in electricity is Gauss's law, developed by the great mathematician Karl Friedrich Gauss (1777–1855). It relates electric charge and electric field, and is a more general and elegant form of Coulomb's law.

Gauss's law involves the concept of **electric flux**, which refers to the electric field passing through a given area. For a uniform electric field $\vec{\mathbf{E}}$ passing through an area A, as shown in Fig. 16–43a, the electric flux Φ_E is defined as

$$\Phi_E = EA\cos\theta,$$

where θ is the angle between the electric field direction and a line drawn perpendicular to the area. The flux can be written equivalently as

$$\Phi_E = E_\perp A = EA_\perp = EA\cos\theta, \qquad \textbf{(16–7)}$$

where $E_\perp = E\cos\theta$ is the component of $\vec{\mathbf{E}}$ perpendicular to the area (Fig. 16–43b) and, similarly, $A_\perp = A\cos\theta$ is the projection of the area A perpendicular to the field $\vec{\mathbf{E}}$ (Fig. 16–43c).

Electric flux can be interpreted in terms of field lines. We mentioned in Section 16–8 that field lines can always be drawn so that the number (N) passing through unit area perpendicular to the field $(A_\perp)$ is proportional to the magnitude of the field (E): that is, $E \propto N/A_\perp$. Hence,

$$N \propto EA_\perp = \Phi_E, \qquad \textbf{(16–8)}$$

so the flux through an area is proportional to the number of lines passing through that area.

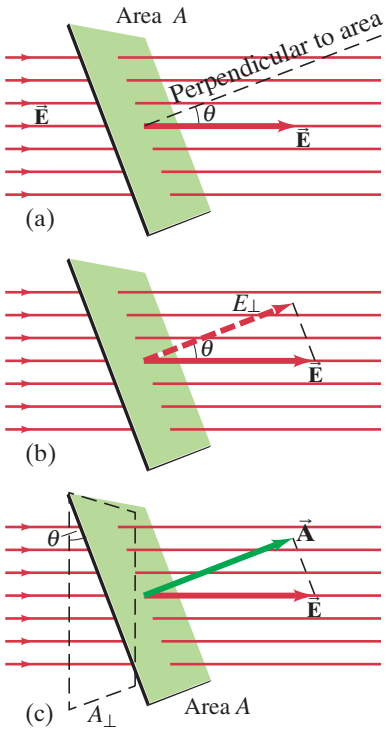


FIGURE 16–43 (a) A uniform electric field $\vec{\mathbf{E}}$ passing through a flat square area A. (b) $E_\perp = E\cos\theta$ is the component of $\vec{\mathbf{E}}$ perpendicular to the plane of area A. (c) $A_\perp = A\cos\theta$ is the projection (dashed) of the area A perpendicular to the field $\vec{\mathbf{E}}$.

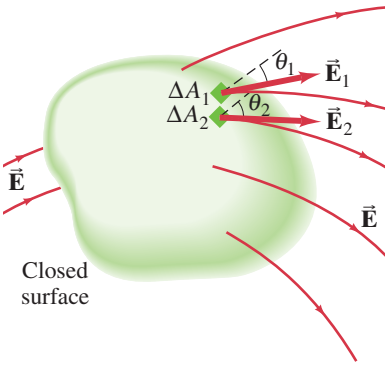


FIGURE 16–44 Electric field lines passing through a closed surface. The surface is divided up into many tiny areas, ΔA_1, ΔA_2, $\cdots$, and so on, of which only two are shown.

Gauss's law involves the *total* flux through a closed surface—a surface of any shape that encloses a volume of space. For any such surface, such as that shown in Fig. 16–44, we divide the surface up into many tiny areas, $\Delta A_1, \Delta A_2, \Delta A_3, \cdots$, and so on. We make the division so that each ΔA is small enough that it can be considered flat and so that the electric field can be considered constant through each ΔA. Then the *total* flux through the entire surface is the sum over all the individual fluxes through each of the tiny areas:

$$\begin{aligned}\Phi_E &= E_1\,\Delta A_1\cos\theta_1 + E_2\,\Delta A_2\cos\theta_2 + \cdots \\ &= \sum E\,\Delta A\cos\theta \;=\; \sum E_\perp\,\Delta A,\end{aligned}$$

where the symbol Σ means "sum of." We saw in Section 16–8 that the number of field lines starting on a positive charge or ending on a negative charge is proportional to the magnitude of the charge. Hence, the *net* number of lines N pointing out of any closed surface (number of lines pointing out minus the number pointing in) must be proportional to the net charge enclosed by the surface, Q_{encl}. But from Eq. 16–8, we have that the net number of lines N is proportional to the total flux Φ_E. Therefore,

$$\Phi_E = \sum_{\substack{\text{closed}\\\text{surface}}} E_\perp\,\Delta A \propto Q_{\text{encl}}.$$

The constant of proportionality, to be consistent with Coulomb's law, is $1/\epsilon_0$, so we have

GAUSS'S LAW

$$\sum_{\substack{\text{closed}\\\text{surface}}} E_\perp\,\Delta A = \frac{Q_{\text{encl}}}{\epsilon_0}, \qquad \textbf{(16–9)}$$

where the sum (Σ) is over any closed surface, and Q_{encl} is the net charge enclosed within that surface. This is **Gauss's law**.

Coulomb's law and Gauss's law can be used to determine the electric field due to a given (static) charge distribution. Gauss's law is useful when the charge distribution is simple and symmetrical. However, we must choose the closed "gaussian" surface very carefully so we can determine $\vec{\mathbf{E}}$. We normally choose a surface that has just the **symmetry** needed so that E will be constant on all or on parts of its surface.

EXAMPLE 16–12 Charged spherical conducting shell. A thin spherical shell of radius r_0 possesses a total net charge Q that is uniformly distributed on it, Fig. 16–45. Determine the electric field at points (*a*) outside the shell, and (*b*) inside the shell.

APPROACH Because the charge is distributed symmetrically, the electric field must be *symmetric*. Thus the field outside the shell must be directed radially outward (inward if $Q < 0$) and must depend only on r.

SOLUTION (*a*) We choose any imaginary gaussian surface as a sphere of radius r ($r > r_0$) concentric with the shell, shown in Fig. 16–45 as the dashed circle A_1. Then, by symmetry, the electric field will have the same magnitude at all points on this gaussian surface. Because $\vec{\mathbf{E}}$ is perpendicular to this surface, Gauss's law gives (with $Q_{\text{encl}} = Q$ in Eq. 16–9)

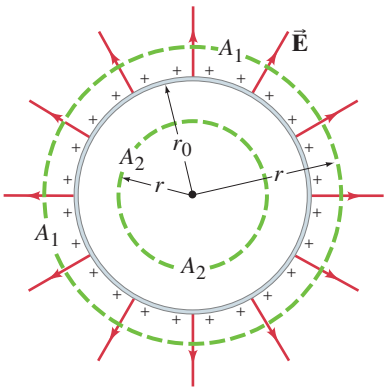
$$\sum E_\perp\,\Delta A = E\sum\Delta A = E(4\pi r^2) = \frac{Q}{\epsilon_0},$$

where $4\pi r^2$ is the surface area of our sphere (gaussian surface) of radius r. Thus

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}. \qquad [r > r_0]$$

We see that the field outside a uniformly charged spherical shell is the same as if all the charge were concentrated at the center as a point charge.

FIGURE 16–45 Cross-sectional drawing of a thin spherical shell (gray) of radius r_0, carrying a net charge Q uniformly distributed. The green circles A_1 and A_2 represent two gaussian surfaces we use to determine $\vec{\mathbf{E}}$. Example 16–12.



(*b*) Inside the shell, the electric field must also be symmetric. So E must again have the same value at all points on a spherical gaussian surface (A_2 in Fig. 16–45) concentric with the shell. Thus, E can be factored out of the sum and, with $Q_{\text{encl}} = 0$ because the charge inside surface A_2 is zero, we have

$$\sum E_\perp \Delta A = E \sum \Delta A$$
$$= E(4\pi r^2) = \frac{Q_{\text{encl}}}{\epsilon_0} = 0.$$

Hence

$$E = 0 \qquad [r < r_0]$$

inside a uniform spherical shell of charge (as claimed in Section 16–9).

The results of Example 16–12 also apply to a uniform *solid* spherical conductor that is charged, since all the charge would lie in a thin layer at the surface (Section 16–9). In particular

$$E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

outside a spherical conductor. Thus, the electric field outside a spherically symmetric distribution of charge is the same as for a point charge of the same magnitude at the center of the sphere. This result applies also outside a uniformly charged nonconductor, because we can use the same gaussian surface A_1 (Fig. 16–45) and the same *symmetry* argument. We can also consider this a demonstration of our statement in Chapter 5 about the **gravitational force**, which is also a perfect $1/r^2$ force: The gravitational force exerted by a uniform sphere is the same as if all the mass were at the center, as stated on page 120.

EXAMPLE 16–13 ***E* near any conducting surface.** Show that the magnitude of the electric field just outside the surface of a good conductor of any shape is given by

$$E = \frac{\sigma}{\epsilon_0},$$

where σ is defined as the surface charge density, Q/A, on the conductor's surface at that point.

APPROACH We choose as our gaussian surface a small cylindrical box, very small in height so that one of its circular ends is just above the conductor (Fig. 16–46). The other end is just below the conductor's surface, and the very short sides are perpendicular to it.

SOLUTION The electric field is zero inside a conductor and is perpendicular to the surface just outside it (Section 16–9), so electric flux passes only through the outside end of our cylindrical box; no flux passes through the very short sides or through the inside end of our gaussian box. We choose the area A (of the flat cylinder end above the conductor surface) small enough so that E is essentially uniform over it. Then Gauss's law gives

$$\sum E_\perp \Delta A = EA = \frac{Q_{\text{encl}}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0},$$

and therefore

$$E = \frac{\sigma}{\epsilon_0}. \qquad [\text{at surface of conductor}]$$

This useful result applies for any shape conductor, including a large, uniformly charged flat sheet: the electric field will be constant and equal to σ/ϵ_0.

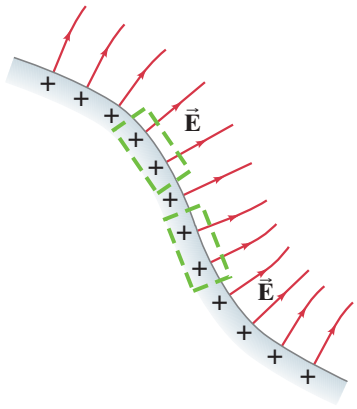


FIGURE 16–46 Electric field near the surface of a conductor. Two small cylindrical boxes are shown dashed. Either one can serve as our gaussian surface. Example 16–13.

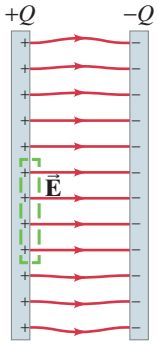


FIGURE 16–47 The electric field between two closely spaced parallel plates is uniform and equal to $E = \sigma/\epsilon_0$.

This last Example also gives us the field between the two parallel plates we discussed in Fig. 16–32d. If the plates are large compared to their separation, then the field lines are perpendicular to the plates and, except near the edges, they are parallel to each other. Therefore the electric field (see Fig. 16–47, which shows a similar very thin gaussian surface as Fig. 16–46) is also

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q/A}{\epsilon_0} = \frac{Q}{\epsilon_0 A}, \quad \left[\begin{array}{l}\text{between two closely spaced,}\\ \text{oppositely charged, parallel plates}\end{array}\right] \qquad \textbf{(16–10)}$$

where $Q = \sigma A$ is the charge on one of the plates.

Summary

There are two kinds of **electric charge**, positive and negative. These designations are to be taken algebraically—that is, any charge is plus or minus so many coulombs (C), in SI units.

Electric charge is **conserved**: if a certain amount of one type of charge is produced in a process, an equal amount of the opposite type is also produced; thus the *net* charge produced is zero.

According to atomic theory, electricity originates in the atom, which consists of a positively charged nucleus surrounded by negatively charged electrons. Each electron has a charge $-e = -1.60 \times 10^{-19}$ C.

Electric **conductors** are those materials in which many electrons are relatively free to move, whereas electric **insulators** or **nonconductors** are those in which very few electrons are free to move.

An object is negatively charged when it has an excess of electrons, and positively charged when it has less than its normal number of electrons. The net charge on any object is a whole number times $+e$ or $-e$. That is, charge is **quantized**.

An object can become charged by rubbing (in which electrons are transferred from one material to another), **by conduction** (which is transfer of charge from one charged object to another by touching), or **by induction** (the separation of charge within an object because of the close approach of another charged object but without touching).

Electric charges exert a force on each other. If two charges are of opposite types, one positive and one negative, they each exert an attractive force on the other. If the two charges are the same type, each repels the other.

The magnitude of the force one point charge exerts on another is proportional to the product of their charges, and inversely proportional to the square of the distance between them:

$$F = k\frac{Q_1 Q_2}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{Q_1 Q_2}{r^2}; \qquad \textbf{(16–1, 16–2)}$$

this is **Coulomb's law**.

We think of an **electric field** as existing in space around any charge or group of charges. The force on another charged object is then said to be due to the electric field present at its location.

The *electric field*, $\vec{\mathbf{E}}$, at any point in space due to one or more charges, is defined as the force per unit charge that would act on a tiny positive test charge q placed at that point:

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q}. \qquad \textbf{(16–3)}$$

The magnitude of the electric field a distance r from a point charge Q is

$$E = k\frac{Q}{r^2}. \qquad \textbf{(16–4a)}$$

The total electric field at a point in space is equal to the vector sum of the individual fields due to each contributing charge. This is the **principle of superposition**.

Electric fields are represented by **electric field lines** that start on positive charges and end on negative charges. Their direction indicates the direction the force would be on a tiny positive test charge placed at each point. The lines can be drawn so that the number per unit area is proportional to the magnitude of E.

The static electric field inside a conductor is zero, and the electric field lines just outside a charged conductor are perpendicular to its surface.

[*In the replication of DNA, the electrostatic force plays a crucial role in selecting the proper molecules so that the genetic information is passed on accurately from generation to generation.]

[*Photocopiers and computer printers use electric charge placed on toner particles and a drum to form an image.]

[*The **electric flux** passing through a small area A for a uniform electric field $\vec{\mathbf{E}}$ is

$$\Phi_E = E_\perp A, \qquad \textbf{(16–7)}$$

where $E_\perp$ is the component of $\vec{\mathbf{E}}$ perpendicular to the surface. The flux through a surface is proportional to the number of field lines passing through it.]

[***Gauss's law** states that the total flux summed over any closed surface (considered as made up of many small areas ΔA) is equal to the net charge Q_{encl} enclosed by the surface divided by ϵ_0:

$$\sum_{\substack{\text{closed}\\ \text{surface}}} E_\perp \, \Delta A = \frac{Q_{\text{encl}}}{\epsilon_0}. \qquad \textbf{(16–9)}$$

Gauss's law can be used to determine the electric field due to given charge distributions, but its usefulness is mainly limited to cases where the charge distribution displays much symmetry. The real importance of Gauss's law is that it is a general and elegant statement of the relation between electric charge and electric field.]

Questions

1. If you charge a pocket comb by rubbing it with a silk scarf, how can you determine if the comb is positively or negatively charged?
2. Why does a shirt or blouse taken from a clothes dryer sometimes cling to your body?
3. Explain why fog or rain droplets tend to form around ions or electrons in the air.
4. Why does a plastic ruler that has been rubbed with a cloth have the ability to pick up small pieces of paper? Why is this difficult to do on a humid day?
5. A positively charged rod is brought close to a neutral piece of paper, which it attracts. Draw a diagram showing the separation of charge in the paper, and explain why attraction occurs.
6. Contrast the *net charge* on a conductor to the "free charges" in the conductor.
7. Figures 16–7 and 16–8 show how a charged rod placed near an uncharged metal object can attract (or repel) electrons. There are a great many electrons in the metal, yet only some of them move as shown. Why not all of them?
8. When an electroscope is charged, its two leaves repel each other and remain at an angle. What balances the electric force of repulsion so that the leaves don't separate further?
9. The balloon in Fig. 16–48 was rubbed on a student's hair. Explain why the water drip curves instead of falling vertically.



FIGURE 16–48 Question 9.

10. The form of Coulomb's law is very similar to that for Newton's law of universal gravitation. What are the differences between these two laws? Compare also gravitational mass and electric charge.
11. When a charged ruler attracts small pieces of paper, sometimes a piece jumps quickly away after touching the ruler. Explain.
12. We are not normally aware of the gravitational or electric force between two ordinary objects. What is the reason in each case? Give an example where we are aware of each one and why.
13. Explain why the test charges we use when measuring electric fields must be small.
14. When determining an electric field, must we use a *positive* test charge, or would a negative one do as well? Explain.
15. Draw the electric field lines surrounding two negative electric charges a distance ℓ apart.
16. Assume that the two opposite charges in Fig. 16–32a are 12.0 cm apart. Consider the magnitude of the electric field 2.5 cm from the positive charge. On which side of this charge—top, bottom, left, or right—is the electric field the strongest? The weakest? Explain.
17. Consider the electric field at the three points indicated by the letters A, B, and C in Fig. 16–49. First draw an arrow at each point indicating the direction of the net force that a positive test charge would experience if placed at that point, then list the letters in order of *decreasing* field strength (strongest first). Explain.

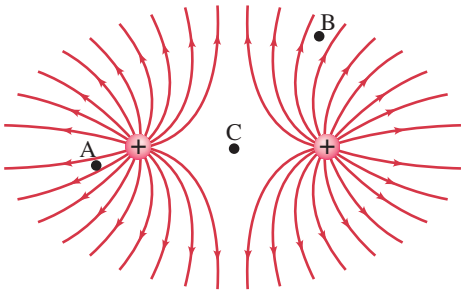


FIGURE 16–49 Question 17.

18. Why can electric field lines never cross?
19. Show, using the three rules for field lines given in Section 16–8, that the electric field lines starting or ending on a single point charge must be symmetrically spaced around the charge.
20. Given two point charges, Q and $2Q$, a distance ℓ apart, is there a point along the straight line that passes through them where $E = 0$ when their signs are (*a*) opposite, (*b*) the same? If yes, state roughly where this point will be.
21. Consider a small positive test charge located on an electric field line at some point, such as point P in Fig. 16–32a. Is the direction of the velocity and/or acceleration of the test charge along this line? Discuss.

*22. A point charge is surrounded by a spherical gaussian surface of radius r. If the sphere is replaced by a cube of side r, will Φ_E be larger, smaller, or the same? Explain.

MisConceptual Questions

1. $Q_1 = -0.10\,\mu C$ is located at the origin. $Q_2 = +0.10\,\mu C$ is located on the positive x axis at $x = 1.0$ m. Which of the following is true of the force on Q_1 due to Q_2?
(*a*) It is attractive and directed in the $+x$ direction.
(*b*) It is attractive and directed in the $-x$ direction.
(*c*) It is repulsive and directed in the $+x$ direction.
(*d*) It is repulsive and directed in the $-x$ direction.

2. Swap the positions of Q_1 and Q_2 of MisConceptual Question 1. Which of the following is true of the force on Q_1 due to Q_2?
(*a*) It does not change.
(*b*) It changes from attractive to repulsive.
(*c*) It changes from repulsive to attractive.
(*d*) It changes from the $+x$ direction to the $-x$ direction.
(*e*) It changes from the $-x$ direction to the $+x$ direction.

3. Fred the lightning bug has a mass m and a charge $+q$. Jane, his lightning-bug wife, has a mass of $\frac{3}{4}m$ and a charge $-2q$. Because they have charges of opposite sign, they are attracted to each other. Which is attracted more to the other, and by how much?
(*a*) Fred, twice as much.
(*b*) Jane, twice as much.
(*c*) Fred, four times as much.
(*d*) Jane, four times as much.
(*e*) They are attracted to each other by the same amount.

4. Figure 16–50 shows electric field lines due to a point charge. What can you say about the field at point 1 compared with the field at point 2?
(*a*) The field at point 2 is larger, because point 2 is on a field line.
(*b*) The field at point 1 is larger, because point 1 is not on a field line.
(*c*) The field at point 1 is zero, because point 1 is not on a field line.
(*d*) The field at point 1 is larger, because the field lines are closer together in that region.

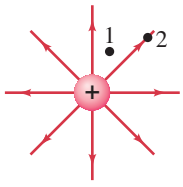


FIGURE 16–50
MisConceptual Question 4.

5. A negative point charge is in an electric field created by a positive point charge. Which of the following is true?
(*a*) The field points toward the positive charge, and the force on the negative charge is in the same direction as the field.
(*b*) The field points toward the positive charge, and the force on the negative charge is in the opposite direction to the field.
(*c*) The field points away from the positive charge, and the force on the negative charge is in the same direction as the field.
(*d*) The field points away from the positive charge, and the force on the negative charge is in the opposite direction to the field.

6. As an object acquires a positive charge, its mass usually
(*a*) decreases.
(*b*) increases.
(*c*) stays the same.
(*d*) becomes negative.

7. Refer to Fig. 16–32d. If the two charged plates were moved until they are half the distance shown without changing the charge on the plates, the electric field near the center of the plates would
(*a*) remain almost exactly the same.
(*b*) increase by a factor of 2.
(*c*) increase, but not by a factor of 2.
(*d*) decrease by a factor of 2.
(*e*) decrease, but not by a factor of 2.

8. We wish to determine the electric field at a point near a positively charged metal sphere (a good conductor). We do so by bringing a small positive test charge, q_0, to this point and measure the force F_0 on it. F_0/q_0 will be _____ the electric field $\vec{\mathbf{E}}$ as it was at that point before the test charge was present.
(*a*) greater than
(*b*) less than
(*c*) equal to

9. We are usually not aware of the electric force acting between two everyday objects because
(*a*) the electric force is one of the weakest forces in nature.
(*b*) the electric force is due to microscopic-sized particles such as electrons and protons.
(*c*) the electric force is invisible.
(*d*) most everyday objects have as many plus charges as minus charges.

10. To be safe during a lightning storm, it is best to be
(*a*) in the middle of a grassy meadow.
(*b*) inside a metal car.
(*c*) next to a tall tree in a forest.
(*d*) inside a wooden building.
(*e*) on a metal observation tower.

11. Which are the worst places in MisConceptual Question 10?

12. Which vector best represents the direction of the electric field at the fourth corner of the square due to the three charges shown in Fig. 16–51?

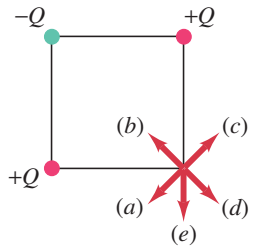


FIGURE 16–51
MisConceptual Question 12.

13. A small metal ball hangs from the ceiling by an insulating thread. The ball is attracted to a positively charged rod held near the ball. The charge of the ball must be
(*a*) positive.
(*b*) negative.
(*c*) neutral.
(*d*) positive or neutral.
(*e*) negative or neutral.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

16–5 and 16–6 Coulomb's Law

$[1\,\text{mC} = 10^{-3}\,\text{C},\ \ 1\,\mu\text{C} = 10^{-6}\,\text{C},\ \ 1\,\text{nC} = 10^{-9}\,\text{C}.]$

1. (I) What is the magnitude of the electric force of attraction between an iron nucleus ($q = +26e$) and its innermost electron if the distance between them is 1.5×10^{-12} m?
2. (I) How many electrons make up a charge of $-48.0\,\mu$C?
3. (I) What is the magnitude of the force a $+25\,\mu$C charge exerts on a $+2.5$ mC charge 16 cm away?
4. (I) What is the repulsive electrical force between two protons 4.0×10^{-15} m apart from each other in an atomic nucleus?
5. (II) When an object such as a plastic comb is charged by rubbing it with a cloth, the net charge is typically a few microcoulombs. If that charge is $3.0\,\mu$C, by what percentage does the mass of a 9.0-g comb change during charging?
6. (II) Two charged dust particles exert a force of 4.2×10^{-2} N on each other. What will be the force if they are moved so they are only one-eighth as far apart?
7. (II) Two small charged spheres are 6.52 cm apart. They are moved, and the force each exerts on the other is found to have tripled. How far apart are they now?
8. (II) A person scuffing her feet on a wool rug on a dry day accumulates a net charge of $-28\,\mu$C. How many excess electrons does she get, and by how much does her mass increase?
9. (II) What is the total charge of all the electrons in a 12-kg bar of gold? What is the net charge of the bar? (Gold has 79 electrons per atom and an atomic mass of 197 u.)
10. (II) Compare the electric force holding the electron in orbit ($r = 0.53 \times 10^{-10}$ m) around the proton nucleus of the hydrogen atom, with the gravitational force between the same electron and proton. What is the ratio of these two forces?
11. (II) Particles of charge $+65$, $+48$, and $-95\,\mu$C are placed in a line (Fig. 16–52). The center one is 0.35 m from each of the others. Calculate the net force on each charge due to the other two.

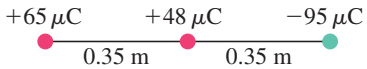


FIGURE 16–52 Problem 11.

12. (II) Three positive particles of equal charge, $+17.0\,\mu$C, are located at the corners of an equilateral triangle of side 15.0 cm (Fig. 16–53). Calculate the magnitude and direction of the net force on each particle due to the other two.

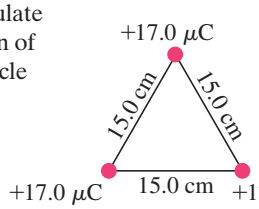


FIGURE 16–53 Problem 12.

13. (II) A charge Q is transferred from an initially uncharged plastic ball to an identical ball 24 cm away. The force of attraction is then 17 mN. How many electrons were transferred from one ball to the other?
14. (II) A charge of 6.15 mC is placed at each corner of a square 0.100 m on a side. Determine the magnitude and direction of the force on each charge.
15. (II) At each corner of a square of side ℓ there are point charges of magnitude Q, $2Q$, $3Q$, and $4Q$ (Fig. 16–54). Determine the magnitude and direction of the force on the charge $2Q$.

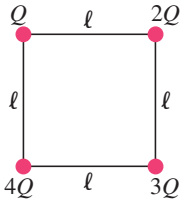


FIGURE 16–54 Problem 15.

16. (II) A large electroscope is made with "leaves" that are 78-cm-long wires with tiny 21-g spheres at the ends. When charged, nearly all the charge resides on the spheres. If the wires each make a 26° angle with the vertical (Fig. 16–55), what total charge Q must have been applied to the electroscope? Ignore the mass of the wires.

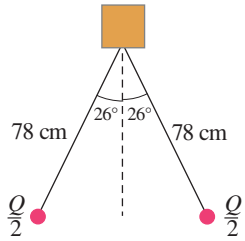


FIGURE 16–55 Problem 16.

17. (III) Two small nonconducting spheres have a total charge of $90.0\,\mu$C. (*a*) When placed 28.0 cm apart, the force each exerts on the other is 12.0 N and is repulsive. What is the charge on each? (*b*) What if the force were attractive?
18. (III) Two charges, $-Q$ and $-3Q$, are a distance ℓ apart. These two charges are free to move but do not because there is a third (fixed) charge nearby. What must be the magnitude of the third charge and its placement in order for the first two to be in equilibrium?

16–7 and 16–8 Electric Field, Field Lines

19. (I) Determine the magnitude and direction of the electric force on an electron in a uniform electric field of strength 2460 N/C that points due east.
20. (I) A proton is released in a uniform electric field, and it experiences an electric force of 1.86×10^{-14} N toward the south. Find the magnitude and direction of the electric field.
21. (I) Determine the magnitude and direction of the electric field 21.7 cm directly above an isolated 33.0×10^{-6} C charge.
22. (I) A downward electric force of 6.4 N is exerted on a $-7.3\,\mu$C charge. Find the magnitude and direction of the electric field at the position of this charge.
23. (II) Determine the magnitude of the acceleration experienced by an electron in an electric field of 756 N/C. How does the direction of the acceleration depend on the direction of the field at that point?
24. (II) Determine the magnitude and direction of the electric field at a point midway between a $-8.0\,\mu$C and a $+5.8\,\mu$C charge 6.0 cm apart. Assume no other charges are nearby.
25. (II) Draw, approximately, the electric field lines about two point charges, $+Q$ and $-3Q$, which are a distance ℓ apart.

26. (II) What is the electric field strength at a point in space where a proton experiences an acceleration of 2.4 million "g's"?

27. (II) An electron is released from rest in a uniform electric field and accelerates to the north at a rate of 105 m/s^2. Find the magnitude and direction of the electric field.

28. (II) The electric field midway between two equal but opposite point charges is 386 N/C, and the distance between the charges is 16.0 cm. What is the magnitude of the charge on each?

29. (II) Calculate the electric field at one corner of a square 1.22 m on a side if the other three corners are occupied by 3.25×10^{-6} C charges.

30. (II) Calculate the electric field at the center of a square 42.5 cm on a side if one corner is occupied by a $-38.6\,\mu$C charge and the other three are occupied by $-27.0\,\mu$C charges.

31. (II) Determine the direction and magnitude of the electric field at the point P in Fig. 16–56. The charges are separated by a distance $2a$, and point P is a distance x from the midpoint between the two charges. Express your answer in terms of Q, x, a, and k.

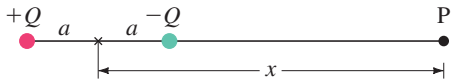


FIGURE 16–56 Problem 31.

32. (II) Two point charges, $Q_1 = -32\,\mu$C and $Q_2 = +45\,\mu$C, are separated by a distance of 12 cm. The electric field at the point P (see Fig. 16–57) is zero. How far from Q_1 is P?

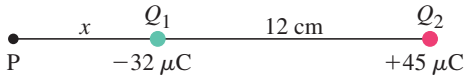


FIGURE 16–57 Problem 32.

33. (II) Determine the electric field $\vec{\mathbf{E}}$ at the origin 0 in Fig. 16–58 due to the two charges at A and B.

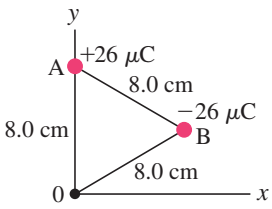


FIGURE 16–58 Problem 33.

34. (II) You are given two unknown point charges, Q_1 and Q_2. At a point on the line joining them, one-third of the way from Q_1 to Q_2, the electric field is zero (Fig. 16–59). What is the ratio Q_1/Q_2?

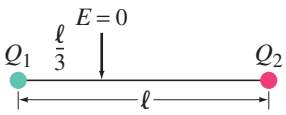


FIGURE 16–59 Problem 34.

35. (III) Use Coulomb's law to determine the magnitude and direction of the electric field at points A and B in Fig. 16–60 due to the two positive charges ($Q = 4.7\,\mu$C) shown. Are your results consistent with Fig. 16–32b?

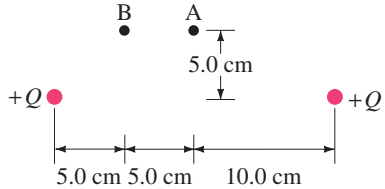


FIGURE 16–60 Problem 35.

36. (III) An electron (mass $m = 9.11 \times 10^{-31}$ kg) is accelerated in the uniform field $\vec{\mathbf{E}}$ ($E = 1.45 \times 10^4$ N/C) between two thin parallel charged plates. The separation of the plates is 1.60 cm. The electron is accelerated from rest near the negative plate and passes through a tiny hole in the positive plate, Fig. 16–61. (*a*) With what speed does it leave the hole? (*b*) Show that the gravitational force can be ignored.

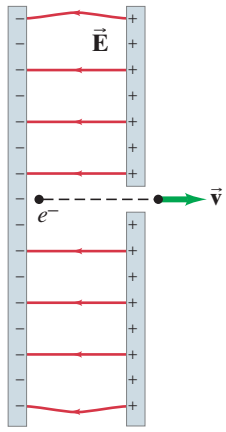


FIGURE 16–61 Problem 36.

*16–10 DNA

***37.** (III) The two strands of the helix-shaped DNA molecule are held together by electrostatic forces as shown in Fig. 16–39. Assume that the net average charge (due to electron sharing) indicated on H and N atoms has magnitude $0.2e$ and on the indicated C and O atoms is $0.4e$. Assume also that atoms on each molecule are separated by 1.0×10^{-10} m. Estimate the net force between (*a*) a thymine and an adenine; and (*b*) a cytosine and a guanine. For each bond (red dots) consider only the three atoms in a line (two atoms on one molecule, one atom on the other). (*c*) Estimate the total force for a DNA molecule containing 10^5 pairs of such molecules. Assume half are A–T pairs and half are C–G pairs.

*16–12 Gauss's Law

***38.** (I) The total electric flux from a cubical box of side 28.0 cm is 1.85×10^3 N·m^2/C. What charge is enclosed by the box?

***39.** (II) In Fig. 16–62, two objects, O_1 and O_2, have charges $+1.0\,\mu$C and $-2.0\,\mu$C, respectively, and a third object, O_3, is electrically neutral. (*a*) What is the electric flux through the surface A_1 that encloses all three objects? (*b*) What is the electric flux through the surface A_2 that encloses the third object only?

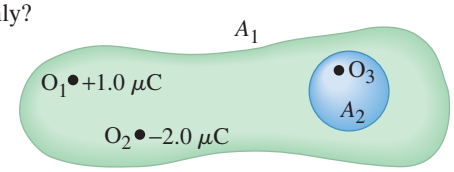


FIGURE 16–62 Problem 39.

***40.** (II) A cube of side 8.50 cm is placed in a uniform field $E = 7.50 \times 10^3$ N/C with edges parallel to the field lines. (*a*) What is the net flux through the cube? (*b*) What is the flux through each of its six faces?

***41.** (II) The electric field between two parallel square metal plates is 130 N/C. The plates are 0.85 m on a side and are separated by 3.0 cm. What is the charge on each plate (assume equal and opposite)? Neglect edge effects.

***42.** (II) The field just outside a 3.50-cm-radius metal ball is 3.75×10^2 N/C and points toward the ball. What charge resides on the ball?

***43.** (III) A point charge Q rests at the center of an uncharged thin spherical conducting shell. (See Fig. 16–34.) What is the electric field E as a function of r (*a*) for r less than the inner radius of the shell, (*b*) inside the shell, and (*c*) beyond the shell? (*d*) How does the shell affect the field due to Q alone? How does the charge Q affect the shell?

General Problems

44. How close must two electrons be if the magnitude of the electric force between them is equal to the weight of either at the Earth's surface?

45. Given that the human body is mostly made of water, estimate the total amount of positive charge in a 75-kg person.

46. A 3.0-g copper penny has a net positive charge of 32 μC. What fraction of its electrons has it lost?

47. Measurements indicate that there is an electric field surrounding the Earth. Its magnitude is about 150 N/C at the Earth's surface and points inward toward the Earth's center. What is the magnitude of the electric charge on the Earth? Is it positive or negative? [*Hint*: The electric field outside a uniformly charged sphere is the same as if all the charge were concentrated at its center.]

48. (*a*) The electric field near the Earth's surface has magnitude of about 150 N/C. What is the acceleration experienced by an electron near the surface of the Earth? (*b*) What about a proton? (*c*) Calculate the ratio of each acceleration to $g = 9.8\ \text{m/s}^2$.

49. A water droplet of radius 0.018 mm remains stationary in the air. If the downward-directed electric field of the Earth is 150 N/C, how many excess electron charges must the water droplet have?

50. Estimate the net force between the CO group and the HN group shown in Fig. 16–63. The C and O have charges $\pm 0.40e$, and the H and N have charges $\pm 0.20e$, where $e = 1.6 \times 10^{-19}$ C. [*Hint*: Do not include the "internal" forces between C and O, or between H and N.]

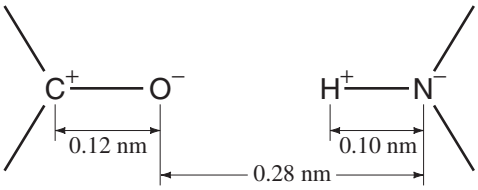


FIGURE 16–63 Problem 50.

51. In a simple model of the hydrogen atom, the electron revolves in a circular orbit around the proton with a speed of 2.2×10^6 m/s. Determine the radius of the electron's orbit. [*Hint*: See Chapter 5 on circular motion.]

52. Two small charged spheres hang from cords of equal length ℓ as shown in Fig. 16–64 and make small angles θ_1 and θ_2 with the vertical. (*a*) If $Q_1 = Q$, $Q_2 = 2Q$, and $m_1 = m_2 = m$, determine the ratio θ_1/θ_2. (*b*) Estimate the distance between the spheres.

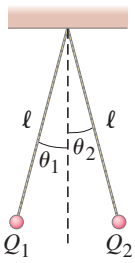


FIGURE 16–64
Problem 52.

53. A positive point charge $Q_1 = 2.5 \times 10^{-5}$ C is fixed at the origin of coordinates, and a negative point charge $Q_2 = -5.0 \times 10^{-6}$ C is fixed to the x axis at $x = +2.4$ m. Find the location of the place(s) along the x axis where the electric field due to these two charges is zero.

54. Dry air will break down and generate a spark if the electric field exceeds about 3×10^6 N/C. How much charge could be packed onto a green pea (diameter 0.75 cm) before the pea spontaneously discharges? [*Hint*: Eqs. 16–4 work outside a sphere if r is measured from its center.]

55. Two point charges, $Q_1 = -6.7\ \mu$C and $Q_2 = 1.8\ \mu$C, are located between two oppositely charged parallel plates, as shown in Fig. 16–65. The two charges are separated by a distance of $x = 0.47$ m. Assume that the electric field produced by the charged plates is uniform and equal to $E = 53{,}000$ N/C. Calculate the net electrostatic force on Q_1 and give its direction.

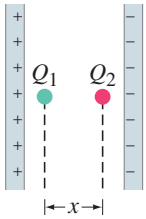


FIGURE 16–65
Problem 55.

56. Packing material made of pieces of foamed polystyrene can easily become charged and stick to each other. Given that the density of this material is about 35 kg/m^3, estimate how much charge might be on a 2.0-cm-diameter foamed polystyrene sphere, assuming the electric force between two spheres stuck together is equal to the weight of one sphere.

57. A point charge ($m = 1.0$ gram) at the end of an insulating cord of length 55 cm is observed to be in equilibrium in a uniform horizontal electric field of 9500 N/C, when the pendulum's position is as shown in Fig. 16–66, with the charge 12 cm above the lowest (vertical) position. If the field points to the right in Fig. 16–66, determine the magnitude and sign of the point charge.

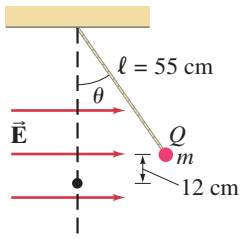


FIGURE 16–66
Problem 57.

58. Two small, identical conducting spheres A and B are a distance R apart; each carries the same charge Q. (*a*) What is the force sphere B exerts on sphere A? (*b*) An identical sphere with zero charge, sphere C, makes contact with sphere B and is then moved very far away. What is the net force now acting on sphere A? (*c*) Sphere C is brought back and now makes contact with sphere A and is then moved far away. What is the force on sphere A in this third case?

59. For an experiment, a colleague of yours says he smeared toner particles uniformly over the surface of a sphere 1.0 m in diameter and then measured an electric field of 5000 N/C near its surface. (*a*) How many toner particles (Example 16–6) would have to be on the surface to produce these results? (*b*) What is the total mass of the toner particles?

60. A proton ($m = 1.67 \times 10^{-27}$ kg) is suspended at rest in a uniform electric field $\vec{\mathbf{E}}$. Take into account gravity at the Earth's surface, and determine $\vec{\mathbf{E}}$.

61. A point charge of mass 0.185 kg, and net charge +0.340 μC, hangs at rest at the end of an insulating cord above a large sheet of charge. The horizontal sheet of fixed uniform charge creates a uniform vertical electric field in the vicinity of the point charge. The tension in the cord is measured to be 5.18 N. Calculate the magnitude and direction of the electric field due to the sheet of charge (Fig. 16–67).

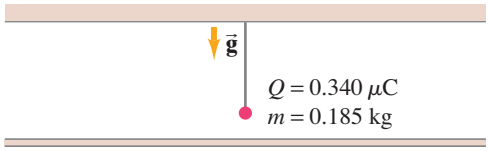


FIGURE 16–67 Problem 61.

62. An electron with speed $v_0 = 5.32 \times 10^6$ m/s is traveling parallel to an electric field of magnitude $E = 9.45 \times 10^3$ N/C. (*a*) How far will the electron travel before it stops? (*b*) How much time will elapse before it returns to its starting point?

63. Given the two charges shown in Fig. 16–68, at what position(s) x is the electric field zero?

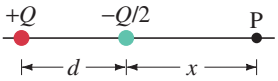


FIGURE 16–68 Problem 63.

64. What is the total charge of all the electrons in a 25-kg bar of aluminum? (Aluminum has 13 electrons per atom and an atomic mass of 27 u.)

65. Two point charges, $+Q$ and $-Q$ of mass m, are placed on the ends of a massless rod of length ℓ, which is fixed to a table by a pin through its center. If the apparatus is then subjected to a uniform electric field E parallel to the table and perpendicular to the rod, find the net torque on the system of rod plus charges.

66. Determine the direction and magnitude of the electric field at point P, Fig. 16–69. The two charges are separated by a distance of 2a. Point P is on the perpendicular bisector of the line joining the charges, a distance x from the midpoint between them. Express your answers in terms of Q, x, a, and k.

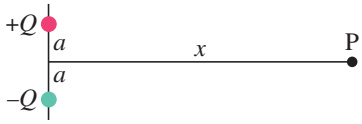


FIGURE 16–69 Problem 66.

67. A mole of carbon contains 7.22×10^{24} electrons. Two electrically neutral carbon spheres, each containing 1 mole of carbon, are separated by 15.0 cm (center to center). What fraction of electrons would have to be transferred from one sphere to the other for the electric force and the gravitational force between the spheres to be equal?

Search and Learn

1. Referring to Section 16–4 and Figs. 16–11 and 16–12, what happens to the separation of the leaves of an electroscope when the charging object is removed from an electroscope (*a*) charged by induction and (*b*) charged by conduction? (*c*) Is it possible to tell whether the electroscope in Fig. 16–12a has been charged by induction or by conduction? If so, which way was it charged? (*d*) Draw electric field lines (Section 16–8) for the electroscopes in Figs. 16–11a and 16–11b, omitting the fields around the charging rod. How do the fields differ?

2. Four equal positive point charges, each of charge 6.4 μC, are at the corners of a square of side 9.2 cm. What charge should be placed at the center of the square so that all charges are at equilibrium? Is this a stable or an unstable equilibrium (Section 9–4) in the plane?

3. Suppose electrons enter a uniform electric field midway between two plates at an angle θ_0 to the horizontal, as shown in Fig. 16–70. The path is symmetrical, so they leave at the same angle θ_0 and just barely miss the top plate. What is θ_0? Ignore fringing of the field.

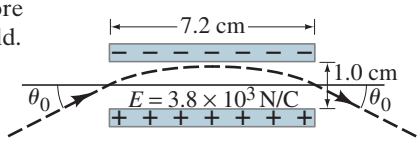


FIGURE 16–70 Search and Learn 3.

4. What experimental observations mentioned in the text rule out the possibility that the numerator in Coulomb's law contains the sum $(Q_1 + Q_2)$ rather than the product $Q_1 Q_2$?

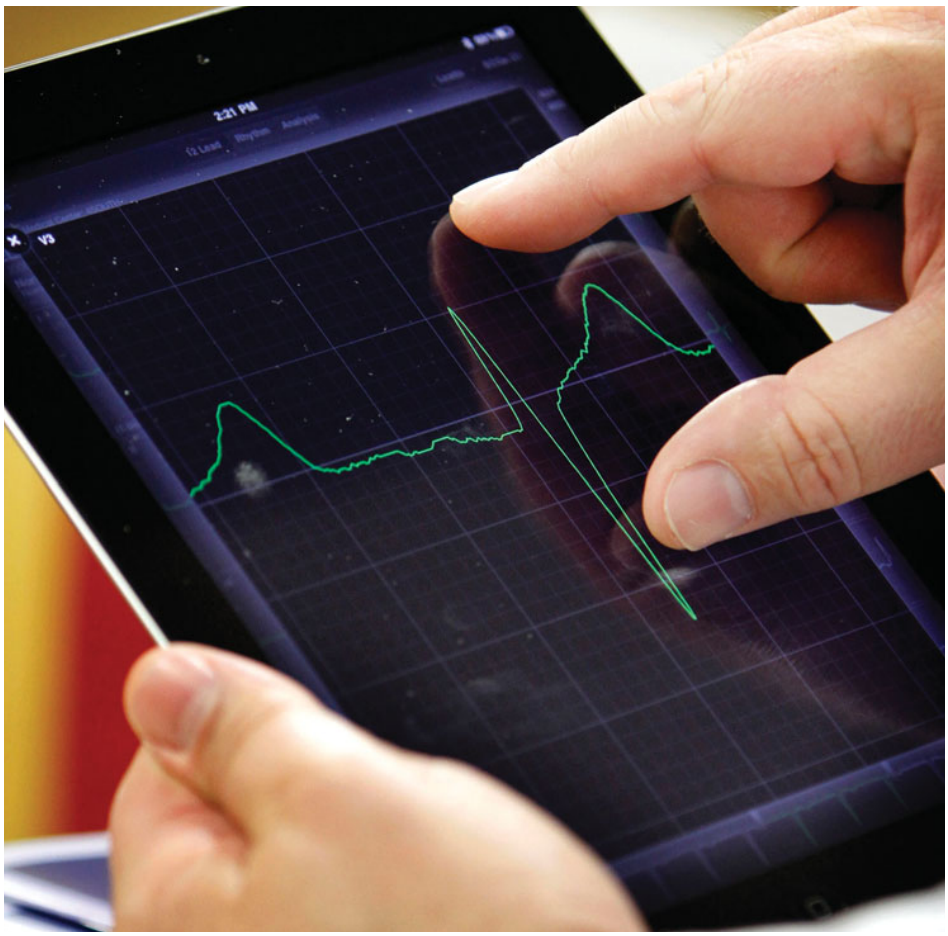
5. Near the surface of the Earth, there is a downward electric field of 150 N/C and a downward gravitational field of 9.8 N/kg. A charged 1.0-kg mass is observed to fall with acceleration 8.0 m/s^2. What are the magnitude and sign of its charge?

6. Identical negative charges $(Q = -e)$ are located at two of the three vertices of an equilateral triangle. The length of a side of the triangle is ℓ. What is the magnitude of the net electric field at the third vertex? If a third identical negative charge was located at the third vertex, then what would be the net electrostatic force on it due to the other two charges? Use symmetry and explain how you used it.

7. Suppose that electrical attraction, rather than gravity, were responsible for holding the Moon in orbit around the Earth. If equal and opposite charges Q were placed on the Earth and the Moon, what should be the value of Q to maintain the present orbit? Use data given on the inside front cover of this book. Treat the Earth and Moon as point particles.

ANSWERS TO EXERCISES

A: (*d*).
B: Opposite.
C: 0.3 N, to the right.
D: 0.32 m.
E: (*a*) No; (*b*) yes, midway between them.
F: 9.0×10^4 N/C, vertically upward.
G: (*d*), if the two + charges are not at opposite corners (use symmetry).



We are used to voltage in our lives—a 12-volt car battery, 110 V or 220 V at home, 1.5-volt flashlight batteries, and so on. Here we see displayed the voltage produced across a human heart, known as an electrocardiogram. Voltage is the same as electric potential difference between two points. Electric potential is defined as the potential energy per unit charge.

We discuss voltage and its relation to electric field, as well as electric energy storage, capacitors, and applications including the ECG shown here, binary numbers and digital electronics, TV and computer monitors, and digital TV.

CHAPTER 17

Electric Potential

CHAPTER-OPENING QUESTION—Guess now!

When two positively charged small spheres are pushed toward each other, what happens to their potential energy?

(a) It remains unchanged.
(b) It decreases.
(c) It increases.
(d) There is no potential energy in this situation.

We saw in Chapter 6 that the concept of energy was extremely valuable in dealing with the subject of mechanics. For one thing, energy is a conserved quantity and is thus an important tool for understanding nature. Furthermore, we saw that many Problems could be solved using the energy concept even though a detailed knowledge of the forces involved was not possible, or when a calculation involving Newton's laws would have been too difficult.

The energy point of view can be used in electricity, and it is especially useful. It not only extends the law of conservation of energy, but it gives us another way to view electrical phenomena. The energy concept is also a tool in solving Problems more easily in many cases than by using forces and electric fields.

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*17–6 Potential Due to Electric Dipole; Dipole Moment
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*17–11 TV and Computer Monitors: CRTs, Flat Screens
*17–12 Electrocardiogram (ECG or EKG)

17–1 Electric Potential Energy and Potential Difference

Electric Potential Energy

To apply conservation of energy, we need to define electric potential energy as we did for other types of potential energy. As we saw in Chapter 6, potential energy can be defined only for a conservative force. The work done by a conservative force in moving an object between any two positions is independent of the path taken. The electrostatic force between any two charges (Eq. 16–1, $F = kQ_1Q_2/r^2$) is conservative because the dependence on position is just like the gravitational force (Eq. 5–4), which is conservative. Hence we can define potential energy PE for the electrostatic force.

We saw in Chapter 6 that the change in potential energy between any two points, a and b, equals the negative of the work done by the conservative force on an object as it moves from point a to point b: $\Delta PE = -W$.

Thus we define the change in electric potential energy, $PE_b - PE_a$, when a point charge q moves from some point a to another point b, as the negative of the work done by the electric force on the charge as it moves from point a to point b. For example, consider the electric field between two equally but oppositely charged parallel plates; we assume their separation is small compared to their width and height, so the field $\vec{\mathbf{E}}$ will be uniform over most of the region, Fig. 17–1. Now consider a tiny positive point charge q placed at the point "a" very near the positive plate as shown. This charge q is so small that it has no effect on $\vec{\mathbf{E}}$. If this charge q at point a is released, the electric force will do work on the charge and accelerate it toward the negative plate. The work W done by the electric field E to move the charge a distance d is (using Eq. 16–5, $F = qE$)

$$W = Fd = qEd. \qquad \text{[uniform } \vec{\mathbf{E}}\text{]}$$

The change in electric potential energy equals the negative of the work done by the electric force:

$$PE_b - PE_a = -qEd \qquad \text{[uniform } \vec{\mathbf{E}}\text{]} \quad \textbf{(17–1)}$$

for this case of uniform electric field $\vec{\mathbf{E}}$. In the case illustrated, the potential energy decreases (ΔPE is negative); and as the charged particle accelerates from point a to point b in Fig. 17–1, the particle's kinetic energy KE increases—by an equal amount. In accord with the conservation of energy, electric potential energy is transformed into kinetic energy, and the total energy is conserved. Note that the positive charge q has its greatest potential energy at point a, near the positive plate.[†] The reverse is true for a negative charge: its potential energy is greatest near the negative plate.

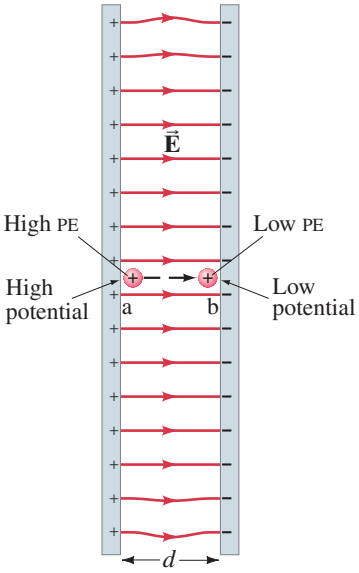


FIGURE 17–1 Work is done by the electric field $\vec{\mathbf{E}}$ in moving the positive charge from position a to position b.

Electric Potential and Potential Difference

In Chapter 16, we found it useful to define the electric field as the force per unit charge. Similarly, it is useful to define the **electric potential** (or simply the **potential** when "electric" is understood) as the *electric potential energy per unit charge*. Electric potential is given the symbol V. If a positive test charge q in an electric field has electric potential energy PE_a at some point a (relative to some zero potential energy), the electric potential V_a at this point is

$$V_a = \frac{PE_a}{q}. \qquad \textbf{(17–2a)}$$

As we discussed in Chapter 6, only differences in potential energy are physically meaningful. Hence only the **difference in potential**, or the **potential difference**, between two points a and b (such as those shown in Fig. 17–1) is measurable.

[†]At point a, the positive charge q has its greatest ability to do work (on some other object or system).

When the electric force does positive work on a charge, the kinetic energy increases and the potential energy decreases. The difference in potential energy, $\text{PE}_\text{b} - \text{PE}_\text{a}$, is equal to the negative of the work, W_ba, done by the electric field to move the charge from a to b; so the potential difference V_ba is

$$V_\text{ba} = V_\text{b} - V_\text{a} = \frac{\text{PE}_\text{b} - \text{PE}_\text{a}}{q} = -\frac{W_\text{ba}}{q}. \qquad \textbf{(17–2b)}$$

Note that electric potential, like electric field, does not depend on our test charge q. V depends on the other charges that create the field, not on the test charge q; q acquires potential energy by being in the potential V due to the other charges.

We can see from our definition that the positive plate in Fig. 17–1 is at a higher potential than the negative plate. Thus a positively charged object moves naturally from a high potential to a low potential. A negative charge does the reverse.

The unit of electric potential, and of potential difference, is joules/coulomb and is given a special name, the **volt**, in honor of Alessandro Volta (1745–1827) who is best known for inventing the electric battery. The volt is abbreviated V, so 1 V = 1 J/C. Potential difference, since it is measured in volts, is often referred to as **voltage**. (Be careful not to confuse V for volts, with italic V for voltage.)

If we wish to speak of the potential V_a at some point a, we must be aware that V_a depends on where the potential is chosen to be zero. The zero for electric potential in a given situation can be chosen arbitrarily, just as for potential energy, because only differences in potential energy can be measured. Often the ground, or a conductor connected directly to the ground (the Earth), is taken as zero potential, and other potentials are given with respect to ground. (Thus, a point where the voltage is 50 V is one where the difference of potential between it and ground is 50 V.) In other cases, as we shall see, we may choose the potential to be zero at an infinite distance.

CONCEPTUAL EXAMPLE 17–1 **A negative charge.** Suppose a negative charge, such as an electron, is placed near the negative plate in Fig. 17–1, at point b, shown here in Fig. 17–2. If the electron is free to move, will its electric potential energy increase or decrease? How will the electric potential change?

RESPONSE An electron released at point b will be attracted to the positive plate. As the electron accelerates toward the positive plate, its kinetic energy increases, so its potential energy *decreases*: $\text{PE}_\text{a} < \text{PE}_\text{b}$ and $\Delta\text{PE} = \text{PE}_\text{a} - \text{PE}_\text{b} < 0$. But note that the electron moves from point b at low potential to point a at higher potential: $\Delta V = V_\text{a} - V_\text{b} > 0$. (Potentials V_a and V_b are due to the charges on the plates, not due to the electron.) The signs of ΔPE and ΔV are opposite because of the negative charge of the electron.

NOTE A positive charge placed next to the negative plate at b would stay there, with no acceleration. A positive charge tends to move from high potential to low.

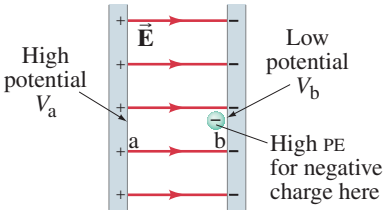


FIGURE 17–2 Central part of Fig. 17–1, showing a negative point charge near the negative plate. Example 17–1.

CAUTION
A negative charge has high PE when potential V is low

Because the electric potential difference is defined as the potential energy difference per unit charge, then the change in potential energy of a charge q when it moves from point a to point b is

$$\Delta\text{PE} = \text{PE}_\text{b} - \text{PE}_\text{a} = q(V_\text{b} - V_\text{a}) = qV_\text{ba}. \qquad \textbf{(17–3)}$$

That is, if an object with charge q moves through a potential difference V_ba, its potential energy changes by an amount qV_ba. For example, if the potential difference between the two plates in Fig. 17–1 is 6 V, then a +1 C charge moved from point b to point a will gain (1 C)(6 V) = 6 J of electric potential energy. (And it will lose 6 J of electric potential energy if it moves from a to b.) Similarly, a +2 C charge will gain $\Delta\text{PE} = (2\text{ C})(6\text{ V}) = 12$ J, and so on. Thus, electric potential difference is a measure of how much energy an electric charge can acquire in a given situation. And, since energy is the ability to do work, the electric potential difference is also a measure of how much *work* a given charge can do. The exact amount of energy or work depends both on the potential difference and on the charge.

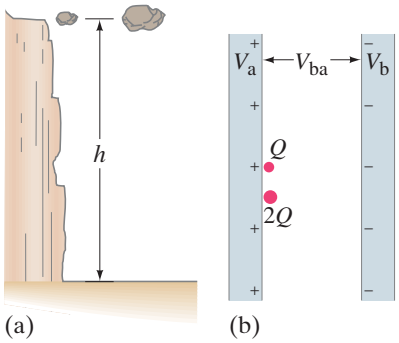
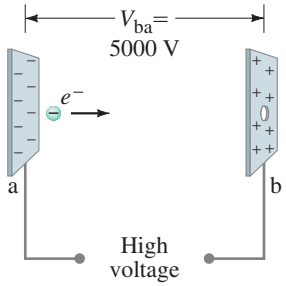


FIGURE 17–3 (a) Two rocks are at the same height. The larger rock has more potential energy. (b) Two positive charges have the same electric potential. The $2Q$ charge has more potential energy.

TABLE 17–1 Some Typical Potential Differences (Voltages)

Source	Voltage (approx.)
Thundercloud to ground	10^8 V
High-voltage power line	10^5–10^6 V
Automobile ignition	10^4 V
Household outlet	10^2 V
Automobile battery	12 V
Flashlight battery (AA, AAA, C, D)	1.5 V
Resting potential across nerve membrane	10^{-1} V
Potential changes on skin (ECG and EEG)	10^{-4} V

FIGURE 17–4 Electron accelerated, Example 17–2.



To better understand electric potential, let's make a comparison to the gravitational case when a rock falls from the top of a cliff. The greater the height, h, of a cliff, the more potential energy ($= mgh$) the rock has at the top of the cliff relative to the bottom, and the more kinetic energy it will have when it reaches the bottom. The actual amount of kinetic energy it will acquire, and the amount of work it can do, depends both on the height of the cliff and the mass m of the rock. A large rock and a small rock can be at the same height h (Fig. 17–3a) and thus have the same "gravitational potential," but the larger rock has the greater potential energy (it has more mass). The electrical case is similar (Fig. 17–3b): the potential energy change, or the work that can be done, depends both on the potential difference (corresponding to the height of the cliff) and on the charge (corresponding to mass), Eq. 17–3. But note a significant difference: electric charge comes in two types, + and −, whereas gravitational mass is always +.

Sources of electrical energy such as batteries and electric generators are meant to maintain a potential difference. The actual amount of energy transformed by such a device depends on how much charge flows, as well as the potential difference (Eq. 17–3). For example, consider an automobile headlight connected to a 12.0-V battery. The amount of energy transformed (into light and thermal energy) is proportional to how much charge flows, which in turn depends on how long the light is on. If over a given period of time 5.0 C of charge flows through the light, the total energy transformed is $(5.0\text{ C})(12.0\text{ V}) = 60\text{ J}$. If the headlight is left on twice as long, 10.0 C of charge will flow and the energy transformed is $(10.0\text{ C})(12.0\text{ V}) = 120\text{ J}$. Table 17–1 presents some typical voltages.

EXAMPLE 17–2 Electron in TV tube. Suppose an electron is accelerated from rest through a potential difference $V_b - V_a = V_{ba} = +5000$ V (Fig. 17–4). (*a*) What is the change in electric potential energy of the electron? What is (*b*) the kinetic energy, and (*c*) the speed of the electron $(m = 9.1 \times 10^{-31}\text{ kg})$ as a result of this acceleration?

APPROACH The electron, accelerated toward the positive plate, will change in potential energy by an amount $\Delta\text{PE} = qV_{ba}$ (Eq. 17–3). The loss in potential energy will equal its gain in kinetic energy (energy conservation).

SOLUTION (*a*) The charge on an electron is $q = -e = -1.6 \times 10^{-19}$ C. Therefore its change in potential energy is

$$\Delta\text{PE} = qV_{ba} = (-1.6 \times 10^{-19}\text{ C})(+5000\text{ V}) = -8.0 \times 10^{-16}\text{ J}.$$

The minus sign indicates that the potential energy decreases. The potential difference, V_{ba}, has a positive sign because the final potential V_b is higher than the initial potential V_a. Negative electrons are attracted toward a positive electrode (or plate) and repelled away from a negative electrode.

(*b*) The potential energy lost by the electron becomes kinetic energy KE. From conservation of energy (Eq. 6–11a), $\Delta\text{KE} + \Delta\text{PE} = 0$, so

$$\Delta\text{KE} = -\Delta\text{PE}$$
$$\tfrac{1}{2}mv^2 - 0 = -q(V_b - V_a) = -qV_{ba},$$

where the initial kinetic energy is zero since we are given that the electron started from rest. So the final $\text{KE} = -qV_{ba} = 8.0 \times 10^{-16}$ J.

(*c*) In the equation just above we solve for v:

$$v = \sqrt{-\frac{2qV_{ba}}{m}} = \sqrt{-\frac{2(-1.6 \times 10^{-19}\text{ C})(5000\text{ V})}{9.1 \times 10^{-31}\text{ kg}}} = 4.2 \times 10^7\text{ m/s}.$$

NOTE The electric potential energy does not depend on the mass, only on the charge and voltage. The speed *does* depend on m.

EXERCISE A Instead of the electron in Example 17–2, suppose a proton $(m = 1.67 \times 10^{-27}\text{ kg})$ was accelerated from rest by a potential difference $V_{ba} = -5000$ V. What would be the proton's (*a*) change in PE, and (*b*) final speed?

17–2 Relation between Electric Potential and Electric Field

The effects of any charge distribution can be described either in terms of electric field or in terms of electric potential. Electric potential is often easier to use because it is a scalar, whereas electric field is a vector. There is an intimate connection between the potential and the field. Let us consider the case of a uniform electric field, such as that between the parallel plates of Fig. 17–1 whose difference of potential is V_{ba}. The work done by the electric field to move a positive charge q from point a to point b is equal to the negative of the change in potential energy (Eq. 17–2b), so

$$W = -q(V_b - V_a) = -qV_{ba}.$$

We can also write the work done as the force times distance, where the force on q is $F = qE$, so

$$W = Fd = qEd,$$

where d is the distance (parallel to the field lines) between points a and b. We now set these two expressions for W equal and find $qV_{ba} = -qEd$, or

$$V_{ba} = -Ed. \qquad \text{[uniform } \vec{\mathbf{E}}\text{]} \quad \textbf{(17–4a)}$$

If we solve for E, we find

$$E = -\frac{V_{ba}}{d}. \qquad \text{[uniform } \vec{\mathbf{E}}\text{]} \quad \textbf{(17–4b)}$$

From Eq. 17–4b we see that the units for electric field can be written as volts per meter (V/m), as well as newtons per coulomb (N/C, from $E = F/q$). These are equivalent because $1\,\text{N/C} = 1\,\text{N}\cdot\text{m/C}\cdot\text{m} = 1\,\text{J/C}\cdot\text{m} = 1\,\text{V/m}$. The minus sign in Eq. 17–4b tells us that $\vec{\mathbf{E}}$ points in the direction of decreasing potential V.

EXAMPLE 17–3 **Electric field obtained from voltage.** Two parallel plates are charged to produce a potential difference of 50 V. If the separation between the plates is 0.050 m, calculate the magnitude of the electric field in the space between the plates (Fig. 17–5).

APPROACH We apply Eq. 17–4b to obtain the magnitude of E, assumed uniform.

SOLUTION The magnitude of the electric field is

$$E = V_{ba}/d = (50\,\text{V}/0.050\,\text{m}) = 1000\,\text{V/m}.$$

NOTE Equations 17–4 apply only for a uniform electric field. The general relationship between $\vec{\mathbf{E}}$ and V is more complicated.

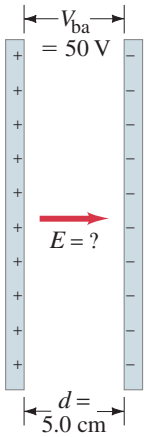


FIGURE 17–5 Example 17–3.

* General Relation between $\vec{\mathbf{E}}$ and V

In a region where $\vec{\mathbf{E}}$ is not uniform, the connection between $\vec{\mathbf{E}}$ and V takes on a different form than Eqs. 17–4. In general, it is possible to show that the electric field in a given direction at any point in space is equal to the *rate at which the electric potential decreases over distance in that direction*. For example, the x component of the electric field is given by $E_x = -\Delta V/\Delta x$, where ΔV is the change in potential over a very short distance Δx.

Breakdown Voltage

When very high voltages are present, air can become ionized due to the high electric fields. Any odd free electron can be accelerated to sufficient kinetic energy to knock electrons out of O_2 and N_2 molecules of the air. This **breakdown** of air occurs when the electric field exceeds about 3×10^6 V/m. When electrons recombine with their molecules, light is emitted. Such breakdown of air is the source of lightning, the spark of a car's spark plug, and even short sparks between your fingers and a doorknob after you walk across a synthetic rug or slide across a car seat (which can result in a significant transfer of charge to you).

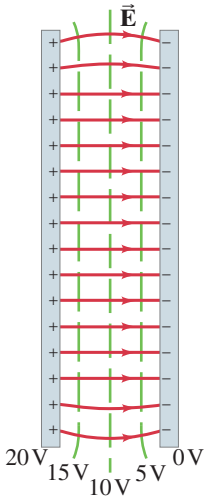
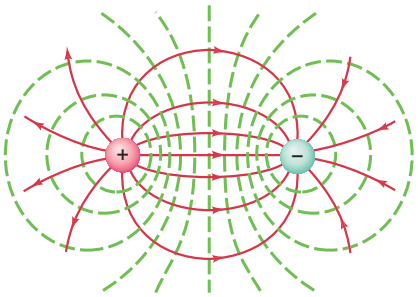


FIGURE 17–6 Equipotential lines (the green dashed lines) between two charged parallel plates are always perpendicular to the electric field (solid red lines).

FIGURE 17–7 Equipotential lines (green, dashed) are always perpendicular to the electric field lines (solid red), shown here for two equal but oppositely charged particles (an "electric dipole").



17–3 Equipotential Lines and Surfaces

The electric potential can be represented by drawing **equipotential lines** or, in three dimensions, **equipotential surfaces**. An equipotential surface is one on which all points are at the same potential. That is, the potential difference between any two points on the surface is zero, so no work is required to move a charge from one point on the surface to the other. An *equipotential surface must be perpendicular to the electric field* at any point. If this were not so—that is, if there were a component of $\vec{\mathbf{E}}$ parallel to the surface—it would require work to move the charge along the surface against this component of $\vec{\mathbf{E}}$; and this would contradict the idea that it is an *equi*potential surface.

The fact that the electric field lines and equipotential surfaces are mutually perpendicular helps us locate the equipotentials when the electric field lines are known. In a normal two-dimensional drawing, we show equipotential *lines*, which are the intersections of equipotential surfaces with the plane of the drawing. In Fig. 17–6, a few of the equipotential lines are drawn (dashed green lines) for the electric field (red lines) between two parallel plates at a potential difference of 20 V. The negative plate is arbitrarily chosen to be zero volts and the potential of each equipotential line is indicated. Note that $\vec{\mathbf{E}}$ points toward lower values of V. The equipotential lines for the case of two equal but oppositely charged particles are shown in Fig. 17–7 as green dashed lines. (This combination of equal + and − charges is called an "electric dipole," as we saw in Section 16–8; see Fig. 16–32a.)

Unlike electric field lines, which start and end on electric charges, equipotential lines and surfaces are always continuous and never end, and so continue beyond the borders of Figs. 17–6 and 17–7. A useful analogy for equipotential lines is a topographic map: the contour lines are gravitational equipotential lines (Fig. 17–8).

We saw in Section 16–9 that there can be no electric field within a conductor in the static case, for otherwise the free electrons would feel a force and would move. Indeed the *entire volume of a conductor must be entirely at the same potential in the static case*. The surface of a conductor is thus an equipotential surface. (If it weren't, the free electrons at the surface would move, because whenever there is a potential difference between two points, free charges will move.) This is fully consistent with our result in Section 16–9 that the electric field at the surface of a conductor must be perpendicular to the surface.



FIGURE 17–8 A topographic map (here, a portion of the Sierra Nevada in California) shows continuous contour lines, each of which is at a fixed height above sea level. Here they are at 80-ft (25-m) intervals. If you walk along one contour line, you neither climb nor descend. If you cross lines, and especially if you climb perpendicular to the lines, you will be changing your gravitational potential (rapidly, if the lines are close together).

17–4 The Electron Volt, a Unit of Energy

The joule is a very large unit for dealing with energies of electrons, atoms, or molecules. For this purpose, the unit **electron volt** (eV) is used. One electron volt is defined as the energy acquired by a particle carrying a charge whose magnitude equals that on the electron ($q = e$) as a result of moving through a potential difference of 1 V. The charge on an electron has magnitude 1.6022×10^{-19} C, and the change in potential energy equals qV. So 1 eV is equal to $(1.6022 \times 10^{-19}\,\text{C})(1.00\,\text{V}) = 1.6022 \times 10^{-19}$ J:

Electron volt

$$1\,\text{eV} = 1.6022 \times 10^{-19} \approx 1.60 \times 10^{-19}\,\text{J}.$$

An electron that accelerates through a potential difference of 1000 V will lose 1000 eV of potential energy and thus gain 1000 eV or 1 keV (kiloelectron volt) of kinetic energy.

On the other hand, if a particle with a charge equal to twice the magnitude of the charge on the electron $(= 2e = 3.2 \times 10^{-19}\,\text{C})$ moves through a potential difference of 1000 V, its kinetic energy will increase by $2000\,\text{eV} = 2\,\text{keV}$.

Although the electron volt is handy for *stating* the energies of molecules and elementary particles, it is *not* a proper SI unit. For calculations, electron volts should be converted to joules using the conversion factor just given. In Example 17–2, for example, the electron acquired a kinetic energy of $8.0 \times 10^{-16}\,\text{J}$. We can quote this energy as 5000 eV $(= 8.0 \times 10^{-16}\,\text{J}/1.6 \times 10^{-19}\,\text{J/eV})$, but when determining the speed of a particle in SI units, we must use the KE in joules (J).

EXERCISE B What is the kinetic energy of a He^{2+} ion released from rest and accelerated through a potential difference of 2.5 kV? (*a*) 2500 eV, (*b*) 500 eV, (*c*) 5000 eV, (*d*) 10,000 eV, (*e*) 250 eV.

17–5 Electric Potential Due to Point Charges

The electric potential at a distance r from a single point charge Q can be derived from the expression for its electric field (Eq. 16–4, $E = kQ/r^2$) using calculus. The potential in this case is usually taken to be zero at infinity ($= \infty$, which means extremely, indefinitely, far away); this is also where the electric field $(E = kQ/r^2)$ is zero. The result is

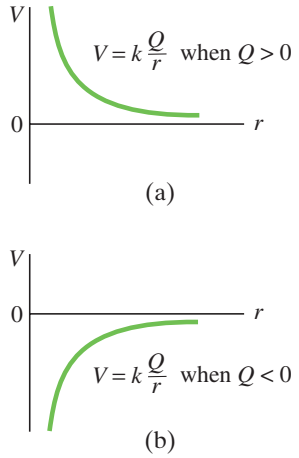
$$V = k\frac{Q}{r} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}, \qquad \left[\begin{array}{l}\text{single point charge}\\ V = 0 \text{ at } r = \infty\end{array}\right] \quad \textbf{(17–5)}$$

where $k = 8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2 \approx 9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2$. We can think of V here as representing the absolute potential at a distance r from the charge Q, where $V = 0$ at $r = \infty$; or we can think of V as the potential difference between r and infinity. (The symbol ∞ means infinitely far away.) Notice that the potential V decreases with the first power of the distance, whereas the electric field (Eq. 16–4) decreases as the *square* of the distance. The potential near a positive charge is large and positive, and it decreases toward zero at very large distances, Fig. 17–9a. The potential near a negative charge is negative and increases toward zero at large distances, Fig. 17–9b. Equation 17–5 is sometimes called the **Coulomb potential** (it has its origin in Coulomb's law).

CAUTION

$V \propto \frac{1}{r}$, $E \propto \frac{1}{r^2}$ for a point charge

FIGURE 17–9 Potential V as a function of distance r from a single point charge Q when the charge is (a) positive, (b) negative.



EXAMPLE 17–4 **Potential due to a positive or a negative charge.** Determine the potential at a point 0.50 m (*a*) from a $+20\,\mu\text{C}$ point charge, (*b*) from a $-20\,\mu\text{C}$ point charge.

APPROACH The potential due to a point charge is given by Eq. 17–5, $V = kQ/r$.

SOLUTION (*a*) At a distance of 0.50 m from a positive 20 μC charge, the potential is

$$V = k\frac{Q}{r} = (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{20 \times 10^{-6}\,\text{C}}{0.50\,\text{m}}\right) = 3.6 \times 10^5\,\text{V}.$$

(*b*) For the negative charge,

$$V = (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{-20 \times 10^{-6}\,\text{C}}{0.50\,\text{m}}\right) = -3.6 \times 10^5\,\text{V}.$$

NOTE Potential can be positive or negative, and we always include a charge's sign when we find electric potential.

PROBLEM SOLVING

Keep track of charge signs for electric potential

EXAMPLE 17–5 **Work required to bring two positive charges close together.** What minimum work must be done by an external force to bring a charge $q = 3.00\,\mu\text{C}$ from a great distance away (take $r = \infty$) to a point 0.500 m from a charge $Q = 20.0\,\mu\text{C}$?

CAUTION
We cannot use $W = Fd$ if F is not constant

APPROACH To find the work we cannot simply multiply the force times distance because the force is proportional to $1/r^2$ and so is not constant. Instead we can set the change in potential energy equal to the (positive of the) work required of an *external* force (Chapter 6, Eq. 6–7a), and Eq. 17–3: $W_{\text{ext}} = \Delta\text{PE} = q(V_b - V_a)$. We get the potentials V_b and V_a using Eq. 17–5.

SOLUTION The external work required is equal to the change in potential energy:

$$W_{\text{ext}} = q(V_b - V_a) = q\left(\frac{kQ}{r_b} - \frac{kQ}{r_a}\right),$$

where $r_b = 0.500\,\text{m}$ and $r_a = \infty$. The right-hand term within the parentheses is zero $(1/\infty = 0)$ so

$$W_{\text{ext}} = (3.00 \times 10^{-6}\,\text{C})\frac{(8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(2.00 \times 10^{-5}\,\text{C})}{(0.500\,\text{m})} = 1.08\,\text{J}.$$

NOTE We could not use Eqs. 17–4 here because they apply *only* to uniform fields. But we did use Eq. 17–3 because it is always valid.

EXERCISE C What work is required to bring a charge $q = 3.00\,\mu\text{C}$ originally a distance of 1.50 m from a charge $Q = 20.0\,\mu\text{C}$ until it is 0.50 m away?

To determine the electric field at points near a collection of two or more point charges requires adding up the electric fields due to each charge. Since the electric field is a vector, this can be time consuming or complicated. To find the electric potential at a point due to a collection of point charges is far easier, because the electric potential is a scalar, and hence you only need to add numbers (with appropriate signs) without concern for direction.

EXAMPLE 17–6 **Potential above two charges.** Calculate the electric potential (*a*) at point A in Fig. 17–10 due to the two charges shown, and (*b*) at point B. [This is the same situation as Examples 16–9 and 16–10, Fig. 16–29, where we calculated the electric field at these points.]

CAUTION
Potential is a scalar and has no components

APPROACH The total potential at point A (or at point B) is the algebraic sum of the potentials at that point due to each of the two charges Q_1 and Q_2. The potential due to each single charge is given by Eq. 17–5. We do not have to worry about directions because electric potential is a scalar quantity. But we do have to keep track of the signs of charges.

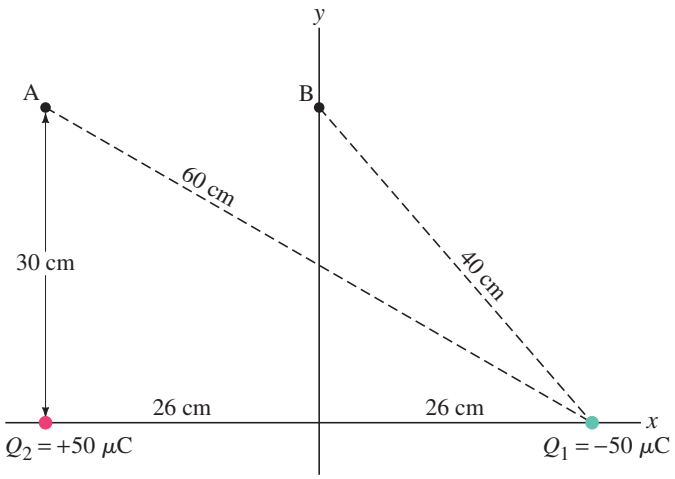


FIGURE 17–10 Example 17–6. (See also Examples 16–9 and 16–10, Fig. 16–29.)

SOLUTION (*a*) We add the potentials at point A due to each charge Q_1 and Q_2, and we use Eq. 17–5 for each:

$$V_A = V_{A2} + V_{A1}$$
$$= k\frac{Q_2}{r_{2A}} + k\frac{Q_1}{r_{1A}}$$

where $r_{1A} = 60\,\text{cm}$ and $r_{2A} = 30\,\text{cm}$. Then

$$V_A = \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(5.0 \times 10^{-5}\,\text{C})}{0.30\,\text{m}} + \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(-5.0 \times 10^{-5}\,\text{C})}{0.60\,\text{m}}$$
$$= 1.50 \times 10^6\,\text{V} - 0.75 \times 10^6\,\text{V}$$
$$= 7.5 \times 10^5\,\text{V}.$$

(*b*) At point B, $r_{1B} = r_{2B} = 0.40\,\text{m}$, so

$$V_B = V_{B2} + V_{B1}$$
$$= \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(5.0 \times 10^{-5}\,\text{C})}{0.40\,\text{m}} + \frac{(9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(-5.0 \times 10^{-5}\,\text{C})}{0.40\,\text{m}}$$
$$= 0\,\text{V}.$$

NOTE The two terms in the sum in (*b*) cancel for any point equidistant from Q_1 and Q_2 ($r_{1B} = r_{2B}$). Thus the potential will be zero everywhere on the plane equidistant between the two opposite charges. This plane is an equipotential surface with $V = 0$.

Simple summations like these can be performed for any number of point charges.

CONCEPTUAL EXAMPLE 17–7 **Potential energies.** Consider the three pairs of charges shown in Fig. 17–11. Call them Q_1 and Q_2. (*a*) Which set has a positive potential energy? (*b*) Which set has the most negative potential energy? (*c*) Which set requires the most work to separate the charges to infinity? Assume the charges all have the same magnitude.

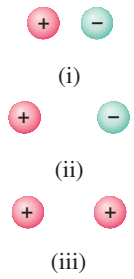


FIGURE 17–11 Example 17–7.

RESPONSE The potential energy equals the work required to bring the two charges near each other, starting at a great distance (∞). Assume the left (+) charge Q_1 is already there. To bring a second charge Q_2 close to the first from a great distance away (∞) requires external work

$$W_{\text{ext}} = Q_2V = k\frac{Q_1Q_2}{r}$$

where r is the final distance between them. Thus the potential energy of the two charges is

$$\text{PE} = k\frac{Q_1Q_2}{r}.$$

(*a*) Set (iii) has a positive potential energy because the charges have the same sign. (*b*) Both (i) and (ii) have opposite signs of charge and negative PE. Because r is smaller in (i), the PE is most negative for (i). (*c*) Set (i) will require the most work for separation to infinity. The more negative the potential energy, the more work required to separate the charges and bring the PE up to zero ($r = \infty$), as in Fig. 17–9b.

EXERCISE D Return to the Chapter-Opening Question, page 473, and answer it again now. Try to explain why you may have answered differently the first time.

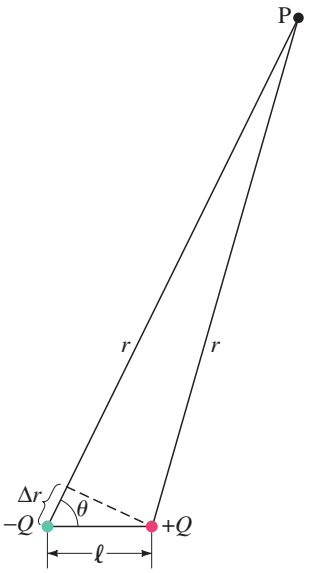


FIGURE 17–12 Electric dipole. Calculation of potential V at point P.

*17–6 Potential Due to Electric Dipole; Dipole Moment

Two equal point charges Q, of opposite sign, separated by a distance ℓ, are called an **electric dipole**. The electric field lines and equipotential surfaces for a dipole were shown in Fig. 17–7. Because electric dipoles occur often in physics, as well as in other disciplines such as molecular biology, it is useful to examine them more closely.

The electric potential at an arbitrary point P due to a dipole, Fig. 17–12, is the sum of the potentials due to each of the two charges:

$$V = \frac{kQ}{r} + \frac{k(-Q)}{r + \Delta r} = kQ\left(\frac{1}{r} - \frac{1}{r + \Delta r}\right) = kQ\frac{\Delta r}{r(r + \Delta r)},$$

where r is the distance from P to the positive charge and $r + \Delta r$ is the distance to the negative charge. This equation becomes simpler if we consider points P whose distance from the dipole is much larger than the separation of the two charges—that is, for $r \gg \ell$. From Fig. 17–12 we see that $\Delta r = \ell \cos\theta$; since $r \gg \Delta r = \ell \cos\theta$, we can neglect Δr in the denominator as compared to r. Then we obtain

$$V \approx \frac{kQ\ell \cos\theta}{r^2}. \qquad \text{[dipole; } r \gg \ell\text{]} \quad \textbf{(17–6a)}$$

We see that the potential decreases as the *square* of the distance from the dipole, whereas for a single point charge the potential decreases with the first power of the distance (Eq. 17–5). It is not surprising that the potential should fall off faster for a dipole: when you are far from a dipole, the two equal but opposite charges appear so close together as to tend to neutralize each other.

The product $Q\ell$ in Eq. 17–6a is referred to as the **dipole moment**, p, of the dipole. Equation 17–6a in terms of the dipole moment is

$$V \approx \frac{kp \cos\theta}{r^2}. \qquad \text{[dipole; } r \gg \ell\text{]} \quad \textbf{(17–6b)}$$

A dipole moment has units of coulomb-meters (C·m), although for molecules a smaller unit called a *debye* is sometimes used: 1 debye $= 3.33 \times 10^{-30}$ C·m.

In many molecules, even though they are electrically neutral, the electrons spend more time in the vicinity of one atom than another, which results in a separation of charge. Such molecules have a dipole moment and are called **polar molecules**. We already saw that water (Fig. 16–4) is a polar molecule, and we have encountered others in our discussion of molecular biology (Section 16–10). Table 17–2 gives the dipole moments for several molecules. The + and − signs indicate on which atoms these charges lie. The last two entries are a part of many organic molecules and play an important role in molecular biology.

PHYSICS APPLIED
Dipoles in molecular biology

TABLE 17–2 Dipole Moments of Selected Molecules

Molecule	Dipole Moment (C · m)
$H_2^{(+)}O^{(-)}$	6.1×10^{-30}
$H^{(+)}Cl^{(-)}$	3.4×10^{-30}
$N^{(-)}H_3^{(+)}$	5.0×10^{-30}
$>N^{(-)}—H^{(+)}$	$\approx 3.0 \times 10^{-30}$ ‡
$>C^{(+)}=O^{(-)}$	$\approx 8.0 \times 10^{-30}$ ‡

‡ These last two groups often appear on larger molecules; hence the value for the dipole moment will vary somewhat, depending on the rest of the molecule.

17–7 Capacitance

A **capacitor** is a device that can store electric charge, and normally consists of two conducting objects (usually plates or sheets) placed near each other but not touching. Capacitors are widely used in electronic circuits and sometimes are called **condensers**. Capacitors store charge for later use, such as in a camera flash, and as energy backup in devices like computers if the power fails. Capacitors also block surges of charge and energy to protect circuits. Very tiny capacitors serve as memory for the "ones" and "zeros" of the binary code in the random access memory (RAM) of computers and other electronic devices (as in Fig. 17–35). Capacitors serve many other applications as well, some of which we will discuss.

PHYSICS APPLIED
Uses of capacitors

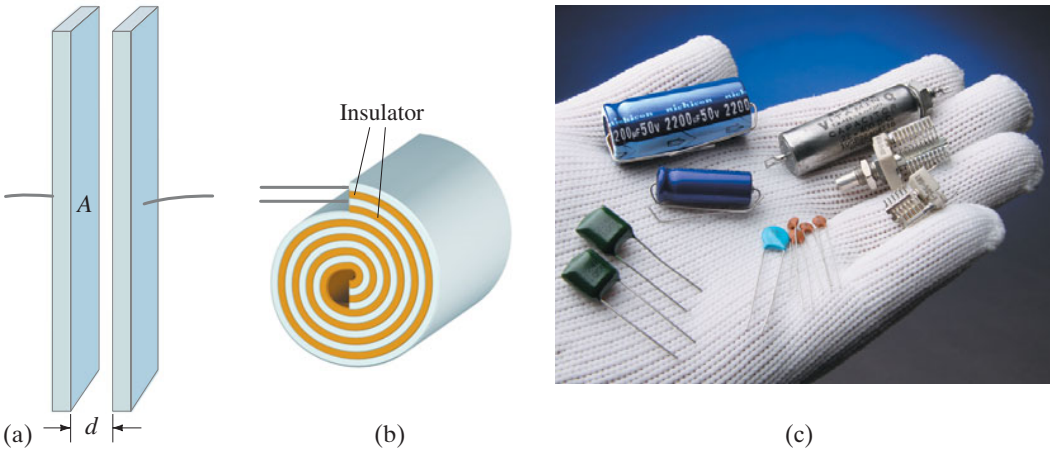


FIGURE 17–13
Capacitors: diagrams of (a) parallel plate, (b) cylindrical (rolled up parallel plate). (c) Photo of some real capacitors.

A simple capacitor consists of a pair of parallel plates of area A separated by a small distance d (Fig. 17–13a). Often the two plates are rolled into the form of a cylinder with paper or other insulator separating the plates, Fig. 17–13b; Fig. 17–13c is a photo of some actual capacitors used for various applications. In circuit diagrams, the symbol

⊣⊢ or ⊣([capacitor symbol]

represents a capacitor. A battery, which is a source of voltage, is indicated by the symbol

⊣⊢ (+ and − at the long and short arms) [battery symbol]

with unequal arms.

If a voltage is applied across a capacitor by connecting the capacitor to a battery with conducting wires as in Fig. 17–14, charge flows from the battery to each of the two plates: one plate acquires a negative charge, the other an equal amount of positive charge. Each battery terminal and the plate of the capacitor connected to it are at the same potential; hence the full battery voltage appears across the capacitor. For a given capacitor, it is found that the amount of charge Q acquired by each plate is proportional to the magnitude of the potential difference V between the plates:

$$Q = CV. \qquad \textbf{(17–7)}$$

The constant of proportionality, C, in Eq. 17–7 is called the **capacitance** of the capacitor. The unit of capacitance is coulombs per volt, and this unit is called a **farad** (F). Common capacitors have capacitance in the range of 1 pF (picofarad $= 10^{-12}$ F) to $10^3\,\mu$F (microfarad $= 10^{-6}$ F). The relation, Eq. 17–7, was first suggested by Volta in the late eighteenth century.

In Eq. 17–7 and from now on, we will use simply V (in italics) to represent a potential difference, such as that produced by a battery, rather than V_{ba}, ΔV, or $V_b - V_a$, as previously.

Also, be sure not to confuse *italic* letters V and C which stand for voltage and capacitance, with non-italic V and C which stand for the units volts and coulombs.

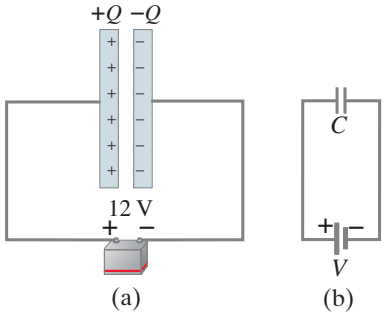
The capacitance C does not in general depend on Q or V. Its value depends only on the size, shape, and relative position of the two conductors, and also on the material that separates them. For a parallel-plate capacitor whose plates have area A and are separated by a distance d of air (Fig. 17–13a), the capacitance is given by

$$C = \epsilon_0 \frac{A}{d}. \qquad \text{[parallel-plate capacitor]} \qquad \textbf{(17–8)}$$

We see that C depends only on geometric factors, A and d, and not on Q or V. We derive this useful relation in the optional subsection at the end of this Section. The constant ϵ_0 is the *permittivity of free space*, which, as we saw in Chapter 16, has the value $8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2$.

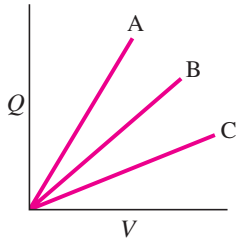
EXERCISE E Graphs for charge versus voltage are shown in Fig. 17–15 for three capacitors, A, B, and C. Which has the greatest capacitance?

FIGURE 17–14 (a) Parallel-plate capacitor connected to a battery. (b) Same circuit shown using symbols.



CAUTION
V = potential difference from here on

FIGURE 17–15 Exercise E.



EXAMPLE 17–8 **Capacitor calculations.** (*a*) Calculate the capacitance of a parallel-plate capacitor whose plates are 20 cm × 3.0 cm and are separated by a 1.0-mm air gap. (*b*) What is the charge on each plate if a 12-V battery is connected across the two plates? (*c*) What is the electric field between the plates? (*d*) Estimate the area of the plates needed to achieve a capacitance of 1 F, assuming the air gap d is 100 times smaller, or 10 microns (1 **micron** = 1 μm = 10^{-6} m).

APPROACH The capacitance is found by using Eq. 17–8, $C = \epsilon_0 A/d$. The charge on each plate is obtained from the definition of capacitance, Eq. 17–7, $Q = CV$. The electric field is uniform, so we can use Eq. 17–4b for the magnitude $E = V/d$. In (*d*) we use Eq. 17–8 again.

SOLUTION (*a*) The area $A = (20 \times 10^{-2}\,\text{m})(3.0 \times 10^{-2}\,\text{m}) = 6.0 \times 10^{-3}\,\text{m}^2$. The capacitance C is then

$$C = \epsilon_0 \frac{A}{d} = (8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)\frac{6.0 \times 10^{-3}\,\text{m}^2}{1.0 \times 10^{-3}\,\text{m}} = 53\,\text{pF}.$$

(*b*) The charge on each plate is

$$Q = CV = (53 \times 10^{-12}\,\text{F})(12\,\text{V}) = 6.4 \times 10^{-10}\,\text{C}.$$

(*c*) From Eq. 17–4b for a uniform electric field, the magnitude of E is

$$E = \frac{V}{d} = \frac{12\,\text{V}}{1.0 \times 10^{-3}\,\text{m}} = 1.2 \times 10^{4}\,\text{V/m}.$$

(*d*) We solve for A in Eq. 17–8 and substitute $C = 1.0\,\text{F}$ and $d = 1.0 \times 10^{-5}\,\text{m}$ to find that we need plates with an area

$$A = \frac{Cd}{\epsilon_0} \approx \frac{(1\,\text{F})(1.0 \times 10^{-5}\,\text{m})}{(9 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)} \approx 10^{6}\,\text{m}^2.$$

NOTE This is the area of a square 10^3 m or 1 km on a side. That is inconveniently large. Large-capacitance capacitors will not be simple parallel plates.

PHYSICS APPLIED
Capacitor as power backup; condenser microphone; computer keyboard

Not long ago, a capacitance greater than a few mF was unusual. Today capacitors are available that are 1 or 2 F, yet they are just a few cm on a side. Such capacitors are used as power backups, for example, in computer memory and electronics where the time and date can be maintained through tiny charge flow. [Capacitors are superior to rechargable batteries for this purpose because they can be recharged more than 10^5 times with no degradation.] Such high-capacitance capacitors can be made of *activated carbon* which has very high porosity, so that the surface area is very large; one-tenth of a gram of activated carbon can have a surface area of 100 m^2. Furthermore, the equal and opposite charges exist in an electric "double layer" about 10^{-9} m thick. Thus, the capacitance of 0.1 g of activated carbon, whose internal area can be $10^2\,\text{m}^2$, is equivalent to a parallel-plate capacitor with $C \approx \epsilon_0 A/d = (8.85 \times 10^{-12}\,\text{C}^2/\text{N}\cdot\text{m}^2)(10^2\,\text{m}^2)/(10^{-9}\,\text{m}) \approx 1\,\text{F}$.

The proportionality, $C \propto A/d$ in Eq. 17–8, is valid also for a parallel-plate capacitor that is rolled up into a spiral cylinder, as in Fig. 17–13b. However, the constant factor, ϵ_0, must be replaced if an insulator such as paper separates the plates, as is usual, as discussed in the next Section.

One type of microphone is a **condenser**, or capacitor, **microphone**, diagrammed in Fig. 17–16. The changing air pressure in a sound wave causes one plate of the capacitor C to move back and forth. The voltage across the capacitor changes at the same frequency as the sound wave.

Some computer keyboards operate by capacitance. As shown in Fig. 17–17, each key is connected to the upper plate of a capacitor. The upper plate moves down when the key is pressed, reducing the spacing between the capacitor plates, and increasing the capacitance (Eq. 17–8: smaller d, larger C). The *change* in capacitance results in an electric signal that is detected by an electronic circuit.

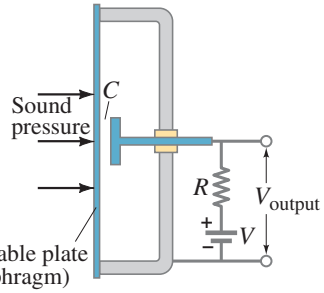
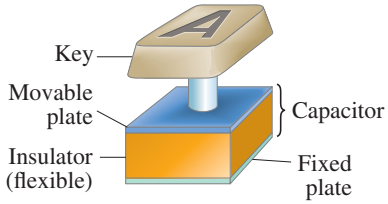


FIGURE 17–16 Diagram of a condenser microphone.

FIGURE 17–17 Key on a computer keyboard. Pressing the key reduces the plate spacing, increasing the capacitance.



* Derivation of Capacitance for Parallel-Plate Capacitor

Equation 17–8 can be derived using the result from Section 16–12 on Gauss's law, namely that the electric field between two parallel plates is given by Eq. 16–10:

$$E = \frac{Q/A}{\epsilon_0}.$$

We combine this with Eq. 17–4a, using magnitudes, $V = Ed$, to obtain

$$V = \left(\frac{Q}{A\epsilon_0}\right)d.$$

Then, from Eq. 17–7, the definition of capacitance,

$$C = \frac{Q}{V} = \frac{Q}{(Q/A\epsilon_0)d} = \epsilon_0\frac{A}{d}$$

which is Eq. 17–8.

17–8 Dielectrics

In most capacitors there is an insulating sheet of material, such as paper or plastic, called a **dielectric** between the plates (Fig. 17–18). This serves several purposes. First, dielectrics break down (allowing electric charge to flow) less readily than air, so higher voltages can be applied without charge passing across the gap. Furthermore, a dielectric allows the plates to be placed closer together without touching, thus allowing an increased capacitance because d is smaller in Eq. 17–8. Thirdly, it is found experimentally that if the dielectric fills the space between the two conductors, it increases the capacitance by a factor K, known as the **dielectric constant**. Thus, for a parallel-plate capacitor,

$$C = K\epsilon_0\frac{A}{d}. \qquad \textbf{(17–9)}$$

This can be written

$$C = \epsilon\frac{A}{d},$$

where $\epsilon = K\epsilon_0$ is called the **permittivity** of the material.

The values of the dielectric constant for various materials are given in Table 17–3. Also shown in Table 17–3 is the **dielectric strength**, the maximum electric field before breakdown (charge flow) occurs.

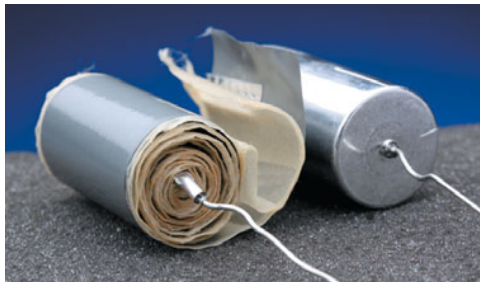


FIGURE 17–18 A cylindrical capacitor, unrolled from its case to show the dielectric between the plates. See also Fig. 17–13b.

CONCEPTUAL EXAMPLE 17–9 **Inserting a dielectric at constant V.** An air-filled capacitor consisting of two parallel plates separated by a distance d is connected to a battery of constant voltage V and acquires a charge Q. While it is still connected to the battery, a slab of dielectric material with $K = 3$ is inserted between the plates of the capacitor. Will Q increase, decrease, or stay the same?

RESPONSE Since the capacitor remains connected to the battery, the voltage stays constant and equal to the battery voltage V. The capacitance C increases when the dielectric material is inserted because K in Eq. 17–9 has increased. From the relation $Q = CV$, if V stays constant, but C increases, Q must increase as well. As the dielectric is inserted, more charge will be pulled from the battery and deposited onto the plates of the capacitor as its capacitance increases.

EXERCISE F If the dielectric in Example 17–9 fills the space between the plates, by what factor does (*a*) the capacitance change, (*b*) the charge on each plate change?

CONCEPTUAL EXAMPLE 17–10 **Inserting a dielectric into an isolated capacitor.** Suppose the air-filled capacitor of Example 17–9 is charged (to Q) and then disconnected from the battery. Next a dielectric is inserted between the plates. Will Q, C, or V change?

RESPONSE The charge Q remains the same—the capacitor is isolated, so there is nowhere for the charge to go. The capacitance increases as a result of inserting the dielectric (Eq. 17–9). The voltage across the capacitor also changes—it *decreases* because, by Eq. 17–7, $Q = CV$, so $V = Q/C$; if Q stays constant and C increases (it is in the denominator), then V decreases.

TABLE 17–3
Dielectric Constants (at 20°C)

Material	Dielectric constant K	Dielectric strength (V/m)
Vacuum	1.0000	
Air (1 atm)	1.0006	3×10^6
Paraffin	2.2	10×10^6
Polystyrene	2.6	24×10^6
Vinyl (plastic)	2–4	50×10^6
Paper	3.7	15×10^6
Quartz	4.3	8×10^6
Oil	4	12×10^6
Glass, Pyrex	5	14×10^6
Rubber, neoprene	6.7	12×10^6
Porcelain	6–8	5×10^6
Mica	7	150×10^6
Water (liquid)	80	
Strontium titanate	300	8×10^6

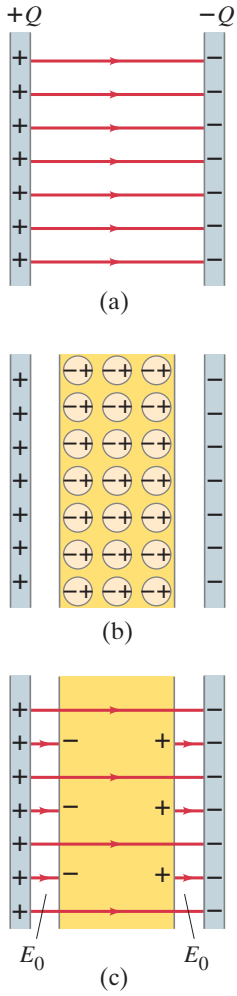


FIGURE 17–19 Molecular view of the effects of a dielectric.

*Molecular Description of Dielectrics

Let us examine, from the molecular point of view, why the capacitance of a capacitor should be larger when a dielectric is between the plates. A capacitor C_0 whose plates are separated by an air gap has a charge $+Q$ on one plate and $-Q$ on the other (Fig. 17–19a). Assume it is isolated (not connected to a battery) so charge cannot flow to or from the plates. The potential difference between the plates, V_0, is given by Eq. 17–7:

$$Q = C_0 V_0,$$

where the subscripts refer to air between the plates. Now we insert a dielectric between the plates (Fig. 17–19b). Because of the electric field between the capacitor plates, the dielectric molecules will tend to become oriented as shown in Fig. 17–19b. If the dielectric molecules are *polar*, the positive end is attracted to the negative plate and vice versa. Even if the dielectric molecules are not polar, electrons within them will tend to move slightly toward the positive capacitor plate, so the effect is the same. The net effect of the aligned dipoles is a net negative charge on the outer edge of the dielectric facing the positive plate, and a net positive charge on the opposite side, as shown in Fig. 17–19c.

Some of the electric field lines, then, do not pass through the dielectric but instead end on charges induced on the surface as shown in Fig. 17–19c. Hence the electric field within the dielectric is less than in air. That is, the electric field in the space between the capacitor plates, assumed filled by the dielectric, has been reduced by some factor K. The voltage across the capacitor is reduced by the same factor K because $V = Ed$ (Eq. 17–4) and hence, by Eq. 17–7, $Q = CV$, the capacitance C must increase by that same factor K to keep Q constant.

17–9 Storage of Electric Energy

A charged capacitor stores electric energy by separating + and − charges. The energy stored in a capacitor will be equal to the work done to charge it. The net effect of charging a capacitor is to remove charge from one plate and add it to the other plate. This is what a battery does when it is connected to a capacitor. A capacitor does not become charged instantly. It takes some time, often very little (Section 19–6). Initially, when the capacitor is uncharged, no work is required to move the first bit of charge over. As more charge is transferred, work is needed to move charge against the increasing voltage V. The work needed to add a small amount of charge Δq, when a potential difference V is across the plates, is $\Delta W = V\,\Delta q$. The total work needed to move total charge Q is equivalent to moving all the charge Q across a voltage equal to the *average* voltage during the process. (This is just like calculating the work done to compress a spring, Section 6–4, page 148.) The average voltage is $(V_f - 0)/2 = V_f/2$, where V_f is the final voltage; so the work to move the total charge Q from one plate to the other is

$$W = Q\frac{V_f}{2}.$$

Thus we can say that the electric potential energy, PE, stored in a capacitor is

$$\text{PE} = \text{energy} = \tfrac{1}{2}QV,$$

where V is the potential difference between the plates (we dropped the subscript), and Q is the charge on each plate. Since $Q = CV$, we can also write

$$\text{PE} = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \tfrac{1}{2}\frac{Q^2}{C}. \qquad \textbf{(17–10)}$$

PHYSICS APPLIED
Camera flash

EXAMPLE 17–11 **Energy stored in a capacitor.** A camera flash unit (Fig. 17–20) stores energy in a 660-μF capacitor at 330 V. (*a*) How much electric energy can be stored? (*b*) What is the power output if nearly all this energy is released in 1.0 ms?

APPROACH We use Eq. 17–10 in the form $\text{PE} = \frac{1}{2}CV^2$ because we are given C and V.

SOLUTION (*a*) The energy stored is

$$\text{PE} = \tfrac{1}{2}CV^2 = \tfrac{1}{2}(660 \times 10^{-6}\,\text{F})(330\,\text{V})^2 = 36\,\text{J}.$$

(*b*) If this energy is released in $\frac{1}{1000}$ of a second $\left(= 1.0\,\text{ms} = 1.0 \times 10^{-3}\,\text{s}\right)$, the power output is $P = \text{PE}/t = (36\,\text{J})/(1.0 \times 10^{-3}\,\text{s}) = 36{,}000\,\text{W}$.

EXERCISE G A capacitor stores 0.50 J of energy at 9.0 V. What is its capacitance?

CONCEPTUAL EXAMPLE 17–12 **Capacitor plate separation increased.** A parallel-plate capacitor carries charge Q and is then disconnected from a battery. The two plates are initially separated by a distance d. Suppose the plates are pulled apart until the separation is $2d$. How has the energy stored in this capacitor changed?

RESPONSE If we increase the plate separation d, we decrease the capacitance according to Eq. 17–8, $C = \epsilon_0 A/d$, by a factor of 2. The charge Q hasn't changed. So according to Eq. 17–10, where we choose the form $\text{PE} = \frac{1}{2}Q^2/C$ because we know Q is the same and C has been halved, the reduced C means the PE stored increases by a factor of 2.

NOTE We can see why the energy stored increases from a physical point of view: the two plates are charged equal and opposite, so they attract each other. If we pull them apart, we must do work, so we raise the potential energy.

It is useful to think of the energy stored in a capacitor as being stored in the electric field between the plates. As an example let us calculate the energy stored in a parallel-plate capacitor in terms of the electric field.

We have seen that the electric field $\vec{\mathbf{E}}$ between two close parallel plates is nearly uniform and its magnitude is related to the potential difference by $V = Ed$ (Eq. 17–4), where d is the separation. Also, Eq. 17–8 tells us $C = \epsilon_0 A/d$ for a parallel-plate capacitor. Thus

$$\begin{aligned}\text{PE} &= \tfrac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\epsilon_0 A}{d}\right)(E^2d^2) \\ &= \tfrac{1}{2}\epsilon_0 E^2 A d.\end{aligned}$$

The quantity Ad is the volume between the plates in which the electric field E exists. If we divide both sides of this equation by the volume, we obtain an expression for the energy per unit volume or **energy density**:

$$\text{energy density} = \frac{\text{PE}}{\text{volume}} = \tfrac{1}{2}\epsilon_0 E^2. \qquad \textbf{(17–11)}$$

The *electric energy stored per unit volume in any region of space is proportional to the square of the electric field* in that region. We derived Eq. 17–11 for the special case of a parallel-plate capacitor. But it can be shown to be true for any region of space where there is an electric field. Indeed, we will use this result when we discuss electromagnetic radiation (Chapter 22).

Health Effects

The energy stored in a large capacitance can give you a burn or a shock. One reason you are warned not to touch a circuit, or open an electronic device, is because capacitors may still be carrying charge even if the external power is turned off.

On the other hand, the basis of a heart *defibrillator* is a capacitor charged to a high voltage. A heart attack can be characterized by fast irregular beating of the heart, known as *ventricular* (or *cardiac*) *fibrillation*. The heart then does not pump blood to the rest of the body properly, and if the interruption lasts for long, death results. A sudden, brief jolt of charge through the heart from a defibrillator can cause complete heart stoppage, sometimes followed by a resumption of normal beating. The defibrillator capacitor is charged to a high voltage, typically a few thousand volts, and is allowed to discharge very rapidly through the heart via a pair of wide contacts known as "pads" or "paddles" that spread out the current over the chest (Fig. 17–21).

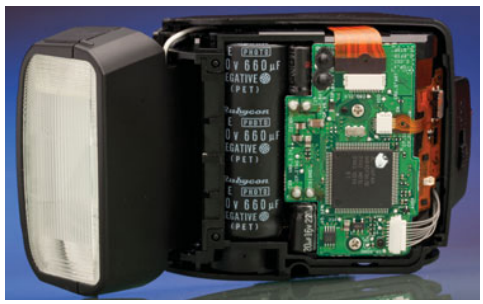


FIGURE 17–20 A camera flash unit. The 660-μF capacitor is the black cylinder.

PHYSICS APPLIED
Shocks, burns, defibrillators

FIGURE 17–21 Heart defibrillator.



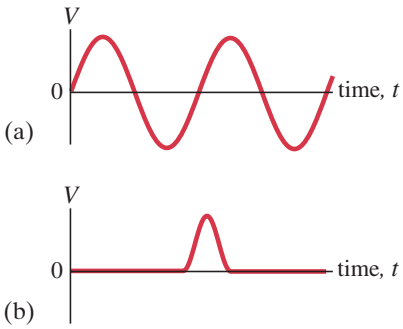


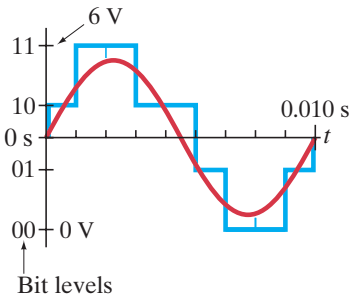
FIGURE 17–22 Two kinds of signal voltage: (a) sinusoidal, (b) a pulse, both analog. Many other shapes are possible.

TABLE 17–4
Binary to Decimal

Binary† number	Decimal number
00000000	0
00000001	1
00000010	2
00000011	3
00000100	4
00000111	7
00001000	8
00100101	37
11111111	255

†Note that we start counting from right to left: the 1's digit is on the far right, then the 2's, the 4's, the 8's, the 16's, the 32's, the 64's, and the 128's.

FIGURE 17–24 The red analog sine wave, which is at a 100-Hz frequency (1 wavelength is done in 0.010 s), has been converted to a 2-bit (4 level) digital signal (blue).



17–10 Digital; Binary Numbers; Signal Voltage

Batteries and a wall plug are meant to provide a constant **supply voltage** as power to operate a flashlight, an electric heater, and other electric and electronic devices.

A **signal voltage**, on the other hand, is a voltage intended to affect something else. A signal voltage varies in time and can also be very brief. For example, a sound such as a pure tone, which may be sinusoidal as we discussed in Chapters 11 and 12 (see Figs. 11–24 and 12–14), will produce an output voltage from a high quality microphone that is also sinusoidal. That signal voltage is amplified and reaches a loudspeaker, making it produce the sound we hear. Signal voltages (see Fig. 17–22) are sometimes a simple pulse (as in Figs. 11–23 and 11–33), and often act to change some aspect of an electronic device.

Signal voltages are sent to cell phones ("I've got signal"), to computers from the Internet, or to TV sets with the information on the picture and sound. Not long ago, signal voltages were **analog**—the voltage varied continuously, as in Fig. 17–22.

Today, television and computer signals are **digital** and use a binary number system to represent a numerical value. In a normal number, such as 609, there are *ten* choices for each digit—from 0 to 9—and normal numbers are called **decimal** (Latin for ten). In a **binary** number, each digit or **bit** has only *two* possibilities, 0 or 1 (sometimes referred to as "off" or "on"). In binary, 0001 means "one," 0010 means 2, 0011 means 3, and 1101 means $8 + 4 + 0 + 1 = 13$ in decimal. See Table 17–4, and note that counting starts from the right, just as in regular decimal (the "ones" digit is last, on the far right, then to the left is the "tens" and then "hundreds": for 609, the "ones" are 9, the "hundreds" are 6). Any value can be represented by a voltage pattern something like that shown in Fig. 17–23.

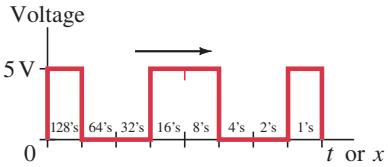


FIGURE 17–23 A traveling digital signal: voltage vs. position x or time t. If standing alone, this sequence would represent 10011001 or 153 $(= 128 + 0 + 0 + 16 + 8 + 0 + 0 + 1)$.

A "1" is a positive voltage such as +5 V, whereas a "0" is 0 V. The brightness signal, for example, that goes to each of the millions of tiny picture elements or "subpixels" of a TV or computer screen (Fig. 17–31, Section 17–11), is contained in a **byte**. One byte is 8 bits, which means

each byte of 8 bits allows $2^8 = 256$ possibilities

(that is, 0 to 255) or 256 shades for each of 3 colors: red, green, blue. The full color of each pixel (the three subpixel colors) has $(256)^3 = 17 \times 10^6$ possibilities. Digital television signals, which we discuss in the next Section, are transmitted at about 19 Mb/s = 19 Megabits per second. So 19×10^6 bits pass a given point per second, or one bit every 53 nanoseconds. We could write this in terms of bytes as 2.4 MB/s, where for bytes we use capital B.

When an analog signal, such as the pure sine wave of Fig. 17–22a, is converted to digital (**analog-to-digital converter**, ADC), the digital signal may look like the blue squared-off curve of Fig. 17–24. The digital signal has a limited number of discrete values. The difference between the original continuous analog signal and its digital approximation is called the **quantization error** or **quantization loss**. To minimize that loss, there are two important factors: (i) the **resolution** or **bit depth**, which is the number of bits or values for the voltage of each sample (= measurement); (ii) the **sampling rate**, which is the number of times per second the original analog voltage is measured ("sampled").

Consider a digital approximation for a 100-Hz sine wave: Figure 17–24 shows (i) a 0 to 6-V, 2-bit depth, measuring only 4 possible voltages (00, 01, 10, 11, or 0, 1, 2, 3 in decimal), and (ii) a sampling rate of (9 samples in one cycle or wavelength) × (100 cycles/s = 100 Hz) which is 900 samples/s or 900 Hz. This is very poor quality. For high quality reproduction, a greater bit depth and higher sampling rate are needed, which requires more memory, and more data to be transmitted.

For audio CDs, the sampling rate is 44.1 kHz (44,100 samplings every second) and 16-bit resolution, meaning each sampled voltage can have $2^8 \times 2^8 = 2^{16} \approx 65{,}000$ different voltage levels between, say, 0 and 5 volts. See Fig. 17–25 for details. Audio recording today typically uses 96 kHz and 24-bit ($2^{24} \approx 17 \times 10^6$ voltage levels) to give a better approximation of the original analog signal (on super-CDs or solid-state memory), but must be transferred down to 44.1 kHz and 16-bit to produce ordinary CDs. (DVDs can use 192 kHz sampling rate for sound.) But iPods and MP3 players have lower sampling rates and much less detail, which many listeners can notice.

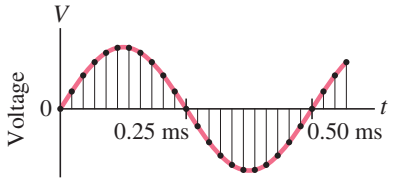
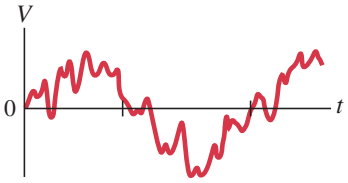


FIGURE 17–25 The sine wave shown could represent the analog electric signal from a microphone due to a pure 2000-Hz tone. (See Chapters 11 and 12.) The analog-to-digital electronics **samples** the signal—that is, measures and records the signal's voltage at intervals, many times per second. Each dot on the curve represents the voltage measured (sampled) at that point. The sampling rate in this diagram is 44,100 each second, or 44.1 kHz, like a CD. That is, a sample is taken every (1 s)/44,100 = 0.000023 s = 0.023 ms. In 0.50 ms, as shown here, 22 samples (black dots) are taken. This is an alternate way to represent sampling compared to Fig. 17-24, and shows that we cannot see any changes that might happen between the samplings (dots).

Figure 17–25 gives some details about a pure 2000-Hz sound sampled at 44.1 kHz. Normal musical sounds are a complex summation of many such sine waves of different frequencies and amplitudes. A simple summation was shown in Fig. 12–14. Another example is shown in Fig. 17–26, where we can see that the fine details may be missed by a digital conversion. Look at Fig. 17–25: if that were 20,000 Hz (highest frequency of human hearing), it would be sampled only about two times per wavelength. Both those samples might be zero volts—obviously missing the entire waveform. Over many wavelengths, it might eventually reproduce the waveform somewhat well. But many sounds only last milliseconds, like the initial attack of a piano note or plucked guitar string. Many audiophiles hear the difference between an original vinyl record and its subsequent release as a CD at 44.1 kHz.

FIGURE 17–26 This type of complex signal is much more normal than the pure sine wave of Fig. 17–25. Sampling may not catch all the details, especially because the waveform is changing very fast in time.



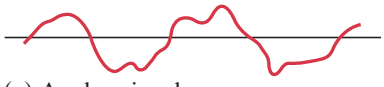
Digital audio signals must be converted back to analog (**digital-to-analog converter**, DAC) before being sent to a loudspeaker or headset. Even in a TV, the digital signals are converted to analog voltages before addressing the pixels (next Section), although the picture itself might be said to be digital since it is made up of separate pixels.

Digital photographs are made up of millions of "pixels" to produce a sharp image that is not "pixelated" or blurry. Also important (and complicated) are the number of bits provided for colors, plus the ability of the sensors (Chapter 25) to sustain a wide range of brightnesses under dim and bright light conditions.

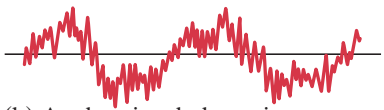
Digital data has some real advantages: for one, it can be **compressed**, in the sense that repeated information can be reduced so that less memory space is required—fewer bits and bytes. For example, adjacent "pixels" on a photograph that includes a blue sky may be essentially identical. If 200 almost identical pixels can be coded as identical, that takes up less memory (or "size") than to specify all the 200 pixels individually. Compression schemes, like **jpeg** for photos, lose some information and may be noticeable. In audio, MP3 players use one-tenth the space that a CD does, but many listeners don't notice. Compression is one reason that more data or "information" can be transmitted digitally for a given **bandwidth**. [Bandwidth is the fixed range of frequencies allotted to each radio or TV station or Internet connection, and limits the number of bits transmitted per second.]

In audio, many listeners claim that digital does not match analog in full sound quality. And what about movies? Will digital ever match Technicolor?

FIGURE 17–27 (a) Original analog signal and (b) the same signal dirtied up by outside signals (= noise). (c) A digital signal is still readable (d) without error if the noise is not too great.



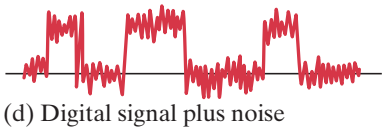
(a) Analog signal



(b) Analog signal plus noise



(c) Digital signal



(d) Digital signal plus noise

*Noise

Digital information transmission has another advantage: any distortion or unwanted (external) electrical signal that intrudes from outside, broadly called **noise**, can badly corrupt an analog signal: Fig. 17–27a shows a time-varying analog signal, and Fig. 17–27b shows nasty outside noise interfering with it. But a digital signal is still readable unless the noise is very large, on the order of half the bit signal itself (Figs. 17–27c and d).

*17–11 TV and Computer Monitors: CRTs, Flat Screens

The first television receivers used a **cathode ray tube** (**CRT**), and as recently as 2008 they accounted for half of all new TV sales. Two years later it was tough to find a new CRT set to buy. Even though new TV sets are flat screen plasma or **liquid crystal displays** (**LCD**), an understanding of how a CRT works is useful.

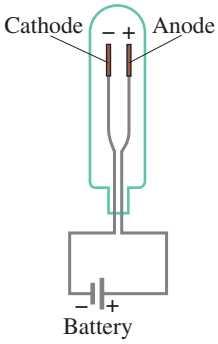


FIGURE 17–28 If the cathode inside the evacuated glass tube is heated to glowing (by an electric current, not shown), negatively charged "cathode rays" (= electrons) are "boiled off" and flow across to the anode (+), to which they are attracted.

*CRT

The operation of a CRT depends on **thermionic emission**, discovered by Thomas Edison (1847–1931). Consider a voltage applied to two small electrodes inside an evacuated glass "tube" as shown in Fig. 17–28: the **cathode** is negative, and the **anode** is positive. If the cathode is heated (usually by an electric current) so that it becomes hot and glowing, it is found that negative charges leave the cathode and flow to the positive anode. These negative charges are now called electrons, but originally they were called **cathode rays** because they seemed to come from the cathode (more detail in Section 27–1 on the discovery of the electron).

Figure 17–29 is a simplified sketch of a CRT which is contained in an evacuated glass tube. A beam of electrons, emitted by the heated cathode, is accelerated by the high-voltage anode and passes through a small hole in that anode. The inside of the tube face on the right (the screen) is coated with a fluorescent material that glows at the spot where the electrons hit. Voltage applied across the horizontal and vertical deflection plates, Fig. 17–29, can be varied to deflect the electron beam to different spots on the screen.

PHYSICS APPLIED
CRT

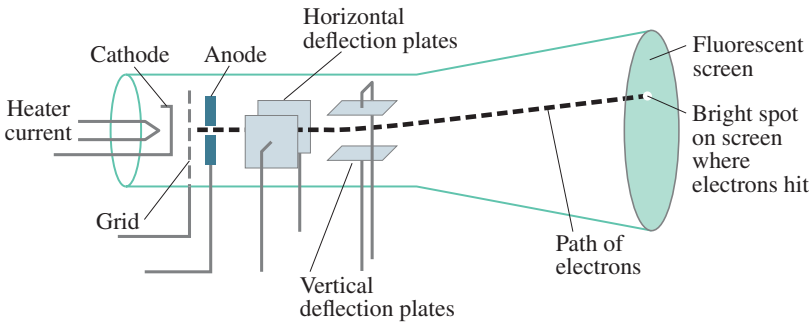
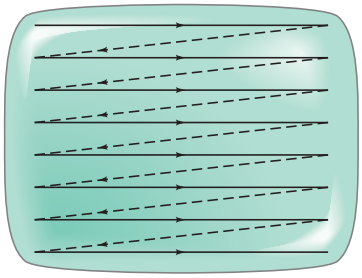


FIGURE 17–29 A cathode-ray tube. Magnetic deflection coils are commonly used in place of the electric deflection plates shown here. The relative positions of the elements have been exaggerated for clarity.

PHYSICS APPLIED
TV and computer monitors

In TV and computer monitors, the CRT electron beam sweeps over the screen in the manner shown in Fig. 17–30 by carefully synchronized voltages applied to the deflection plates (more commonly by magnetic deflection coils—Chapter 20). During each horizontal sweep of the electron beam, the **grid** (Fig. 17–29) receives a signal voltage that limits the flow of electrons at each instant during the sweep; the more negative the grid voltage is, the more electrons are repelled and fewer pass through, producing a less bright spot on the screen. Thus the varying grid voltage is responsible for the brightness of each spot on the screen. At the end of each horizontal sweep of the electron beam, the horizontal deflection voltage changes dramatically to bring the beam back to the opposite side of the screen, and the vertical voltage changes slightly so the beam begins a new horizontal sweep slightly below the previous one. The difference in brightness of the spots on the screen forms the "picture." **Color screens** have red, green, and blue phosphors which glow when struck by the electron beam. The various brightnesses of adjacent red, green, and blue phosphors (so close together we don't distinguish them) produce almost any color. Analog TV for the U.S. provided 480 visible horizontal sweeps[†] to form a complete picture every $\frac{1}{30}$ s. With 30 new frames or pictures every second (25 in countries with 50-Hz line voltage), a "moving picture" is displayed on the TV screen. (Note: commercial movies on film are 24 frames per second.)

FIGURE 17–30 Electron beam sweeps across a CRT television screen in a succession of horizontal lines. Each horizontal sweep is made by varying the voltage on the horizontal deflection plates (Fig. 17–29). Then the electron beam is moved down a short distance by a change in voltage on the vertical deflection plates, and the process (referred to as a raster) is repeated.



[†]525 lines in total, but only 480 form the picture; the other 45 lines contain other information such as synchronization. The sweep is **interlaced**: that is, every $\frac{1}{60}$ s every other line is traced, and in the next $\frac{1}{60}$ s, the lines in between are traced.

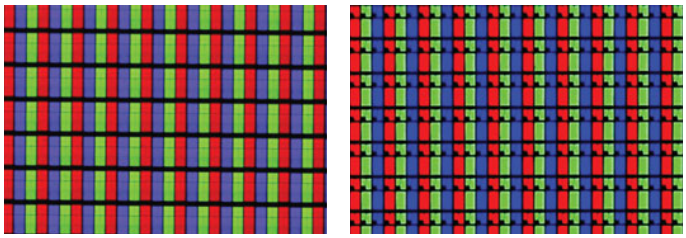


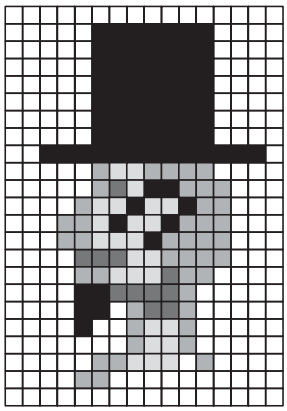
FIGURE 17–31 Close up of a tiny section of two typical LCD screens. You can even make out wires and transistors in the one on the right.

* Flat Screens and Addressing Pixels

PHYSICS APPLIED
How flat screens work

Today's flat screens contain millions of tiny *picture elements*, or **pixels**. Each pixel consists of 3 **subpixels**, a red, a green, and a blue. A close up of a common arrangement of pixels is shown in Fig. 17–31 for an LCD screen. (How liquid crystals work in an LCD screen is described in Section 24–11.) Subpixels are so small that at normal viewing distances we don't distinguish them and the separate red (R), green (G), and blue (B) subpixels blend to produce almost any color, depending on the relative brightnesses of the three subpixels. Liquid crystals act as filters (R, G, and B) that filter the light from a white **backlight**, usually fluorescent lamps or *light-emitting diodes* (LED, Section 29–9).† The picture you see on the screen depends on the level of brightness of each subpixel, as suggested in Fig. 17–32 for a simple black and white picture.

FIGURE 17–32 Example of an image made up of many small squares or *pixels* (picture elements). This one has rather low resolution.



High definition (HD) television screens have 1080 horizontal rows of pixels, each row consisting of 1920 pixels across the screen. That is, there are 1920 vertical columns, for a total of nearly 2 million pixels. Today, television in the U.S. is transmitted digitally at a rate of 60 Hz—that is, 60 frames or pictures per second (50 Hz in many countries) which makes the "moving picture." To form one frame, each subpixel must have the correct brightness. We now describe one way of doing this.

The brightness of each LCD subpixel (Section 24–11) depends on the voltage between its front and its back: if this voltage ΔV is zero, that subpixel is at maximum brightness; if ΔV is at its maximum (which might be +5 volts), that subpixel is dark.

Giving the correct voltage (to provide the correct brightness) is called **addressing** the subpixel. Typically the front of the subpixel is maintained at a positive voltage, such as +5 V. On the back of the display, the voltage at each subpixel is provided at the intersection of the 1080 horizontal wires (rows) and 1920 × 3 (colors) ≈ 6000 vertical wires (columns). See Fig. 17–33, which shows the array, or **matrix**, of wires. Each intersection of one vertical and one horizontal wire lies behind one subpixel. Because many frames are shown per second, the signal voltages applied are brief, like a pulse (see Fig. 17-22b or 11–23).

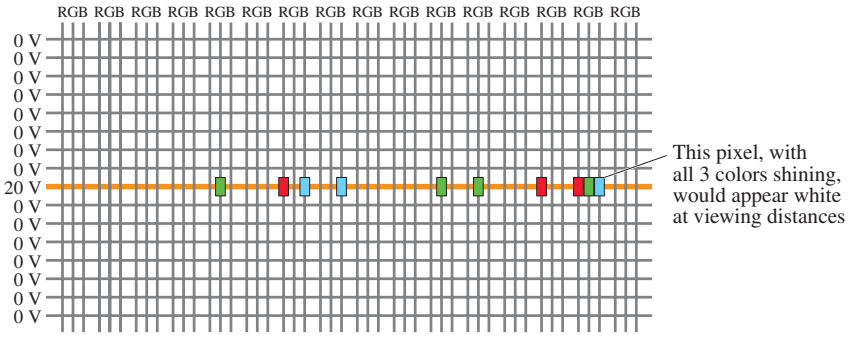


FIGURE 17–33 Array of wires (a matrix) behind all the pixels on an LCD screen. Each intersection of two wires is at a subpixel (red, green, or blue). One horizontal wire is activated at a time (the orange one at the moment shown) meaning it is at a positive voltage (+20 V) which allows that one row of pixels to be addressed at that moment; all other horizontal wires are at 0 V. At this moment, the data stream arrives to all the vertical wires, presenting the needed voltage (between 0 and 5 V) to produce the correct brightness for each of the nearly 6000 subpixels along the activated row.

The video signal that arrives at the display **activates** only one horizontal wire at a time (the orange one in Fig. 17–33): that one horizontal wire has a voltage (let's say +20 V) whereas all the others are at 0 V. That 20 V is not applied directly to the pixels, but *allows* the vertical wires to apply briefly the proper "signal voltage" to each subpixel along that row (via a transistor, see below). These signal voltages, known as the **data stream**, are applied to all the vertical wires just as that one row is activated: they provide the correct brightness for each subpixel in that activated row. A few subpixels are highlighted in Fig. 17–33. Immediately afterward, the other rows are activated, one by one, until the entire frame has been completed (in $\frac{1}{60}$ s).

†LEDs are discussed in Section 29–9. Home TVs advertised as LED generally mean an LCD screen with an LED backlight. LED pixels small enough for home screens are difficult to make, but actual LED screens are found in very large displays such as at stadiums.

Then a new frame is started. The addressing of subpixels for each row of each frame serves the same purpose as the sweep of the electron beam in a CRT, Fig. 17–30.

*Active Matrix (advanced)

High-definition displays use an **active matrix**, meaning that a tiny **thin-film transistor** (**TFT**) is attached to a corner of the back of each subpixel. (Transistors are discussed in Section 29–10.) One electrode of each TFT, called the "source," is connected to the vertical wire which addresses that subpixel, Fig. 17–34, and the "drain" electrode is connected to the back of the subpixel. The horizontal wire that serves the subpixel is connected to the transistor's **gate** electrode. The gate's voltage, by attracting charge or not, functions as a switch to connect or disconnect the source voltage to the drain and to the back of the subpixel (its front is fixed at +5 V). The potential difference ΔV across a subpixel determines if that subpixel will be bright in color ($\Delta V = 0$), black (ΔV = maximum), or something in between. See Fig. 17–35. All the subpixel TFTs along the one activated horizontal wire (the orange one in Fig. 17–33) will have +20 V at the gate: the TFTs are turned "on," like a switch. That allows electric charge to flow, connecting the vertical wire signal voltage at each TFT source to its drain and to the back of the subpixel. Thus all subpixels along one row receive the brightness needed for that line of the frame.

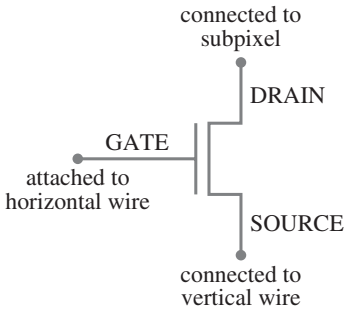


FIGURE 17–34 Thin-film transistor. One is attached to each screen subpixel.

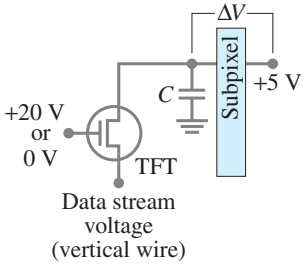


FIGURE 17–35 Circuit diagram for one subpixel. The front of the subpixel is at +5 V. If the TFT gate is at 20 V (horizontal wire activated), the data stream voltage is applied to the back of the subpixel, and determines the brightness of that subpixel. If the gate is at 0 V (horizontal wire not activated), the TFT is "off": no charge passes through, and the capacitance helps maintain ΔV until the subpixel is updated $\frac{1}{60}$ s later.

Within a subpixel's electronics is a capacitance that helps maintain the ΔV until that subpixel is **updated** with a new signal for the next frame, $\frac{1}{60}$ s later (≈ 17 ms). The row below the orange one shown in Fig. 17–33 is activated about 15 μs later $\left[= \left(\frac{1}{60}\text{ s}\right)\left(\frac{1}{1080\text{ lines}}\right)\right]$. The 6000 vertical wires (**data lines**) get their signal voltages (data stream) updated just before each row is activated in order to establish the brightness of each subpixel in that next row. All 1080 rows are activated, one-by-one, within $\frac{1}{60}$ s (≈ 17 ms) to complete that frame. Then a new frame is started.

New TV sets today can often refresh the screen at a higher rate. A **refresh rate** of 120 Hz (or 240 Hz) means that frames are interpolated between the normal ones, by averaging, which produces less blurring in fast action scenes.

Digital TV is transmitted at about 19 MB/s as mentioned in Section 17–10. (This rate is way too slow to do a full refresh every $\frac{1}{60}$ s—try the calculation and see—so a lot of compression is done and the areas where most movement occurs get refreshed.) The TV set or "box" that receives the digital video signal has to decode the signal in order to send analog voltages to the pixels of the screen, and at just the right time. TV stations in the U.S. are allowed to broadcast HD at 1080 × 1920 pixels or at 720 × 1280, or in standard definition (SD) of 480 × 704 pixels.

[When you read 1080p or 1080i for a TV, the "p" stands for "progressive," meaning an entire frame is made in $\frac{1}{60}$ s as described above. The "i" stands for "interlaced," meaning all the odd rows (half the picture) are done in $\frac{1}{60}$ s and then all the even rows are done in the next $\frac{1}{60}$ s, so a full picture is done at 30 per second or 30 Hz, thus reducing the data (or bit) rate. Analog TV (US) was 480i.]

FIGURE 17–36 An electrocardiogram (ECG) trace displayed on a CRT.



PHYSICS APPLIED
Oscilloscope

*Oscilloscopes

An **oscilloscope** is a device for amplifying, measuring, and visually displaying an electrical signal as a function of time on an LCD or CRT monitor, or computer screen. The visible "trace" on the screen, which could be an electrocardiogram (Fig. 17–36), or a signal from an experiment on nerve conduction, is a plot of the signal voltage (vertically) versus time (horizontally). [In a CRT, the electron beam is swept horizontally at a uniform rate in time by the horizontal deflection plates, Figs. 17–29 and 17–30. The signal to be displayed is applied (after amplification) to the vertical deflection plates.]

*17–12 Electrocardiogram (ECG or EKG)

Each time the heart beats, changes in electrical potential occur on its surface that can be detected using *electrodes* (metal contacts), which are attached to the skin. The changes in potential are small, on the order of millivolts (mV), and must be amplified. They are displayed with a chart recorder on paper, or on a monitor (CRT or LCD), Fig. 17–36. An **electrocardiogram** (ECG or EKG) is the record of the potential changes for a given person's heart. An example is shown in Fig. 17–37. We now look at the source of these potential changes and their relation to heart activity.

PHYSICS APPLIED
Electrocardiogram

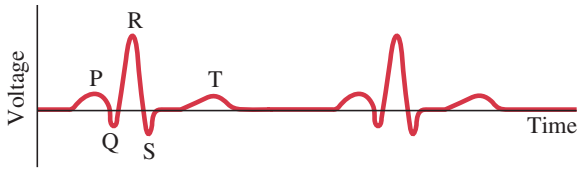
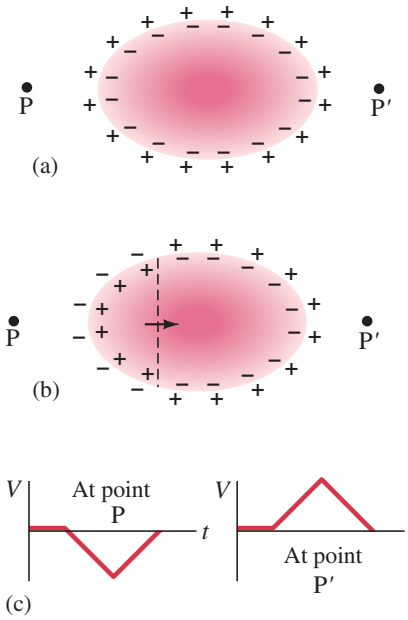


FIGURE 17–37 Typical ECG. Two heart beats are shown.

Both muscle and nerve cells have an electric dipole layer across the cell wall. That is, in the normal situation there is a net positive charge on the exterior surface and a net negative charge on the interior surface, Fig. 17–38a. The amount of charge depends on the size of the cell, but is approximately 10^{-3} C/m^2 of surface. For a cell whose surface area is 10^{-5} m^2, the total charge on either surface is thus $\approx 10^{-8}$ C. Just before the contraction of heart muscles, changes occur in the cell wall, so that positive ions on the exterior of the cell are able to pass through the wall and neutralize charge on the inside, or even make the inside surface slightly positive compared to the exterior. This "depolarization" starts at one end of the cell and progresses toward the opposite end, as indicated by the arrow in Fig. 17–38b, until the whole muscle is depolarized; the muscle then repolarizes to its original state (Fig. 17–38a), all in less than a second. Figure 17–38c shows rough graphs of the potential V as a function of time at the two points P and P′ (on either side of this cell) as the depolarization moves across the cell. The path of depolarization within the heart as a whole is more complicated, and produces the complex potential difference as a function of time, Fig. 17–37.

FIGURE 17–38 Heart muscle cell showing (a) charge dipole layer in resting state; (b) depolarization of cell progressing as muscle begins to contract; and (c) potential V at points P and P′ as a function of time.



It is standard procedure to divide a typical electrocardiogram into regions corresponding to the various deflections (or "waves"), as shown in Fig. 17–37. Each of the deflections corresponds to the activity of a particular part of the heart beat (Fig. 10–42). The P wave corresponds to contraction of the atria. The QRS group corresponds to contraction of the ventricles as the depolarization follows a very complicated path. The T wave corresponds to recovery (repolarization) of the heart in preparation for the next cycle.

The ECG is a powerful tool in identifying heart defects. For example, the right side of the heart enlarges if the right ventricle must push against an abnormally large load (as when blood vessels become hardened or clogged). This problem is readily observed on an ECG, because the S wave becomes very large (negatively). *Infarcts*, which are dead regions of the heart muscle that result from heart attacks, are also detected on an ECG because they reflect the depolarization wave.

Summary

The **electric potential** V at any point in space is defined as the electric potential energy per unit charge:

$$V_a = \frac{PE_a}{q}. \qquad \textbf{(17–2a)}$$

The **electric potential difference** between any two points is defined as the work done to move a 1 C electric charge between the two points. Potential difference is measured in volts (1 V = 1 J/C) and is often referred to as **voltage**.

The change in potential energy when a charge q moves through a potential difference V_{ba} is

$$\Delta PE = qV_{ba}. \qquad \textbf{(17–3)}$$

The potential difference V_{ba} between two points a and b where a uniform electric field E exists is given by

$$V_{ba} = -Ed, \qquad \textbf{(17–4a)}$$

where d is the distance between the two points.

An **equipotential line** or **surface** is all at the same potential, and is perpendicular to the electric field at all points.

The electric potential at a position P due to a single point charge Q, relative to zero potential at infinity, is given by

$$V = \frac{kQ}{r}, \qquad \textbf{(17–5)}$$

where r is the distance from Q to position P and $k = 1/4\pi\epsilon_0$.

[*The potential due to an **electric dipole** drops off as $1/r^2$. The **dipole moment** is $p = Q\ell$, where ℓ is the distance between the two equal but opposite charges of magnitude Q.]

A **capacitor** is a device used to store charge (and electric energy), and consists of two nontouching conductors. The two conductors hold equal and opposite charges, of magnitude Q. The ratio of this charge Q to the potential difference V between the conductors is called the **capacitance**, C:

$$C = \frac{Q}{V}, \quad \text{or} \quad Q = CV. \qquad \textbf{(17–7)}$$

The capacitance of a parallel-plate capacitor is proportional to the area A of each plate and inversely proportional to their separation d:

$$C = \epsilon_0 \frac{A}{d}. \qquad \textbf{(17–8)}$$

The space between the two conductors of a capacitor contains a nonconducting material such as air, paper, or plastic. These materials are referred to as **dielectrics**, and the capacitance is proportional to a property of dielectrics called the **dielectric constant**, K (equal to 1 for air).

A charged capacitor stores an amount of electric energy given by

$$\text{PE} = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \tfrac{1}{2}\frac{Q^2}{C}. \qquad \textbf{(17–10)}$$

This energy can be thought of as stored in the electric field between the plates.

The energy stored in any electric field E has a density

$$\frac{\text{electric PE}}{\text{volume}} = \tfrac{1}{2}\epsilon_0 E^2. \qquad \textbf{(17–11)}$$

Digital electronics converts an analog **signal voltage** into an approximate digital voltage based on a **binary code**: each **bit** has two possibilities, 1 or 0 (also "on" or "off"). The binary number 1101 equals 13. A **byte** is 8 bits and provides $2^8 = 256$ voltage levels. **Sampling rate** is the number of voltage measurements done on the analog signal per second. The **bit depth** is the number of digital voltage levels available at each sampling. CDs are 44.1 kHz, 16-bit.

[*TV and computer monitors traditionally used a **cathode ray tube** (CRT) which accelerates electrons by high voltage, and sweeps them across the screen in a regular way using magnetic coils or electric deflection plates. **LCD flat screens** contain millions of **pixels**, each with a red, green, and blue **subpixel** whose brightness is addressed every $\frac{1}{60}$ s via a **matrix** of horizontal and vertical wires using a **digital** (**binary**) code.]

[*An **electrocardiogram** (ECG or EKG) records the potential changes of each heart beat as the cells depolarize and repolarize.]

Questions

1. If two points are at the same potential, does this mean that no net work is done in moving a test charge from one point to the other? Does this imply that no force must be exerted? Explain.
2. If a negative charge is initially at rest in an electric field, will it move toward a region of higher potential or lower potential? What about a positive charge? How does the potential energy of the charge change in each instance? Explain.
3. State clearly the difference (*a*) between electric potential and electric field, (*b*) between electric potential and electric potential energy.
4. An electron is accelerated from rest by a potential difference of 0.20 V. How much greater would its final speed be if it is accelerated with four times as much voltage? Explain.
5. Is there a point along the line joining two equal positive charges where the electric field is zero? Where the electric potential is zero? Explain.
6. Can a particle ever move from a region of low electric potential to one of high potential and yet have its electric potential energy decrease? Explain.
7. If $V = 0$ at a point in space, must $\vec{\mathbf{E}} = 0$? If $\vec{\mathbf{E}} = 0$ at some point, must $V = 0$ at that point? Explain. Give examples for each.
8. Can two equipotential lines cross? Explain.
9. Draw in a few equipotential lines in Fig. 16–32b and c.
10. When a battery is connected to a capacitor, why do the two plates acquire charges of the same magnitude? Will this be true if the two plates are different sizes or shapes?
11. A conducting sphere carries a charge Q and a second identical conducting sphere is neutral. The two are initially isolated, but then they are placed in contact. (*a*) What can you say about the potential of each when they are in contact? (*b*) Will charge flow from one to the other? If so, how much?
12. The parallel plates of an isolated capacitor carry opposite charges, Q. If the separation of the plates is increased, is a force required to do so? Is the potential difference changed? What happens to the work done in the pulling process?
13. If the electric field $\vec{\mathbf{E}}$ is uniform in a region, what can you infer about the electric potential V? If V is uniform in a region of space, what can you infer about $\vec{\mathbf{E}}$?
14. Is the electric potential energy of two isolated unlike charges positive or negative? What about two like charges? What is the significance of the sign of the potential energy in each case?
15. If the voltage across a fixed capacitor is doubled, the amount of energy it stores (*a*) doubles; (*b*) is halved; (*c*) is quadrupled; (*d*) is unaffected; (*e*) none of these. Explain.
16. How does the energy stored in a capacitor change when a dielectric is inserted if (*a*) the capacitor is isolated so Q does not change; (*b*) the capacitor remains connected to a battery so V does not change? Explain.
17. A dielectric is pulled out from between the plates of a capacitor which remains connected to a battery. What changes occur to (*a*) the capacitance, (*b*) the charge on the plates, (*c*) the potential difference, (*d*) the energy stored in the capacitor, and (*e*) the electric field? Explain your answers.
18. We have seen that the capacitance C depends on the size and position of the two conductors, as well as on the dielectric constant K. What then did we mean when we said that C is a constant in Eq. 17–7?

MisConceptual Questions

1. A $+0.2\ \mu$C charge is in an electric field. What happens if that charge is replaced by a $+0.4\ \mu$C charge?
(*a*) The electric potential doubles, but the electric potential energy stays the same.
(*b*) The electric potential stays the same, but the electric potential energy doubles.
(*c*) Both the electric potential and electric potential energy double.
(*d*) Both the electric potential and electric potential energy stay the same.

2. Two identical positive charges are placed near each other. At the point halfway between the two charges,
(*a*) the electric field is zero and the potential is positive.
(*b*) the electric field is zero and the potential is zero.
(*c*) the electric field is not zero and the potential is positive.
(*d*) the electric field is not zero and the potential is zero.
(*e*) None of these statements is true.

3. Four identical point charges are arranged at the corners of a square [*Hint*: Draw a figure]. The electric field E and potential V at the center of the square are
(*a*) $E = 0,\ V = 0$.
(*b*) $E = 0,\ V \neq 0$.
(*c*) $E \neq 0,\ V \neq 0$.
(*d*) $E \neq 0,\ V = 0$.
(*e*) $E = V$ regardless of the value.

4. Which of the following statements is valid?
(*a*) If the potential at a particular point is zero, the field at that point must be zero.
(*b*) If the field at a particular point is zero, the potential at that point must be zero.
(*c*) If the field throughout a particular region is constant, the potential throughout that region must be zero.
(*d*) If the potential throughout a particular region is constant, the field throughout that region must be zero.

5. If it takes an amount of work W to move two $+q$ point charges from infinity to a distance d apart from each other, then how much work should it take to move three $+q$ point charges from infinity to a distance d apart from each other?
(*a*) $2W$.
(*b*) $3W$.
(*c*) $4W$.
(*d*) $6W$.

6. A proton ($Q = +e$) and an electron ($Q = -e$) are in a constant electric field created by oppositely charged plates. You release the proton from near the positive plate and the electron from near the negative plate. Which feels the larger electric force?
(*a*) The proton.
(*b*) The electron.
(*c*) Neither—there is no force.
(*d*) The magnitude of the force is the same for both and in the same direction.
(*e*) The magnitude of the force is the same for both but in opposite directions.

7. When the proton and electron in MisConceptual Question 6 strike the opposite plate, which one has more kinetic energy?
(*a*) The proton.
(*b*) The electron.
(*c*) Both acquire the same kinetic energy.
(*d*) Neither—there is no change in kinetic energy.
(*e*) They both acquire the same kinetic energy but with opposite signs.

8. Which of the following do not affect capacitance?
(*a*) Area of the plates.
(*b*) Separation of the plates.
(*c*) Material between the plates.
(*d*) Charge on the plates.
(*e*) Energy stored in the capacitor.

9. A battery establishes a voltage V on a parallel-plate capacitor. After the battery is disconnected, the distance between the plates is doubled without loss of charge. Accordingly, the capacitance _____ and the voltage between the plates _____.
(*a*) increases; decreases.
(*b*) decreases; increases.
(*c*) increases; increases.
(*d*) decreases; decreases.
(*e*) stays the same; stays the same.

10. Which of the following is a vector?
(*a*) Electric potential.
(*b*) Electric potential energy.
(*c*) Electric field.
(*d*) Equipotential lines.
(*e*) Capacitance.

11. A $+0.2\ \mu$C charge is in an electric field. What happens if that charge is replaced by a $-0.2\ \mu$C charge?
(*a*) The electric potential changes sign, but the electric potential energy stays the same.
(*b*) The electric potential stays the same, but the electric potential energy changes sign.
(*c*) Both the electric potential and electric potential energy change sign.
(*d*) Both the electric potential and electric potential energy stay the same.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

17–1 to 17–4 Electric Potential

1. (I) How much work does the electric field do in moving a $-7.7\,\mu$C charge from ground to a point whose potential is +65 V higher?
2. (I) How much work does the electric field do in moving a proton from a point at a potential of +125 V to a point at −45 V? Express your answer both in joules and electron volts.
3. (I) What potential difference is needed to stop an electron that has an initial velocity $v = 6.0 \times 10^5$ m/s?
4. (I) How much kinetic energy will an electron gain (in joules and eV) if it accelerates through a potential difference of 18,500 V?
5. (I) An electron acquires 6.45×10^{-16} J of kinetic energy when it is accelerated by an electric field from plate A to plate B. What is the potential difference between the plates, and which plate is at the higher potential?
6. (I) How strong is the electric field between two parallel plates 6.8 mm apart if the potential difference between them is 220 V?
7. (I) An electric field of 525 V/m is desired between two parallel plates 11.0 mm apart. How large a voltage should be applied?
8. (I) The electric field between two parallel plates connected to a 45-V battery is 1900 V/m. How far apart are the plates?
9. (I) What potential difference is needed to give a helium nucleus ($Q = 2e$) 85.0 keV of kinetic energy?
10. (II) Two parallel plates, connected to a 45-V power supply, are separated by an air gap. How small can the gap be if the air is not to become conducting by exceeding its breakdown value of $E = 3 \times 10^6$ V/m?
11. (II) The work done by an external force to move a $-6.50\,\mu$C charge from point A to point B is 15.0×10^{-4} J. If the charge was started from rest and had 4.82×10^{-4} J of kinetic energy when it reached point B, what must be the potential difference between A and B?
12. (II) What is the speed of an electron with kinetic energy (*a*) 850 eV, and (*b*) 0.50 keV?
13. (II) What is the speed of a proton whose KE is 4.2 keV?
14. (II) An alpha particle (which is a helium nucleus, $Q = +2e$, $m = 6.64 \times 10^{-27}$ kg) is emitted in a radioactive decay with KE = 5.53 MeV. What is its speed?
15. (II) An electric field greater than about 3×10^6 V/m causes air to break down (electrons are removed from the atoms and then recombine, emitting light). See Section 17–2 and Table 17–3. If you shuffle along a carpet and then reach for a doorknob, a spark flies across a gap you estimate to be 1 mm between your finger and the doorknob. Estimate the voltage between your finger and the doorknob. Why is no harm done?
16. (II) An electron starting from rest acquires 4.8 keV of KE in moving from point A to point B. (*a*) How much KE would a proton acquire, starting from rest at B and moving to point A? (*b*) Determine the ratio of their speeds at the end of their respective trajectories.
17. (II) Draw a conductor in the oblong shape of a football. This conductor carries a net negative charge, $-Q$. Draw in a dozen or so electric field lines and equipotential lines.

17–5 Potential Due to Point Charges

[Let $V = 0$ at $x = \infty$.]

18. (I) What is the electric potential 15.0 cm from a 3.00 μC point charge?
19. (I) A point charge Q creates an electric potential of +165 V at a distance of 15 cm. What is Q?
20. (II) A +35 μC point charge is placed 46 cm from an identical +35 μC charge. How much work would be required to move a +0.50 μC test charge from a point midway between them to a point 12 cm closer to either of the charges?
21. (II) (*a*) What is the electric potential 2.5×10^{-15} m away from a proton (charge $+e$)? (*b*) What is the electric potential energy of a system that consists of two protons 2.5×10^{-15} m apart—as might occur inside a typical nucleus?
22. (II) Three point charges are arranged at the corners of a square of side ℓ as shown in Fig. 17–39. What is the potential at the fourth corner (point A)?

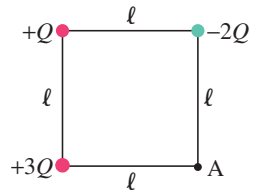


FIGURE 17–39
Problem 22.

23. (II) An electron starts from rest 24.5 cm from a fixed point charge with $Q = -6.50$ nC. How fast will the electron be moving when it is very far away?
24. (II) Two identical +9.5 μC point charges are initially 5.3 cm from each other. If they are released at the same instant from rest, how fast will each be moving when they are very far away from each other? Assume they have identical masses of 1.0 mg.
25. (II) Two point charges, 3.0 μC and −2.0 μC, are placed 4.0 cm apart on the x axis. At what points along the x axis is (*a*) the electric field zero and (*b*) the potential zero?
26. (II) How much work must be done to bring three electrons from a great distance apart to 1.0×10^{-10} m from one another (at the corners of an equilateral triangle)?
27. (II) Point a is 62 cm north of a $-3.8\,\mu$C point charge, and point b is 88 cm west of the charge (Fig. 17–40). Determine (*a*) $V_b - V_a$ and (*b*) $\vec{\mathbf{E}}_b - \vec{\mathbf{E}}_a$ (magnitude and direction).

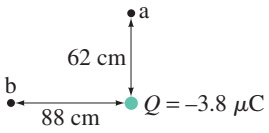


FIGURE 17–40
Problem 27.

28. (II) Many chemical reactions release energy. Suppose that at the beginning of a reaction, an electron and proton are separated by 0.110 nm, and their final separation is 0.100 nm. How much electric potential energy was lost in this reaction (in units of eV)?

29. (III) How much voltage must be used to accelerate a proton (radius 1.2×10^{-15} m) so that it has sufficient energy to just "touch" a silicon nucleus? A silicon nucleus has a charge of $+14e$, and its radius is about 3.6×10^{-15} m. Assume the potential is that for point charges.

30. (III) Two equal but opposite charges are separated by a distance d, as shown in Fig. 17–41. Determine a formula for $V_{BA} = V_B - V_A$ for points B and A on the line between the charges situated as shown.

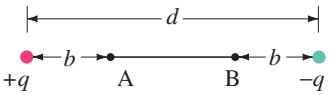


FIGURE 17–41
Problem 30.

31. (III) In the Bohr model of the hydrogen atom, an electron orbits a proton (the nucleus) in a circular orbit of radius 0.53×10^{-10} m. (*a*) What is the electric potential at the electron's orbit due to the proton? (*b*) What is the kinetic energy of the electron? (*c*) What is the total energy of the electron in its orbit? (*d*) What is the *ionization energy*—that is, the energy required to remove the electron from the atom and take it to $r = \infty$, at rest? Express the results of parts (*b*), (*c*), and (*d*) in joules and eV.

*17–6 Electric Dipoles

*32. (I) An electron and a proton are 0.53×10^{-10} m apart. What is their dipole moment if they are at rest?

*33. (II) Calculate the electric potential due to a dipole whose dipole moment is 4.2×10^{-30} C·m at a point 2.4×10^{-9} m away if this point is (*a*) along the axis of the dipole nearer the positive charge; (*b*) 45° above the axis but nearer the positive charge; (*c*) 45° above the axis but nearer the negative charge.

*34. (III) The dipole moment, considered as a vector, points from the negative to the positive charge. The water molecule, Fig. 17–42, has a dipole moment $\vec{\mathbf{p}}$ which can be considered as the vector sum of the two dipole moments, $\vec{\mathbf{p}}_1$ and $\vec{\mathbf{p}}_2$, as shown. The distance between each H and the O is about 0.96×10^{-10} m. The lines joining the center of the O atom with each H atom make an angle of 104°, as shown, and the net dipole moment has been measured to be $p = 6.1 \times 10^{-30}$ C·m. Determine the charge q on each H atom.

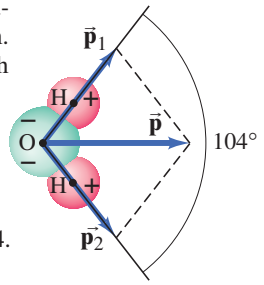


FIGURE 17–42 Problem 34. A water molecule, H_2O.

17–7 Capacitance

35. (I) The two plates of a capacitor hold $+2500\ \mu$C and $-2500\ \mu$C of charge, respectively, when the potential difference is 960 V. What is the capacitance?

36. (I) An 8500-pF capacitor holds plus and minus charges of 16.5×10^{-8} C. What is the voltage across the capacitor?

37. (I) How much charge flows from each terminal of a 12.0-V battery when it is connected to a 5.00-μF capacitor?

38. (I) A 0.20-F capacitor is desired. What area must the plates have if they are to be separated by a 3.2-mm air gap?

39. (II) The charge on a capacitor increases by 15 μC when the voltage across it increases from 97 V to 121 V. What is the capacitance of the capacitor?

40. (II) An electric field of 8.50×10^5 V/m is desired between two parallel plates, each of area 45.0 cm^2 and separated by 2.45 mm of air. What charge must be on each plate?

41. (II) If a capacitor has opposite 4.2 μC charges on the plates, and an electric field of 2.0 kV/mm is desired between the plates, what must each plate's area be?

42. (II) It takes 18 J of energy to move a 0.30-mC charge from one plate of a 15-μF capacitor to the other. How much charge is on each plate?

43. (II) To get an idea how big a farad is, suppose you want to make a 1-F air-filled parallel-plate capacitor for a circuit you are building. To make it a reasonable size, suppose you limit the plate area to 1.0 cm^2. What would the gap have to be between the plates? Is this practically achievable?

44. (II) How strong is the electric field between the plates of a 0.80-μF air-gap capacitor if they are 2.0 mm apart and each has a charge of 62 μC?

45. (III) A 2.50-μF capacitor is charged to 746 V and a 6.80-μF capacitor is charged to 562 V. These capacitors are then disconnected from their batteries. Next the positive plates are connected to each other and the negative plates are connected to each other. What will be the potential difference across each and the charge on each? [*Hint*: Charge is conserved.]

46. (III) A 7.7-μF capacitor is charged by a 165-V battery (Fig. 17–43a) and then is disconnected from the battery. When this capacitor (C_1) is then connected (Fig. 17–43b) to a second (initially uncharged) capacitor, C_2, the final voltage on each capacitor is 15 V. What is the value of C_2? [*Hint*: Charge is conserved.]

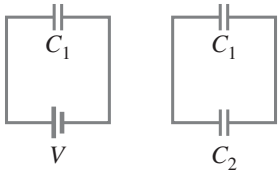


FIGURE 17–43
Problems 46 and 58.

17–8 Dielectrics

47. (I) What is the capacitance of two square parallel plates 6.6 cm on a side that are separated by 1.8 mm of paraffin?

48. (I) What is the capacitance of a pair of circular plates with a radius of 5.0 cm separated by 2.8 mm of mica?

49. (II) An uncharged capacitor is connected to a 21.0-V battery until it is fully charged, after which it is disconnected from the battery. A slab of paraffin is then inserted between the plates. What will now be the voltage between the plates?

50. (II) A 3500-pF air-gap capacitor is connected to a 32-V battery. If a piece of mica is placed between the plates, how much charge will flow from the battery?

51. (II) The electric field between the plates of a paper-separated $(K = 3.75)$ capacitor is 8.24×10^4 V/m. The plates are 1.95 mm apart, and the charge on each is 0.675 μC. Determine the capacitance of this capacitor and the area of each plate.

17–9 Electric Energy Storage

52. (I) 650 V is applied to a 2800-pF capacitor. How much energy is stored?

53. (I) A cardiac defibrillator is used to shock a heart that is beating erratically. A capacitor in this device is charged to 5.0 kV and stores 1200 J of energy. What is its capacitance?

54. (II) How much energy is stored by the electric field between two square plates, 8.0 cm on a side, separated by a 1.5-mm air gap? The charges on the plates are equal and opposite and of magnitude 370 μC.

55. (II) A homemade capacitor is assembled by placing two 9-in. pie pans 4 cm apart and connecting them to the opposite terminals of a 9-V battery. Estimate (*a*) the capacitance, (*b*) the charge on each plate, (*c*) the electric field halfway between the plates, and (*d*) the work done by the battery to charge them. (*e*) Which of the above values change if a dielectric is inserted?

56. (II) A parallel-plate capacitor has fixed charges $+Q$ and $-Q$. The separation of the plates is then halved. (*a*) By what factor does the energy stored in the electric field change? (*b*) How much work must be done to reduce the plate separation from d to $\frac{1}{2}d$? The area of each plate is A.

57. (II) There is an electric field near the Earth's surface whose magnitude is about 150 V/m. How much energy is stored per cubic meter in this field?

58. (III) A 3.70-μF capacitor is charged by a 12.0-V battery. It is disconnected from the battery and then connected to an uncharged 5.00-μF capacitor (Fig. 17–43). Determine the total stored energy (*a*) before the two capacitors are connected, and (*b*) after they are connected. (*c*) What is the change in energy?

17–10 Digital

59. (I) Write the decimal number 116 in binary.

60. (I) Write the binary number 01010101 as a decimal number.

61. (I) Write the binary number 1010101010101010 as a decimal number.

62. (II) Consider a rather coarse 4-bit analog-to-digital conversion where the maximum voltage is 5.0 V. (*a*) What voltage does 1011 represent? (*b*) What is the 4-bit representation for 2.0 V?

63. (II) (*a*) 16-bit sampling provides how many different possible voltages? (*b*) 24-bit sampling provides how many different possible voltages? (*c*) For color TV, 3 subpixels, each 8 bits, provides a total of how many different colors?

64. (II) A few extraterrestrials arrived. They had two hands, but claimed that 3 + 2 = 11. How many fingers did they have on their two hands? Note that our decimal system (and ten characters: 0, 1, 2, ⋯, 9) surely has its origin because we have ten fingers. [*Hint*: 11 is in their system. In our decimal system, the result would be written as 5.]

*17–11 TV and Computer Monitors

***65.** (II) Figure 17–44 is a photograph of a computer screen shot by a camera set at an exposure time of $\frac{1}{4}$ s. During the exposure the cursor arrow was moved around by the mouse, and we see it 15 times. (*a*) Explain why we see the cursor 15 times. (*b*) What is the refresh rate of the screen?



FIGURE 17–44
Problem 65.

***66.** (III) In a given CRT, electrons are accelerated horizontally by 9.0 kV. They then pass through a uniform electric field E for a distance of 2.8 cm, which deflects them upward so they travel 22 cm to the top of the screen, 11 cm above the center. Estimate the value of E.

***67.** (III) Electrons are accelerated by 6.0 kV in a CRT. The screen is 30 cm wide and is 34 cm from the 2.6-cm-long deflection plates. Over what range must the horizontally deflecting electric field vary to sweep the beam fully across the screen?

General Problems

68. A lightning flash transfers 4.0 C of charge and 5.2 MJ of energy to the Earth. (*a*) Across what potential difference did it travel? (*b*) How much water could this boil and vaporize, starting from room temperature? (See also Chapter 14.)

69. In an older television tube, electrons are accelerated by thousands of volts through a vacuum. If a television set were laid on its back, would electrons be able to move upward against the force of gravity? What potential difference, acting over a distance of 2.4 cm, would be needed to balance the downward force of gravity so that an electron would remain stationary? Assume that the electric field is uniform.

70. How does the energy stored in a capacitor change, as the capacitor remains connected to a battery, if the separation of the plates is doubled?

71. How does the energy stored in an isolated capacitor change if (*a*) the potential difference is doubled, or (*b*) the separation of the plates is doubled?

72. A huge 4.0-F capacitor has enough stored energy to heat 2.8 kg of water from 21°C to 95°C. What is the potential difference across the plates?

73. A proton ($q = +e$) and an alpha particle ($q = +2e$) are accelerated by the same voltage V. Which gains the greater kinetic energy, and by what factor?

74. Dry air will break down if the electric field exceeds 3.0×10^6 V/m. What amount of charge can be placed on a parallel-plate capacitor if the area of each plate is 65 cm^2?

75. Three charges are at the corners of an equilateral triangle (side ℓ) as shown in Fig. 17–45. Determine the potential at the midpoint of each of the sides. Let $V = 0$ at $r = \infty$.

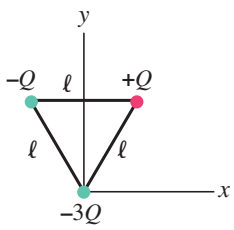


FIGURE 17–45
Problem 75.

76. It takes 15.2 J of energy to move a 13.0-mC charge from one plate of a 17.0-μF capacitor to the other. How much charge is on each plate? Assume constant voltage.

77. A 3.4 μC and a -2.6 μC charge are placed 2.5 cm apart. At what points along the line joining them is (*a*) the electric field zero, and (*b*) the electric potential zero?

78. Near the surface of the Earth there is an electric field of about 150 V/m which points downward. Two identical balls with mass $m = 0.670$ kg are dropped from a height of 2.00 m, but one of the balls is positively charged with $q_1 = 650$ μC, and the second is negatively charged with $q_2 = -650$ μC. Use conservation of energy to determine the difference in the speed of the two balls when they hit the ground. (Neglect air resistance.)

79. The power supply for a pulsed nitrogen laser has a 0.050-μF capacitor with a maximum voltage rating of 35 kV. (*a*) Estimate how much energy could be stored in this capacitor. (*b*) If 12% of this stored electrical energy is converted to light energy in a pulse that is 6.2 microseconds long, what is the power of the laser pulse?

80. In a **photocell**, ultraviolet (UV) light provides enough energy to some electrons in barium metal to eject them from the surface at high speed. To measure the maximum energy of the electrons, another plate above the barium surface is kept at a negative enough potential that the emitted electrons are slowed down and stopped, and return to the barium surface. See Fig. 17–46. If the plate voltage is -3.02 V (compared to the barium) when the fastest electrons are stopped, what was the speed of these electrons when they were emitted?

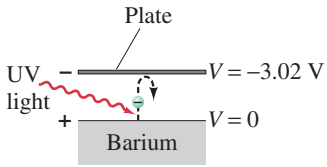


FIGURE 17–46
Problem 80.

81. A $+38$ μC point charge is placed 36 cm from an identical $+38$ μC charge. A -1.5 μC charge is moved from point A to point B as shown in Fig. 17–47. What is the change in potential energy?

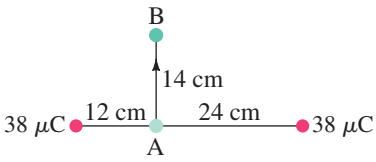


FIGURE 17–47
Problem 81.

82. Paper has a dielectric constant $K = 3.7$ and a dielectric strength of 15×10^6 V/m. Suppose that a typical sheet of paper has a thickness of 0.11 mm. You make a "homemade" capacitor by placing a sheet of 21×14 cm paper between two aluminum foil sheets (Fig. 17–48) of the same size. (*a*) What is the capacitance C of your device? (*b*) About how much charge could you store on your capacitor before it would break down?

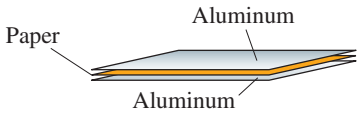


FIGURE 17–48
Problem 82.

83. A capacitor is made from two 1.1-cm-diameter coins separated by a 0.10-mm-thick piece of paper ($K = 3.7$). A 12-V battery is connected to the capacitor. How much charge is on each coin?

84. A $+3.5$ μC charge is 23 cm to the right of a -7.2 μC charge. At the midpoint between the two charges, (*a*) determine the potential and (*b*) the electric field.

85. A parallel-plate capacitor with plate area 3.0 cm^2 and air-gap separation 0.50 mm is connected to a 12-V battery, and fully charged. The battery is then disconnected. (*a*) What is the charge on the capacitor? (*b*) The plates are now pulled to a separation of 0.75 mm. What is the charge on the capacitor now? (*c*) What is the potential difference between the plates now? (*d*) How much work was required to pull the plates to their new separation?

86. A 2.1-μF capacitor is fully charged by a 6.0-V battery. The battery is then disconnected. The capacitor is not ideal and the charge slowly leaks out from the plates. The next day, the capacitor has lost half its stored energy. Calculate the amount of charge lost.

87. Two point charges are fixed 4.0 cm apart from each other. Their charges are $Q_1 = Q_2 = 6.5$ μC, and their masses are $m_1 = 1.5$ mg and $m_2 = 2.5$ mg. (*a*) If Q_1 is released from rest, what will be its speed after a very long time? (*b*) If both charges are released from rest at the same time, what will be the speed of Q_1 after a very long time?

88. Two charges are placed as shown in Fig. 17–49 with $q_1 = 1.2$ μC and $q_2 = -3.3$ μC. Find the potential difference between points A and B.

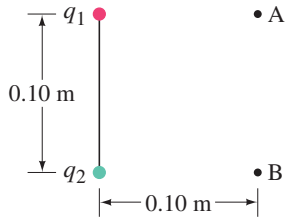


FIGURE 17–49
Problem 88.

89. If the electrons in a single raindrop, 3.5 mm in diameter, could be removed from the Earth (without removing the atomic nuclei), by how much would the potential of the Earth increase?

90. Thunderclouds may develop a voltage difference of about 5×10^7 V. Given that an electric field of 3×10^6 V/m is required to produce an electrical spark within a volume of air, estimate the length of a thundercloud lightning bolt. [Can you see why, when lightning strikes from a cloud to the ground, the bolt has to propagate as a sequence of steps?]

91. A manufacturer claims that a carpet will not generate more than 6.0 kV of static electricity. What magnitude of charge would have to be transferred between a carpet and a shoe for there to be a 6.0-kV potential difference between the shoe and the carpet? Approximate the area of the shoe and assume the shoe and carpet are large sheets of charge separated by a small distance $d = 1.0$ mm.

92. Compact "ultracapacitors" with capacitance values up to several thousand farads are now commercially available. One application for ultracapacitors is in providing power for electrical circuits when other sources (such as a battery) are turned off. To get an idea of how much charge can be stored in such a component, assume a 1200-F ultracapacitor is initially charged to 12.0 V by a battery and is then disconnected from the battery. If charge is then drawn off the plates of this capacitor at a rate of 1.0 mC/s, say, to power the backup memory of some electrical device, how long (in days) will it take for the potential difference across this capacitor to drop to 6.0 V?

93. An electron is accelerated horizontally from rest by a potential difference of 2200 V. It then passes between two horizontal plates 6.5 cm long and 1.3 cm apart that have a potential difference of 250 V (Fig. 17–50). At what angle θ will the electron be traveling after it passes between the plates?

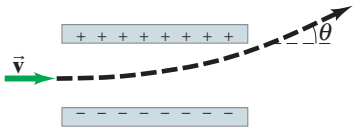


FIGURE 17–50
Problem 93.

94. In the **dynamic random access memory (DRAM)** of a computer, each memory cell contains a capacitor for charge storage. Each of these cells represents a single binary-bit value of "1" when its 35-fF capacitor $(1\ \text{fF} = 10^{-15}\ \text{F})$ is charged at 1.5 V, or "0" when uncharged at 0 V. (*a*) When fully charged, how many excess electrons are on a cell capacitor's negative plate? (*b*) After charge has been placed on a cell capacitor's plate, it slowly "leaks" off at a rate of about 0.30 fC/s. How long does it take for the potential difference across this capacitor to decrease by 2.0% from its fully charged value? (Because of this leakage effect, the charge on a DRAM capacitor is "refreshed" many times per second.) Note: A DRAM cell is shown in Fig. 21–29.

95. In the DRAM computer chip of Problem 94, suppose the two parallel plates of one cell's 35-fF capacitor are separated by a 2.0-nm-thick insulating material with dielectric constant $K = 25$. (*a*) Determine the area A $(\text{in } \mu\text{m}^2)$ of the cell capacitor's plates. (*b*) If the plate area A accounts for half of the area of each cell, estimate how many megabytes of memory can be placed on a 3.0-cm^2 silicon wafer. (1 byte = 8 bits.)

96. A parallel-plate capacitor with plate area $A = 2.0\ \text{m}^2$ and plate separation $d = 3.0$ mm is connected to a 35-V battery (Fig. 17–51a). (*a*) Determine the charge on the capacitor, the electric field, the capacitance, and the energy stored in the capacitor. (*b*) With the capacitor still connected to the battery, a slab of plastic with dielectric strength $K = 3.2$ is placed between the plates of the capacitor, so that the gap is completely filled with the dielectric (Fig. 17–51b). What are the new values of charge, electric field, capacitance, and the energy stored in the capacitor?

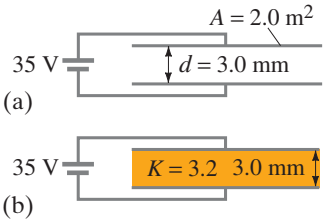


FIGURE 17–51
Problem 96.

Search and Learn

1. Make a list of rules for and properties of equipotential surfaces or lines. You should be able to find eight distinct rules in the text.

2. Figure 17–8 shows contour lines (elevations). Just for fun, assume they are equipotential lines on a flat 2-dimensional surface with the values shown being in volts. Estimate the magnitude and direction of the "electric field" (*a*) between Iceberg Lake and Cecile Lake and (*b*) at the Minaret Mine. Assume that up is $+y$, right is $+x$, and that Cecile Lake is about 1.0 km wide in the middle.

3. In lightning storms, the potential difference between the Earth and the bottom of thunderclouds may be 35,000,000 V. The bottoms of the thunderclouds are typically 1500 m above the Earth, and can have an area of 110 km^2. Modeling the Earth–cloud system as a huge capacitor, calculate (*a*) the capacitance of the Earth–cloud system, (*b*) the charge stored in the "capacitor," and (*c*) the energy stored in the "capacitor."

4. The potential energy stored in a capacitor (Section 17–9) can be written as either $CV^2/2$ or $Q^2/2C$. In the first case the energy is proportional to C; in the second case the energy is proportional to $1/C$. (*a*) Explain how both of these equations can be correct. (*b*) When might you use the first equation and when might you use the second equation? (*c*) If a paper dielectric is inserted into a parallel-plate capacitor that is attached to a battery (V does not change), by what factor will the energy stored in the capacitor change? (*d*) If a quartz dielectric is inserted into a charged parallel-plate capacitor that is isolated from any battery, by what factor will the energy stored in the capacitor change?

5. Suppose it takes 75 kW of power for your car to travel at a constant speed on the highway. (*a*) What is this in horsepower? (*b*) How much energy in joules would it take for your car to travel at highway speed for 5.0 hours? (*c*) Suppose this amount of energy is to be stored in the electric field of a parallel-plate capacitor (Section 17–9). If the voltage on the capacitor is to be 850 V, what is the required capacitance? (*d*) If this capacitor were to be made from activated carbon (Section 17–7), the voltage would be limited to no more than 10 V. In this case, how many grams of activated carbon would be required? (*e*) Is this practical?

6. Capacitors can be used as "electric charge counters." Consider an initially uncharged capacitor of capacitance C with its bottom plate grounded and its top plate connected to a source of electrons. (*a*) If N electrons flow onto the capacitor's top plate, show that the resulting potential difference V across the capacitor is directly proportional to N. (*b*) Assume the voltage-measuring device can accurately resolve voltage changes of about 1 mV. What value of C would be necessary to resolve the arrival of an individual electron? (*c*) Using modern semiconductor technology, a micron-size capacitor can be constructed with parallel conducting plates separated by an insulator of dielectric constant $K = 3$ and thickness $d = 100$ nm. What side length ℓ should the square plates have (in μm)?

ANSWERS TO EXERCISES

A: (*a*) -8.0×10^{-16} J; (*b*) 9.8×10^5 m/s.
B: (*c*).
C: 0.72 J.
D: (*c*).
E: A.
F: (*a*) 3 times greater; (*b*) 3 times greater.
G: 12 mF.